UNIVERSITY OF TRENTO

Data Visualization

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UNIVERSITY OF TRENTO

Lecturer



Head of the unit Digital Commons Lab (DCL) focusing on open data policies and software development.
Expert in data visualization
Lecture of Geospatial Analysis in the Data Science Master - UNITN
He leads different courses for a wide audience on topics like data analysis, geospatial analysis, and effective communication with the data.
Civic hacker who blending his interest for technology, data, and community through maps

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What we will learn



Beautiful does not mean effective
The more a graph needs an explanation, the less the graph is needed.



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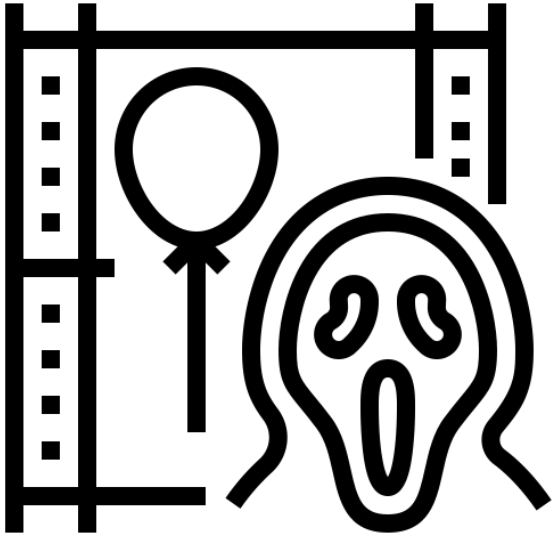
How we will do it



Gallery of horrors
Main errors
Principles of graphic excellence
Knowing how to choose a chart

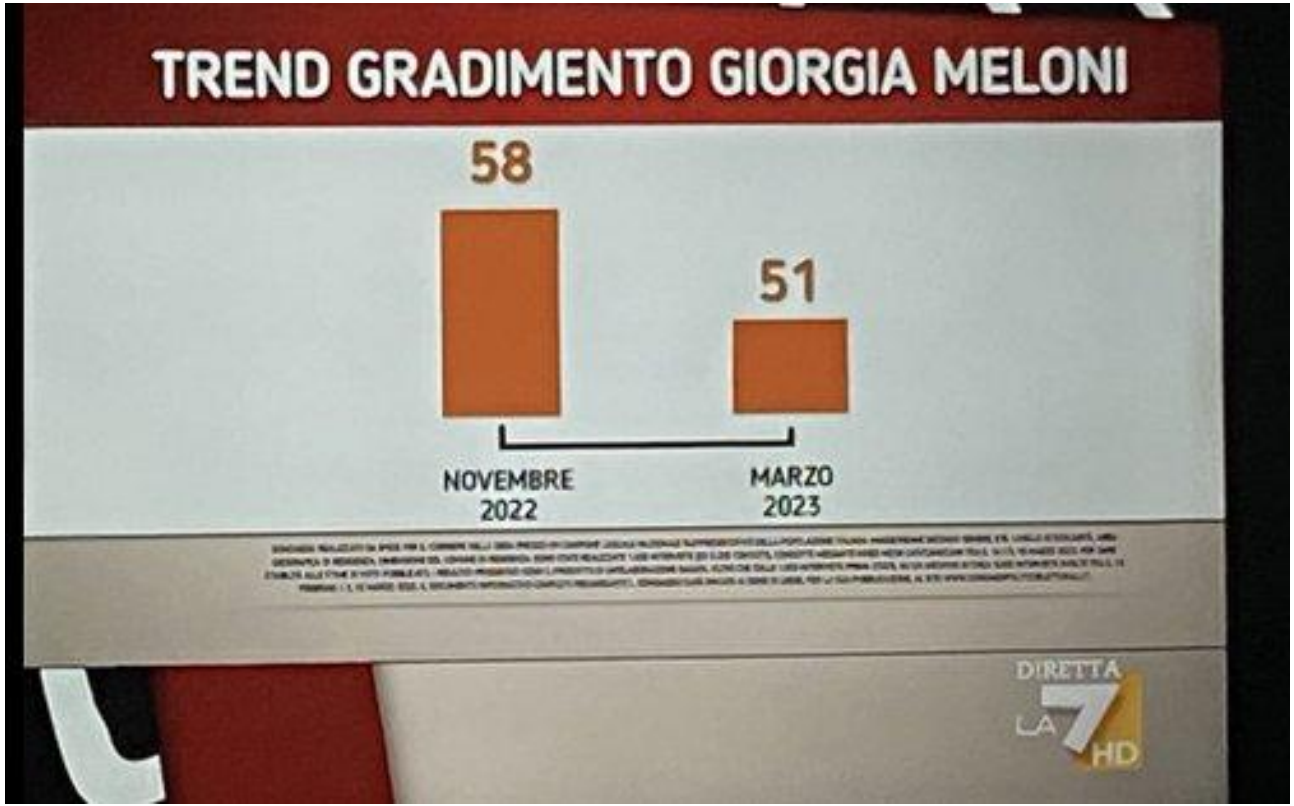
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Gallery of horrors

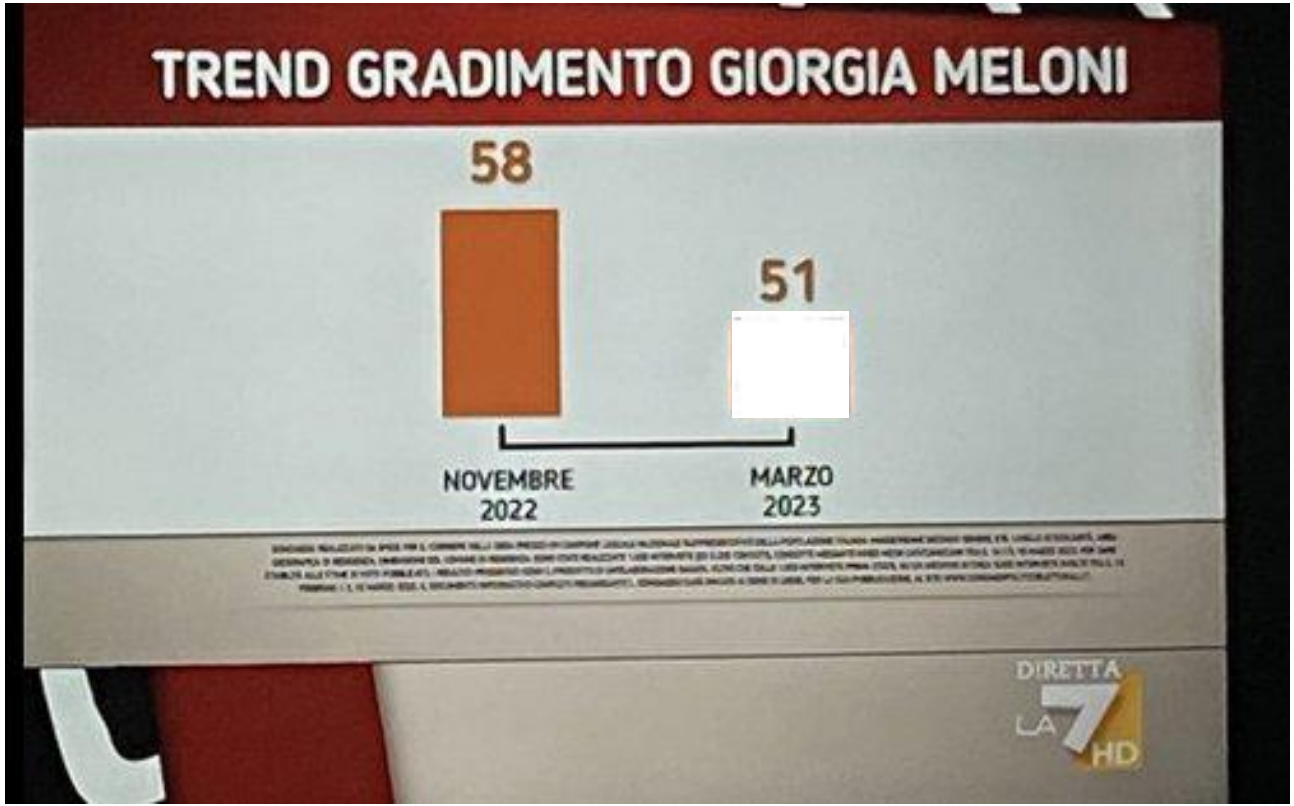


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a 'noticeable' 7 per cent improvement ...

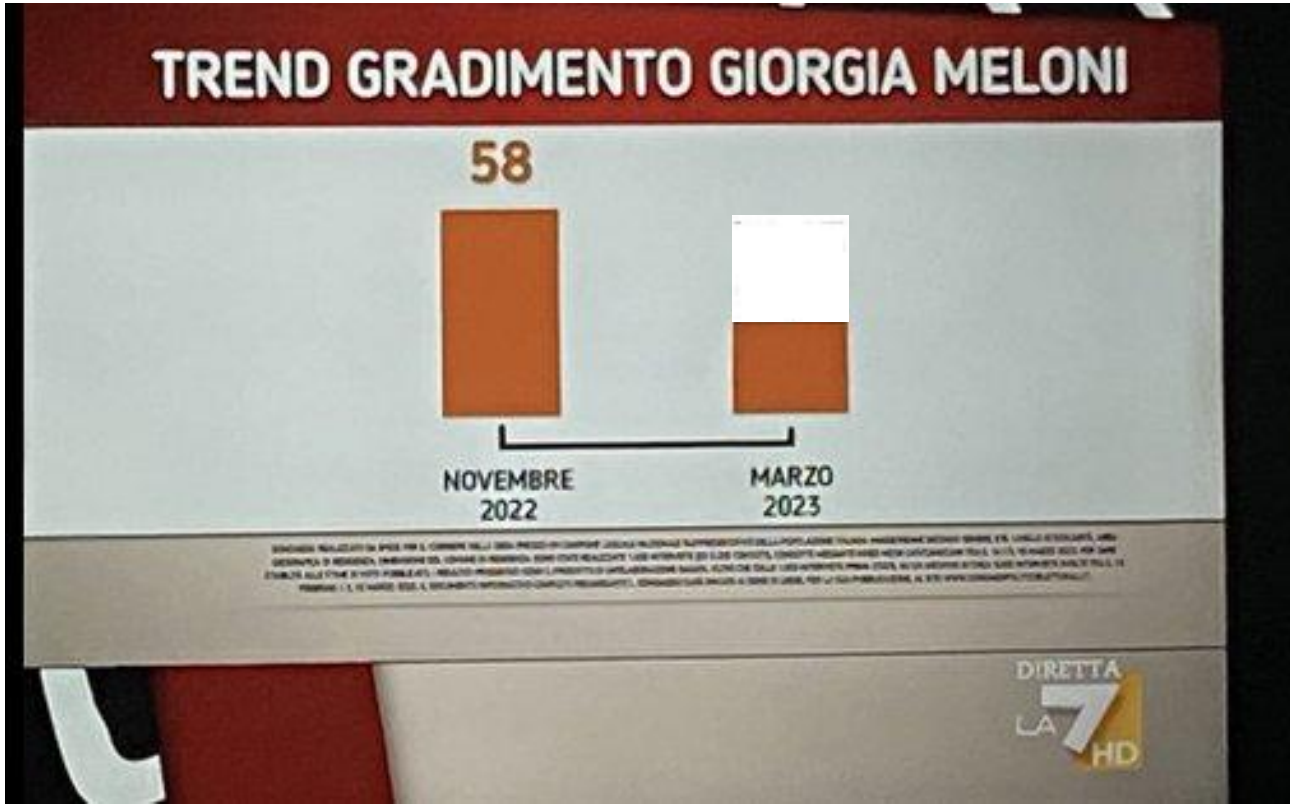


This block is 51% ...

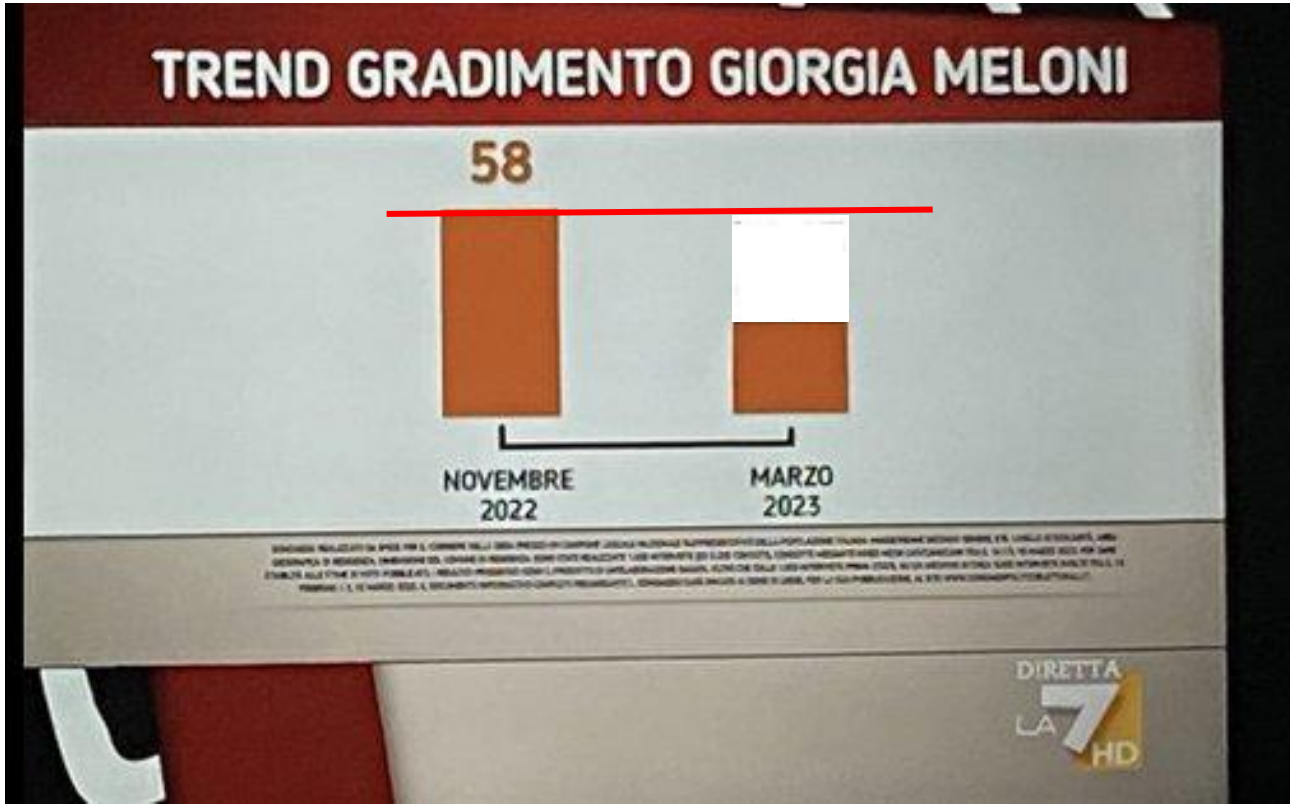


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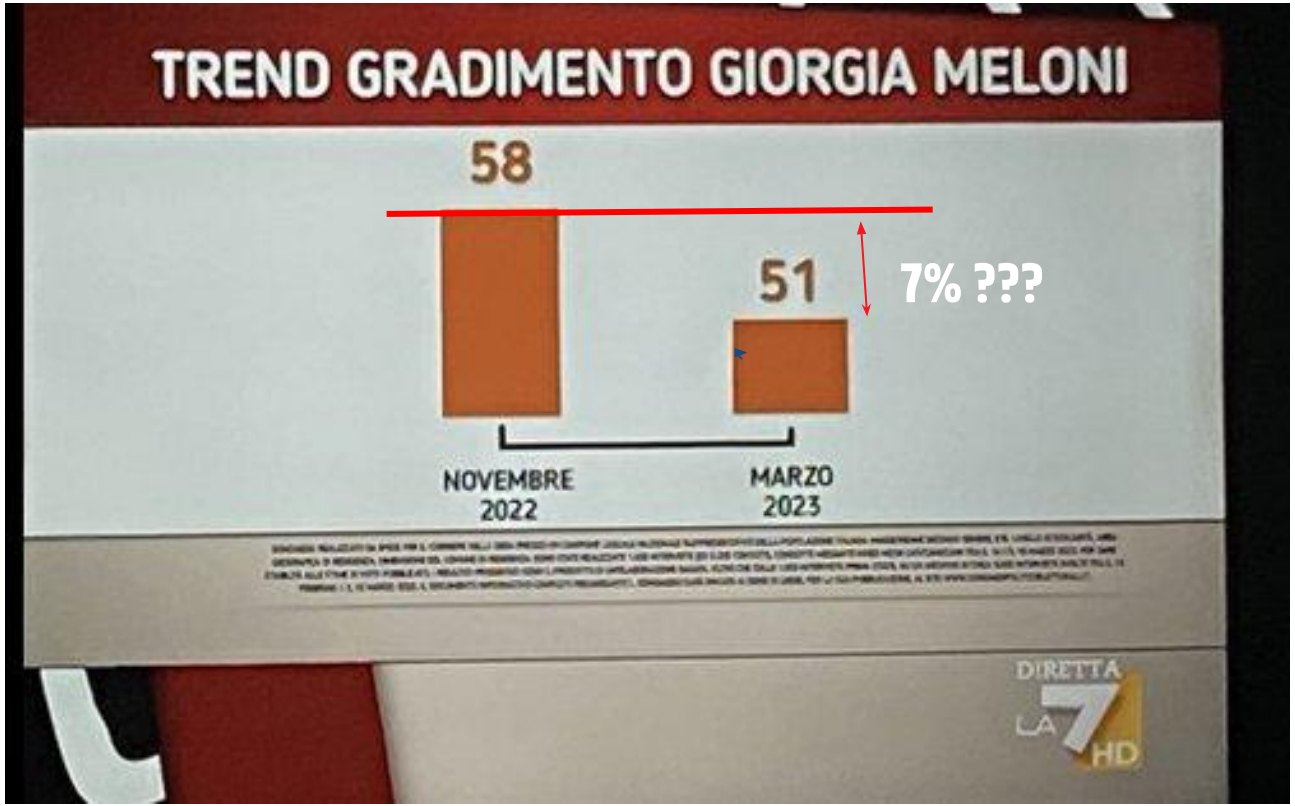
and bring it exactly on top (so I double it)



The red line would be 102% ...



And yet 58-51 = 7 ....



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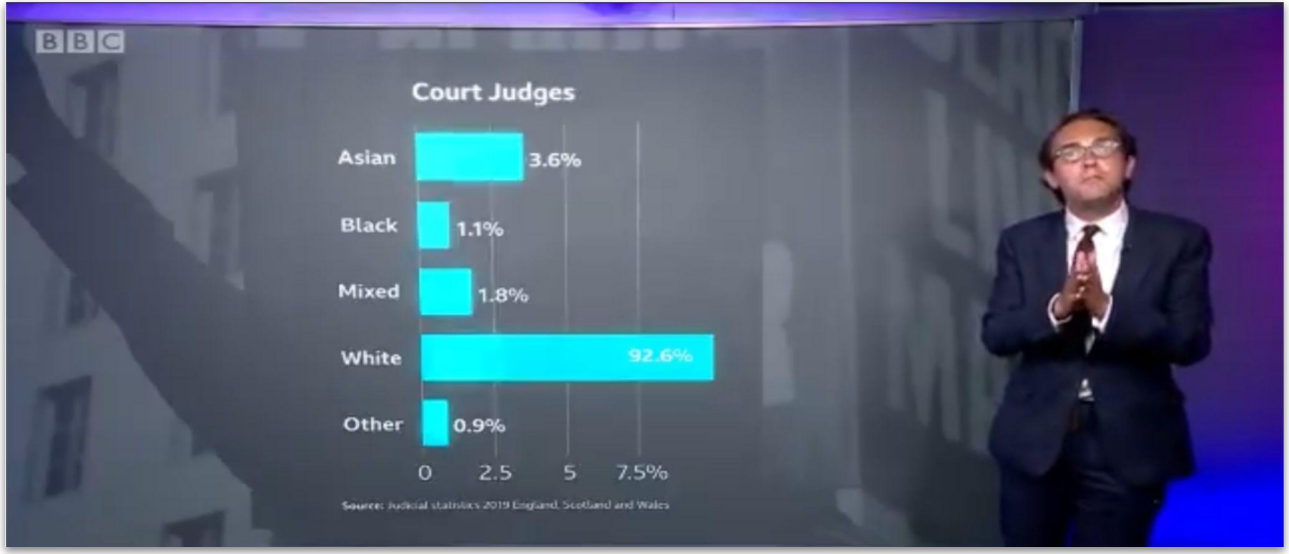
again proportions



https://www.valigiablu.it/dati-manipolazione/

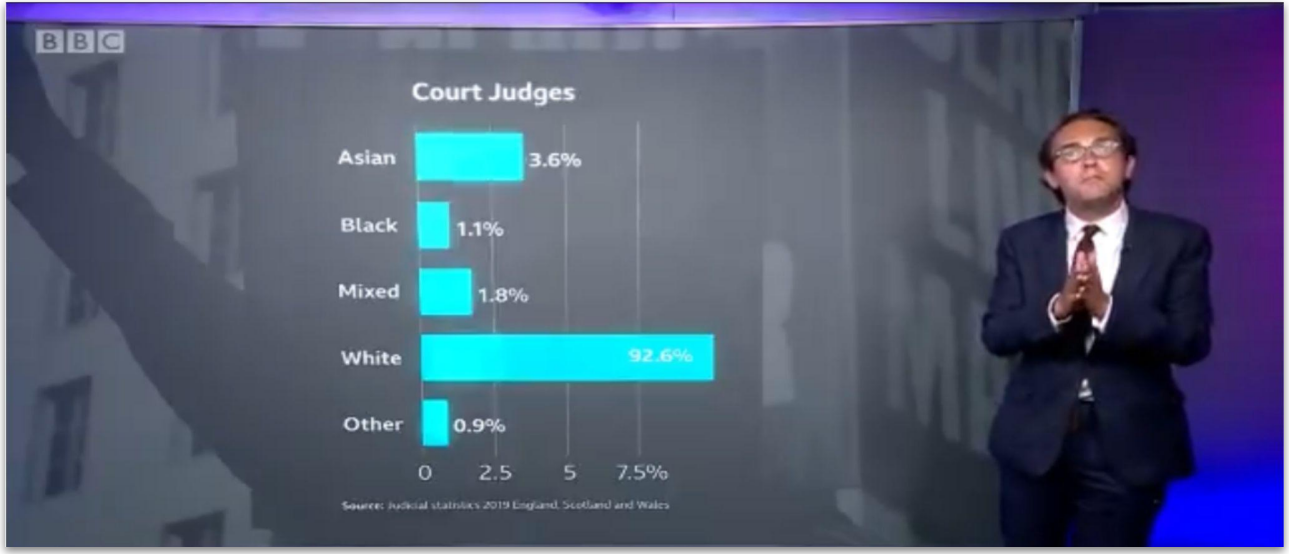
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When it doesn't fit on the screen ....



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When it doesn't fit on the screen ....





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Tourism increased by 75% from 2021 to 2022



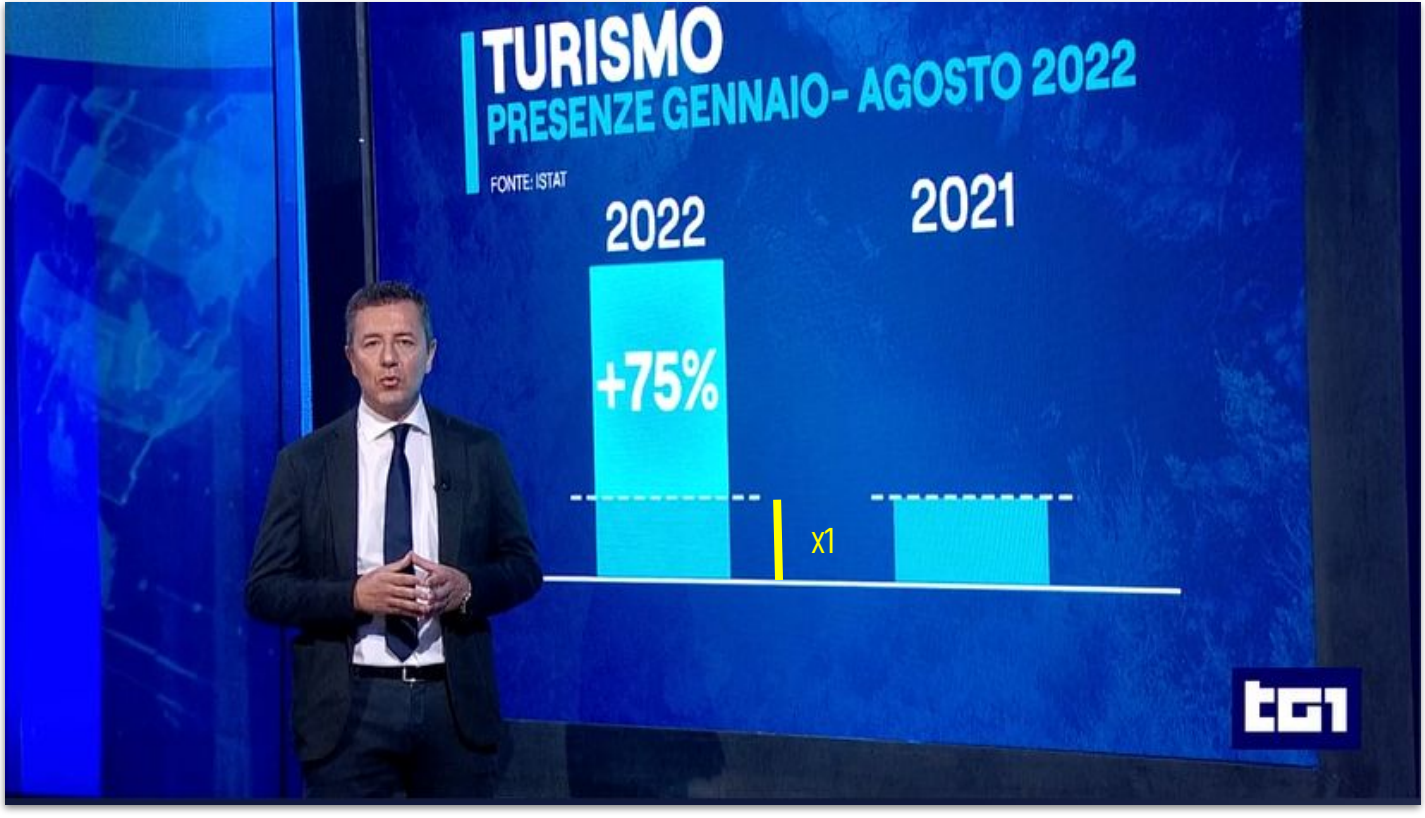
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Tourism increased by 75% from 2021 to 2022 ... so three quarters more



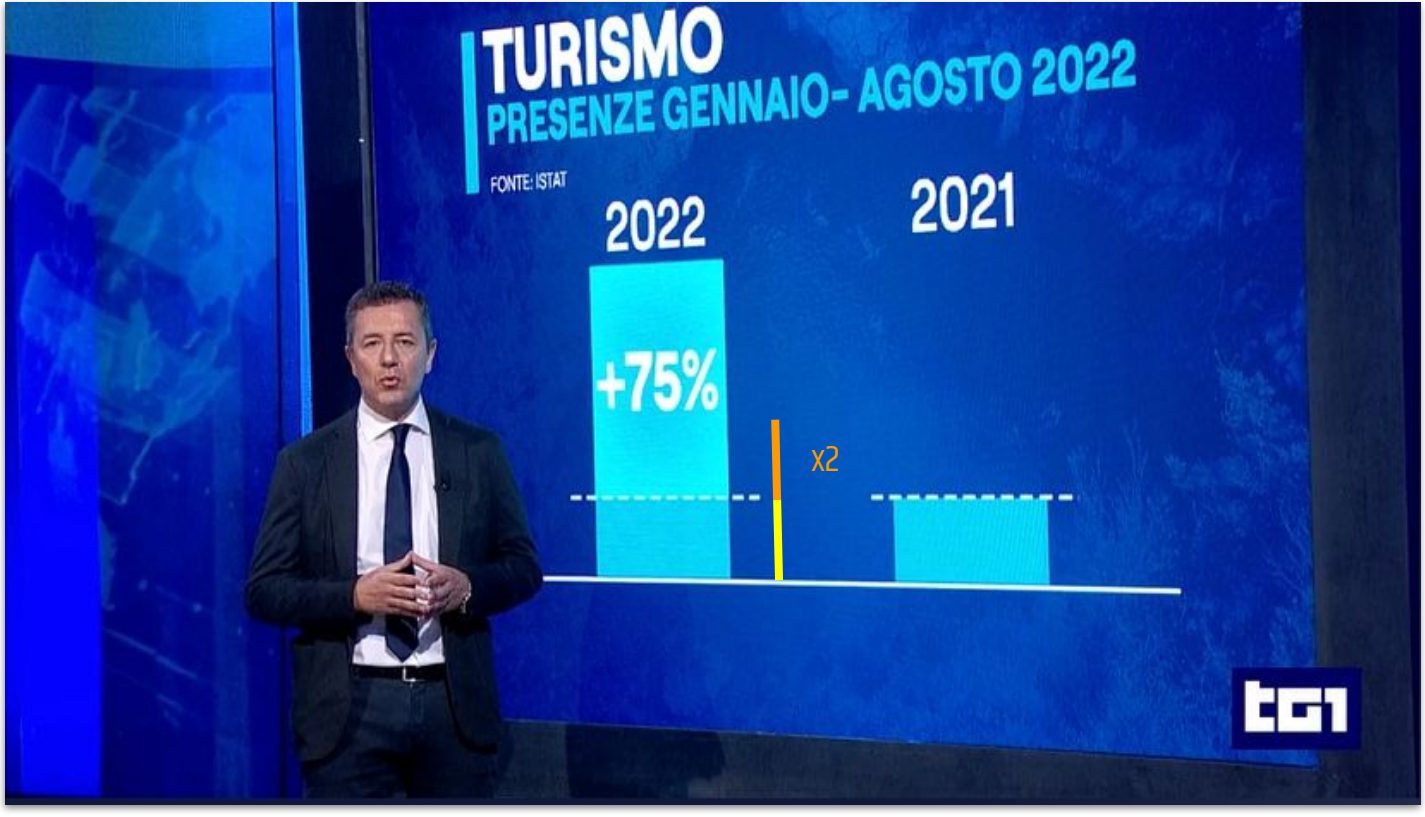
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height to 2021



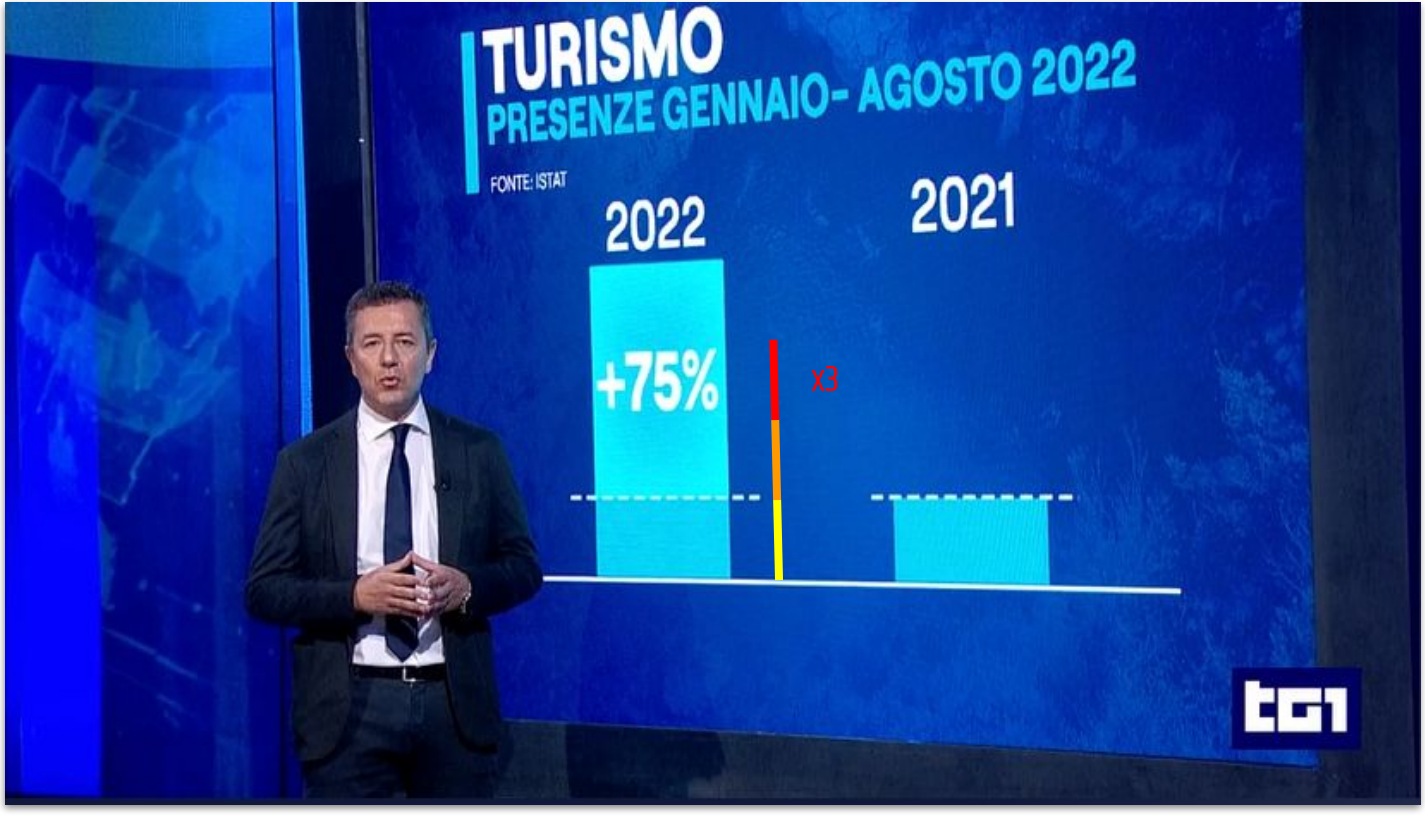
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this one doubles (200%)



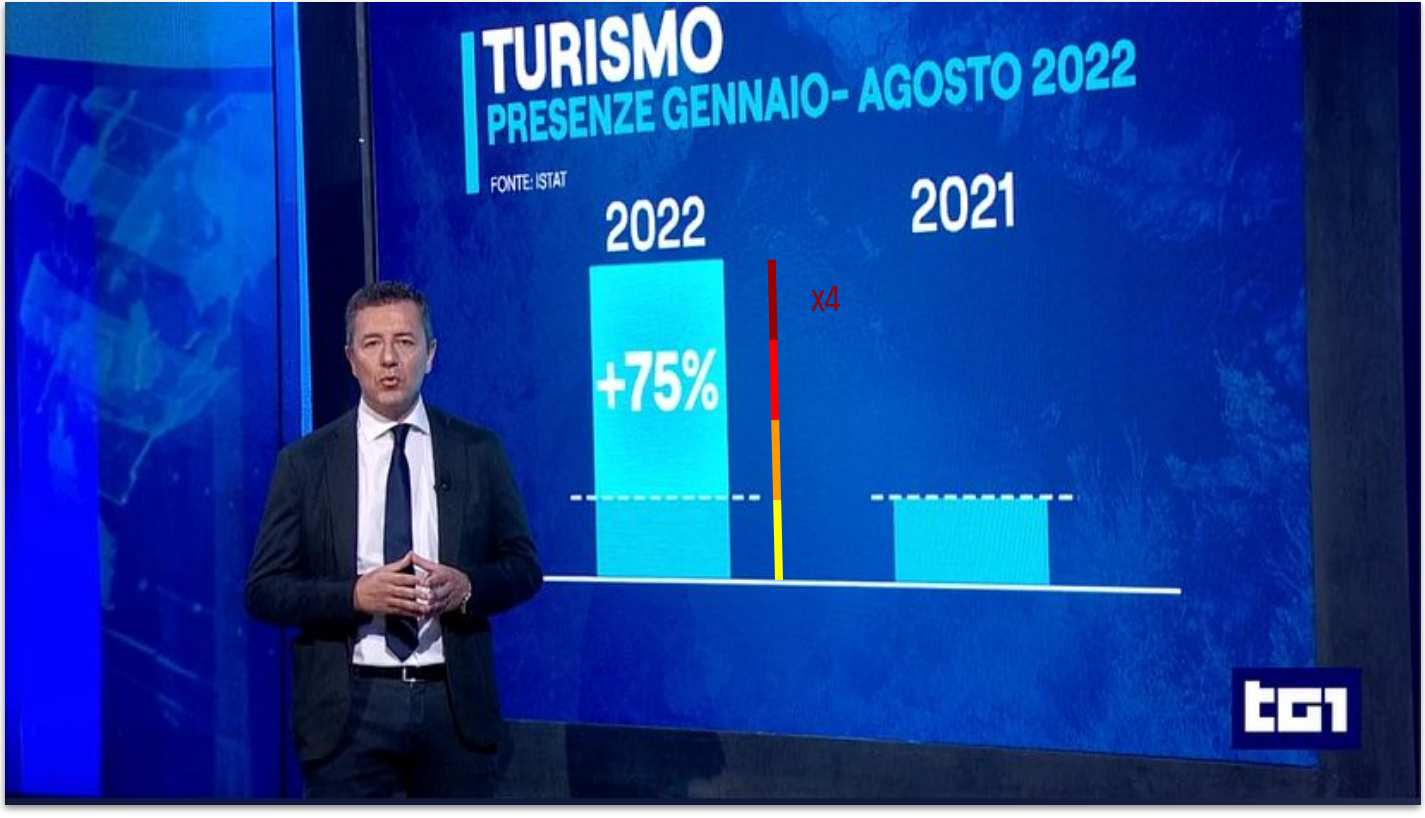
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Urka! Triple (300%)



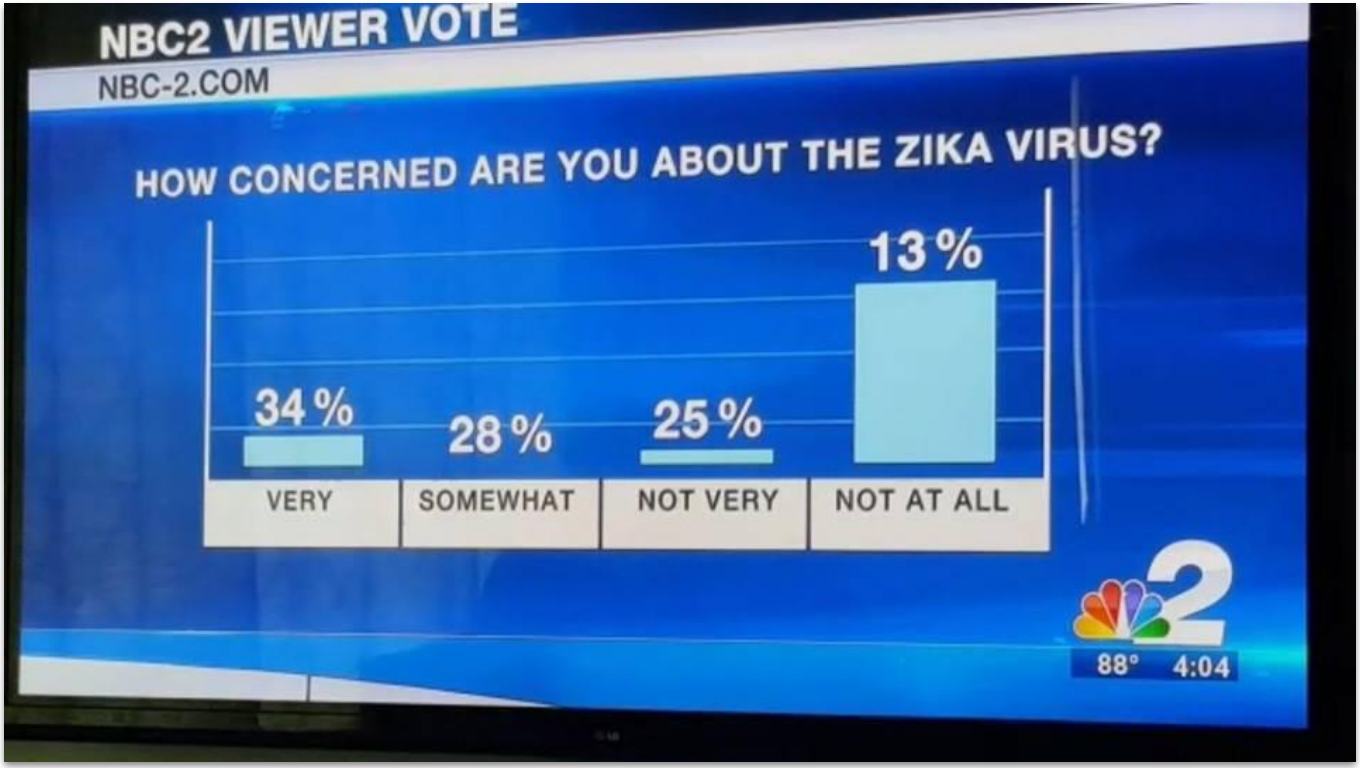
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Um ... 4 times bigger ...



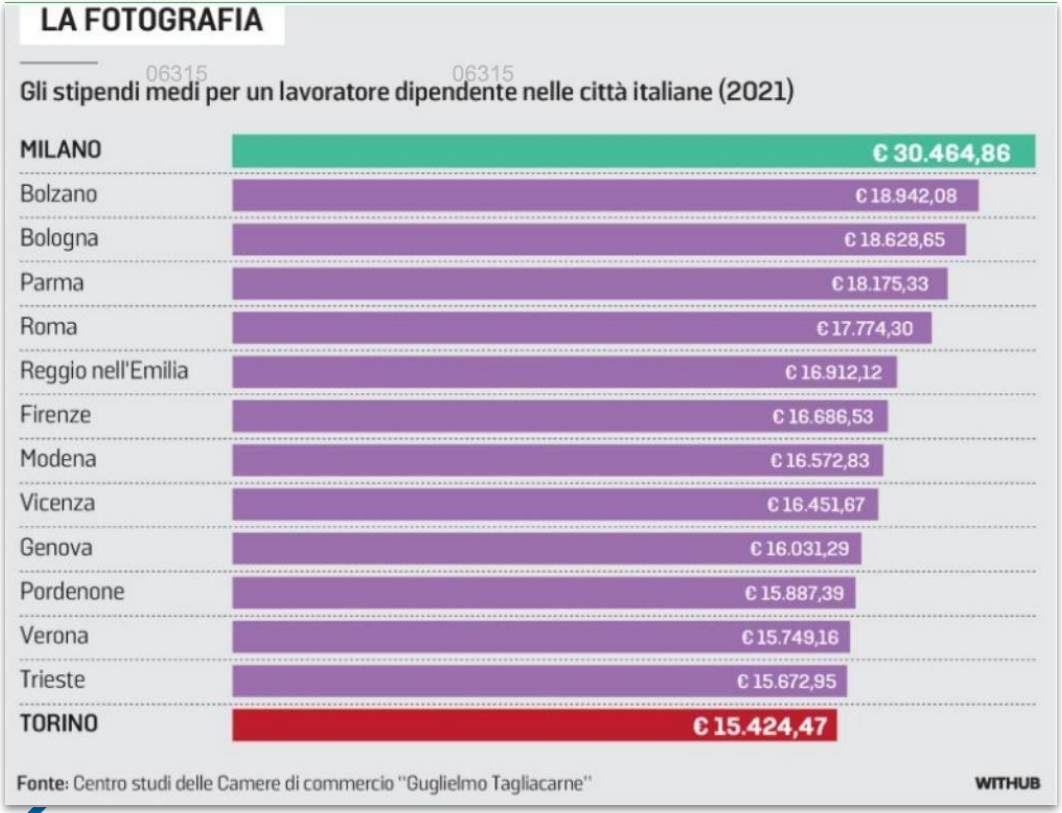
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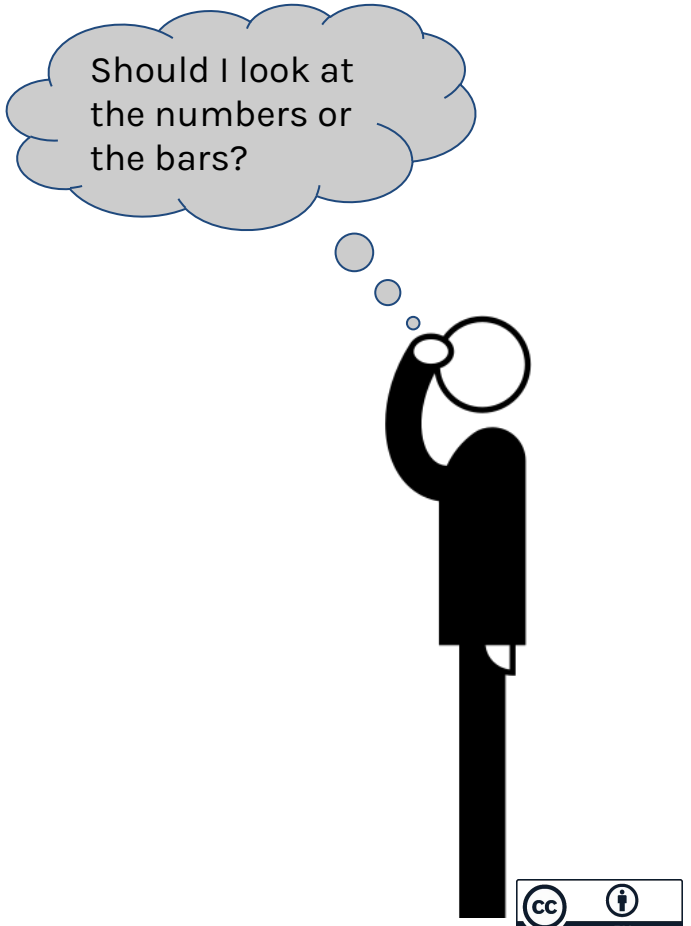
Few ideas well confused ...



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Milan twice as big as Turin!!!

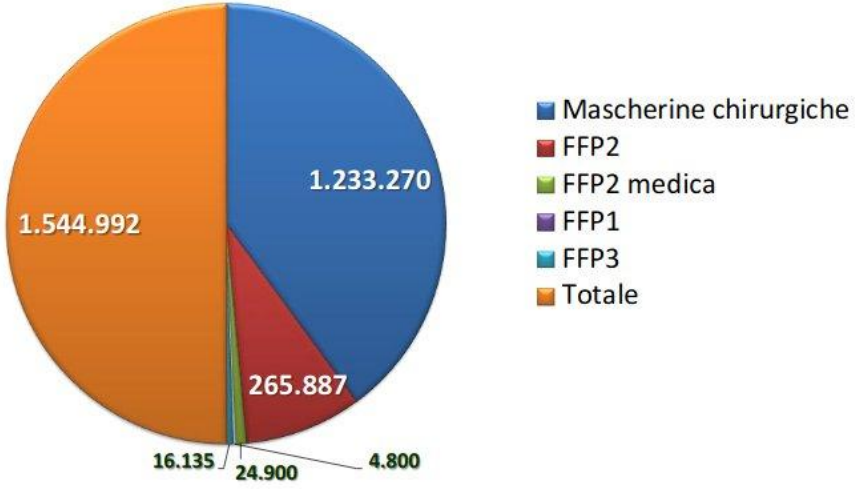


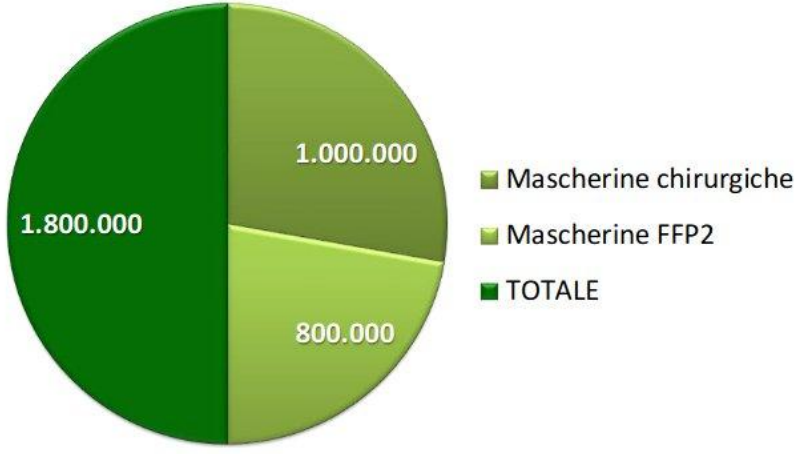


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When 'total' is a type of template/mask



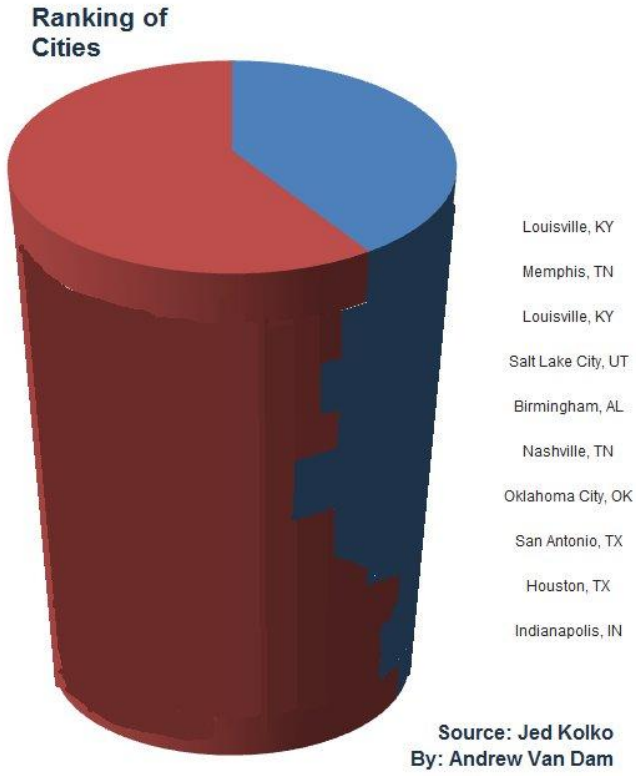


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an orange speck occupying about 12% of the surface???
(about one-eighth)

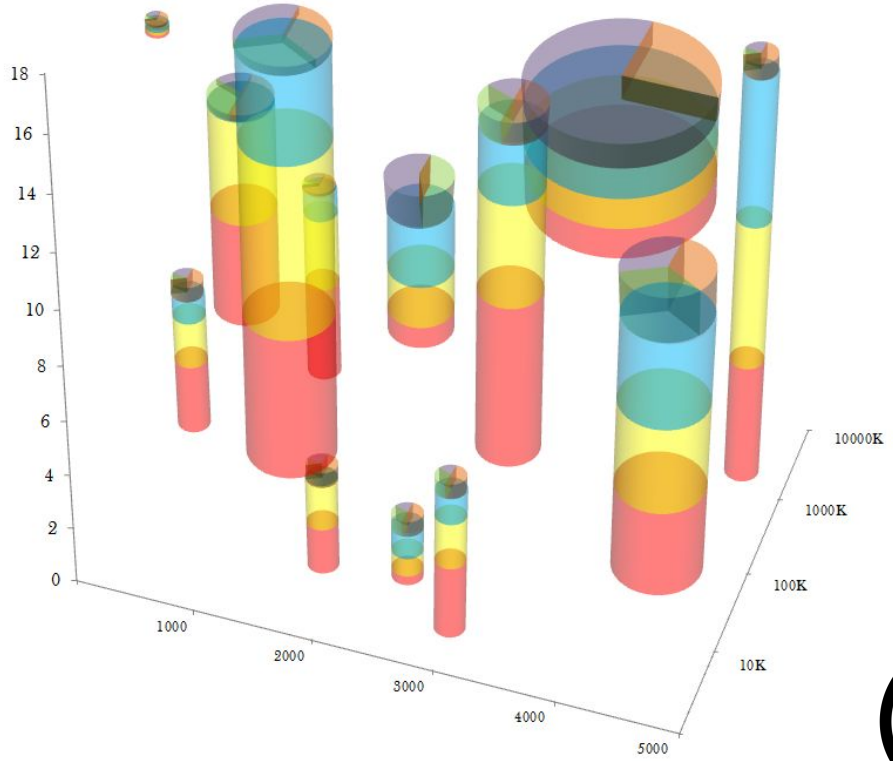


Help me understand the logic of the ranking ...



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When you want to simplify ...



At least seven information-rich variables can be plotted in this new graphical module



he needs holidays

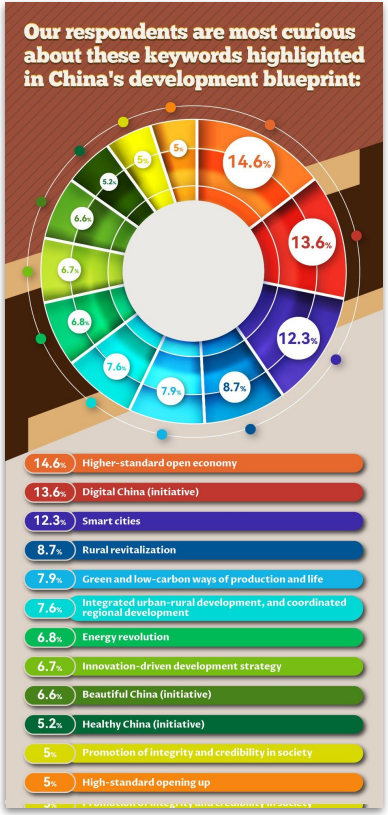


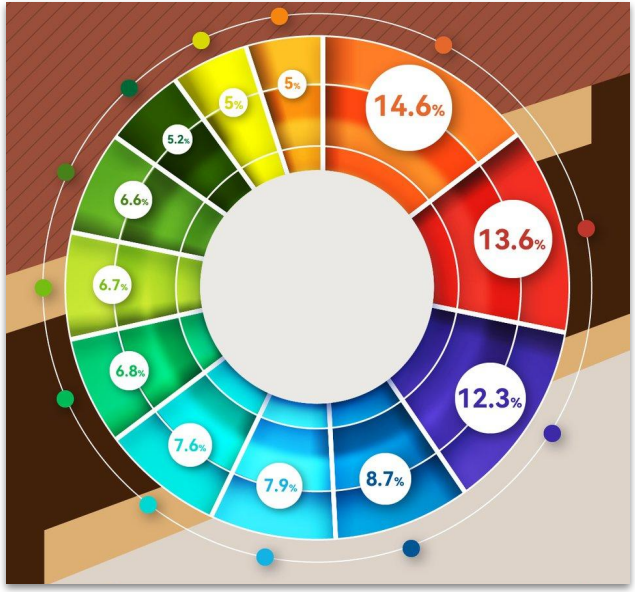
http://eagerpies.com/better-than-minard/

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When the legend serves to ... explain (?)







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Main Mistakes

Respect the proportions

Too much information

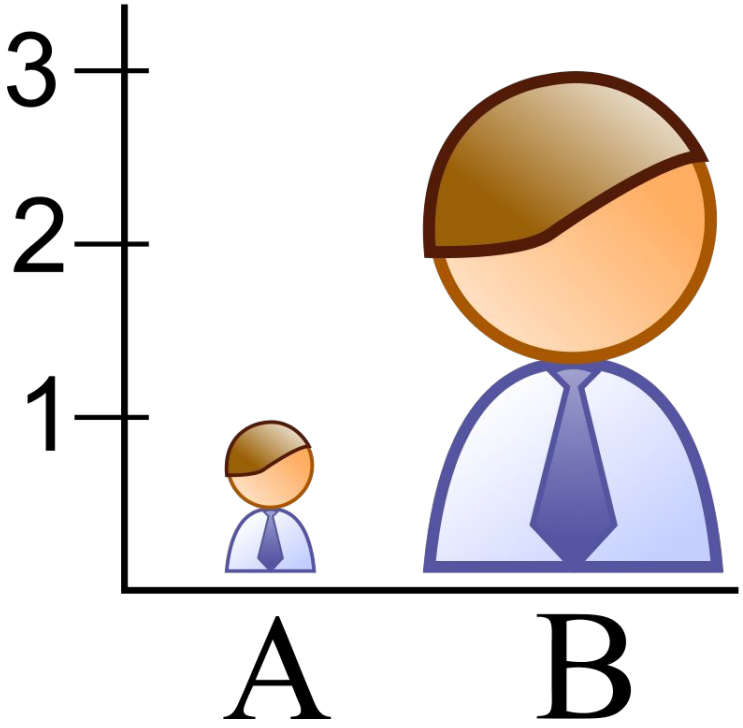


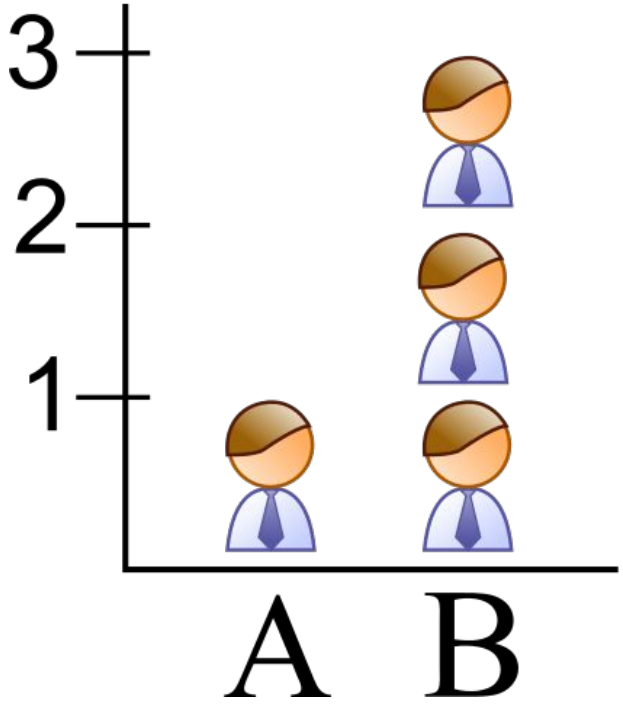
Misuse of colours

Beautiful doesn't mean effective

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Respect the proportions



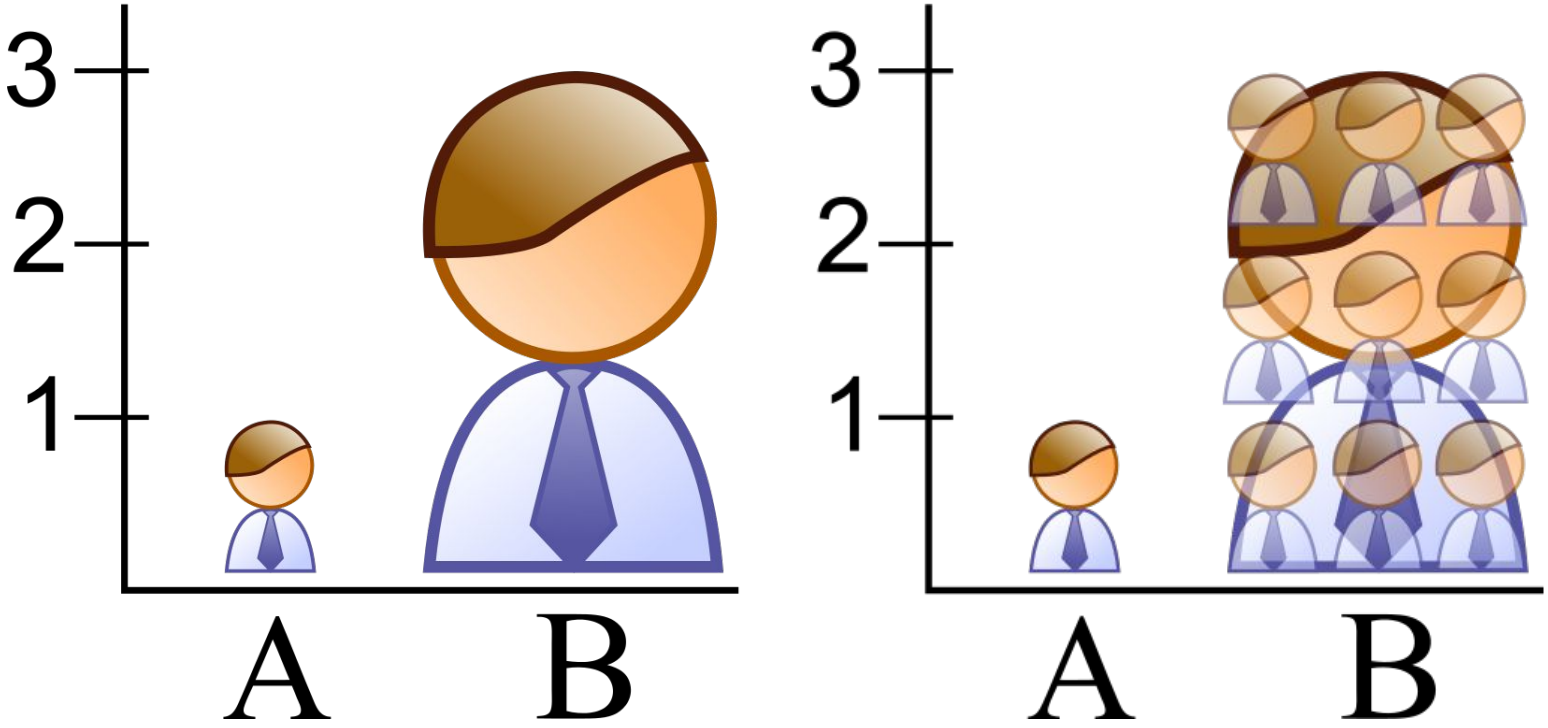


The image on the left should summarise the one on the right

https://en.wikipedia.org/wiki/Misleading_graph

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Respect the proportions

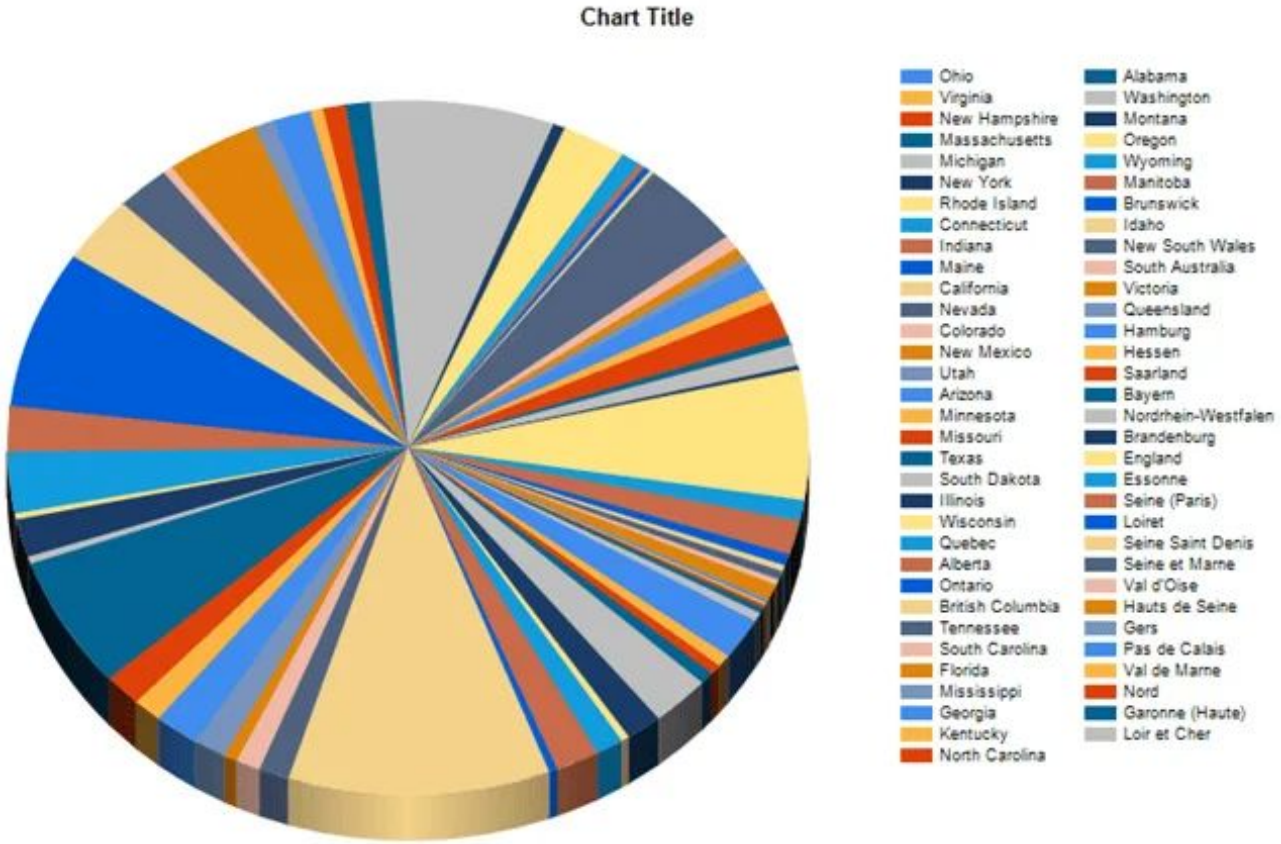


but since it also scales the width ... it is actually representing a wrong value

https://en.wikipedia.org/wiki/Misleading_graph

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Too much information

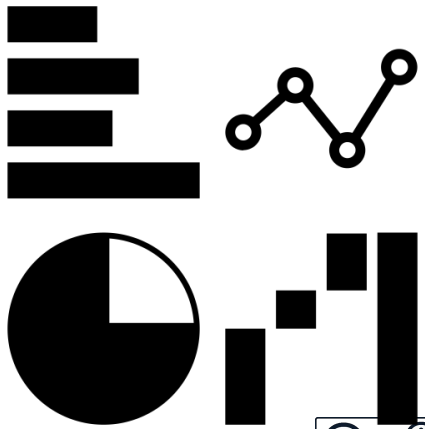


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Misuse of Labels

The use of distorted represented words either in the title of the graph or in the axis labels or caption could mislead the reader

For example, what can you read here? Does it look easy to you without getting a stiff neck?



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Tables and Charts

The use of graphs where they are not needed can lead to unnecessary confusion/interpretation.

The more a graph needs an explanation, the less the graph is needed.

Graphs do not always convey information as tables do.

https://en.wikipedia.org/wiki/Misleading_graph

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Optimising Tables - Dark Horse Analytics

Remove to improve the data tables edition

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Optimising Tables - Dark Horse Analytics



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Optimising Tables

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
Face (The Hero)	Macho Man Randy Savage	Monkey	Feb-2008	157618.00	59.29
Face (The Hero)	Hacksaw Jim Duggan	Pig	Mar-2008	30300.00	53.4332
Face (The Hero)	Superfly Jimmy Snuka	Dragon	Mar-2008	12341.00	52.7
Heel (The Bad Guy)	Rowdy Roddy Piper	Rooster	Jun-1968	71645.00	45.4
Heel (The Bad Guy)	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449342.00	43.7689
Heel (The Bad Guy)	Mr. Perfect Curt Henning	Rat	May-1980	13773.00	38
Heel (The Bad Guy)	Jake the Snake Roberts	Snake	Jul-1975	5609.00	37.99
Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.316
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Optimise tables - remove colours

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Optimise tables - remove grids

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Heel (The Bad Guy)	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449342.00	43.7689
Heel (The Bad Guy)	Mr. Perfect Curt Henning	Rat	May-1980	13773.00	38
Heel (The Bad Guy)	Jake the Snake Roberts	Snake	Jul-1975	5609.00	37.99
Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.316
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Optimise tables - remove borders

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
Face (The Hero)	Macho Man Randy Savage	Monkey	Feb-2008	157618.00	59.29
Face (The Hero)	Hacksaw Jim Duggan	Pig	Mar-2008	30300.00	53.4332
Face (The Hero)	Superfly Jimmy Snuka	Dragon	Mar-2008	12341.00	52.7
Heel (The Bad Guy)	Rowdy Roddy Piper	Rooster	Jun-1968	71645.00	45.4
Heel (The Bad Guy)	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449342.00	43.7689
Heel (The Bad Guy)	Mr. Perfect Curt Henning	Rat	May-1980	13773.00	38
Heel (The Bad Guy)	Jake the Snake Roberts	Snake	Jul-1975	5609.00	37.99
Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.316
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Ottimizzare le tabelle - rimuovere il grassetto

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
Face (The Hero)	Macho Man Randy Savage	Monkey	Feb-2008	157618.00	59.29
Face (The Hero)	Hacksaw Jim Duggan	Pig	Mar-2008	30300.00	53.4332
Face (The Hero)	Superfly Jimmy Snuka	Dragon	Mar-2008	12341.00	52.7
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Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.316
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Optimise tables - remove boldface

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
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Optimise tables - align text to the left

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Optimising Tables - Aligning Headers with Data

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Optimising tables - resizing columns with data

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Optimise Tables - Improve Blanks

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Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Optimise Tables - Improve Blanks

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
Face (The Hero)	Macho Man Randy Savage	Monkey	Feb-2008	157618.00	59.29
Face (The Hero)	Hacksaw Jim Duggan	Pig	Mar-2008	30300.00	53.4332
Face (The Hero)	Superfly Jimmy Snuka	Dragon	Mar-2008	12341.00	52.7
Heel (The Bad Guy)	Rowdy Roddy Piper	Rooster	Jun-1968	71645.00	45.4
Heel (The Bad Guy)	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449342.00	43.7689
Heel (The Bad Guy)	Mr. Perfect Curt Henning	Rat	May-1980	13773.00	38
Heel (The Bad Guy)	Jake the Snake Roberts	Snake	Jul-1975	5609.00	37.99
Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.316
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.61
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Optimise tables - use consistent precision

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551.00	61.978
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Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

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Heel (The Bad Guy)	The Million Dollar Man Ted DiBiase	Rat	Apr-1975	449342.00	43.8
Heel (The Bad Guy)	Mr. Perfect Curt Henning	Rat	May-1980	13773.00	38.0
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Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.6
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0

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Optimising tables - rounding numbers

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320.00	86.2
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Jobber (The Unknown)	Brad Smith	Sheep	Aug-2008	1103.00	36.3
Jobber (The Unknown)	Ted Duncan	Sheep	Aug-2008	200.00	33.6
Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0

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Optimising tables - rounding numbers more

Role	Name	Year of the...	Debut	Number of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97320	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988551	62.0
Face (The Hero)	Macho Man Randy Savage	Monkey	Feb-2008	157618	59.3
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Optimising tables - rounding numbers more

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Optimising tables - rounding numbers more

Role	Name	Year of the...	Debut	Thousands of Fans	Takedown Rate
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Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	0.0	21.0

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Optimising tables - removing redundancies

Role	Name	Year of the...	Debut	Thousands of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97.3	86.2
Face (The Hero)	Hulk Hogan	Oxen	Jan-2008	988.6	62.0
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Optimising tables - removing redundancies

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Optimise tables - please no more Calibri (or Comic Sans)

Role	Name	Year of the...	Debut	Thousands of Fans	Takedown Rate
Face (The Hero)	The Ultimate Warrior	Tiger	May-2011	97.3	86.2
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	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	0.0	21.0

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Optimising Tables - reinserting emphasis

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	Hulk Hogan	Oxen	Jan-2008	988.6	62.0
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	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	0.0	21.0

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Optimising Tables



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Optimising Tables - before

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Jobber (The Unknown)	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	5.00	21.0196

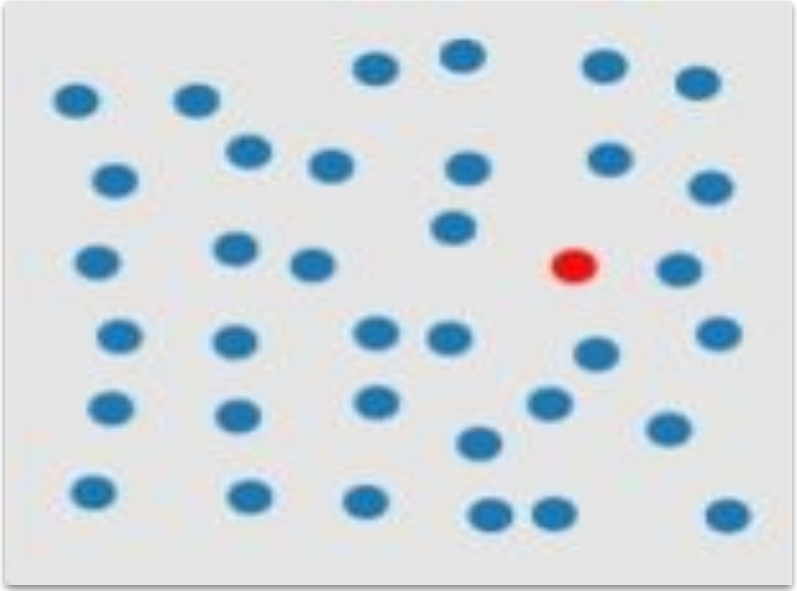
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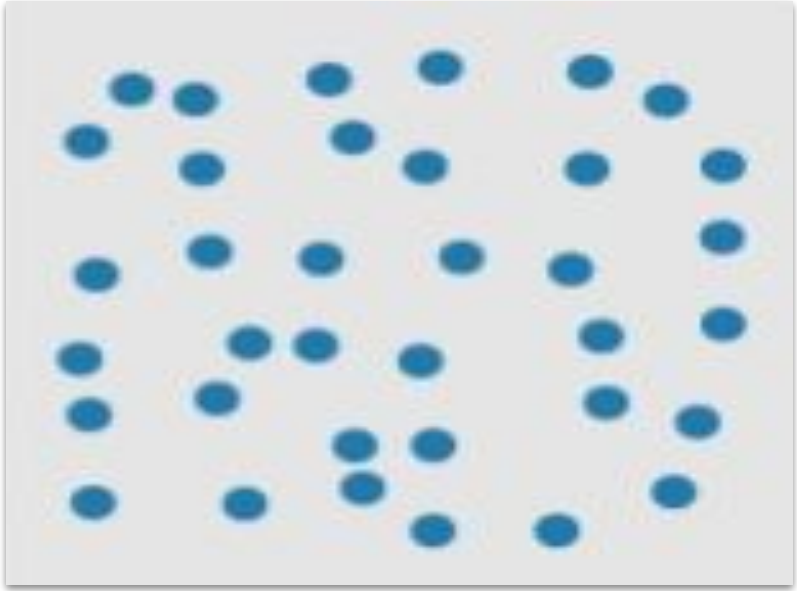
Optimising Tables - after

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	Joey the Uber Nerd Cherdarchuk	Snake	Aug-2008	0.0	21.0

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knowing how to use COLOURS well





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How many times do you see the number three?

4324295044543548576090897942185409054540910204805904390905476
0122097509076486245976654211998677543909855040403102904769442 12
0854087854250854087854250854087854250854087854250854087854250 8

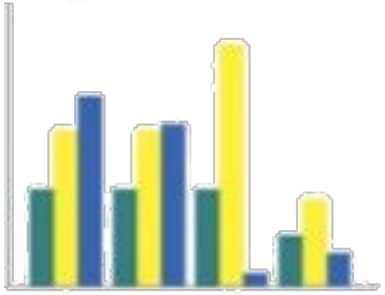
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How many times do you see the number three?

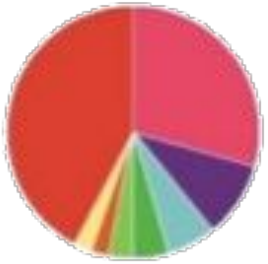
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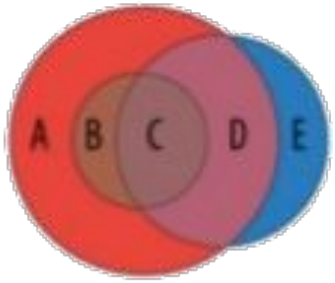
Use of a dominant colour



difficulty in distinguishing

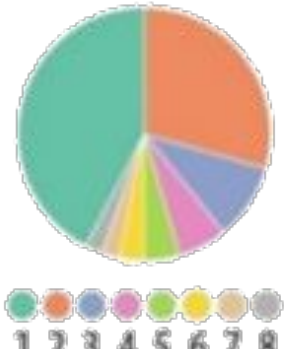


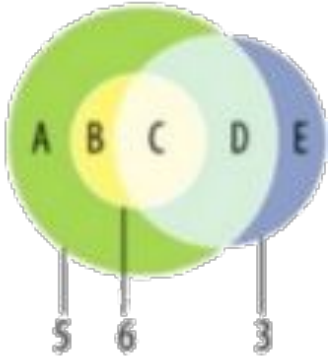
dark colours



after having been to http://colorbrewer2.org

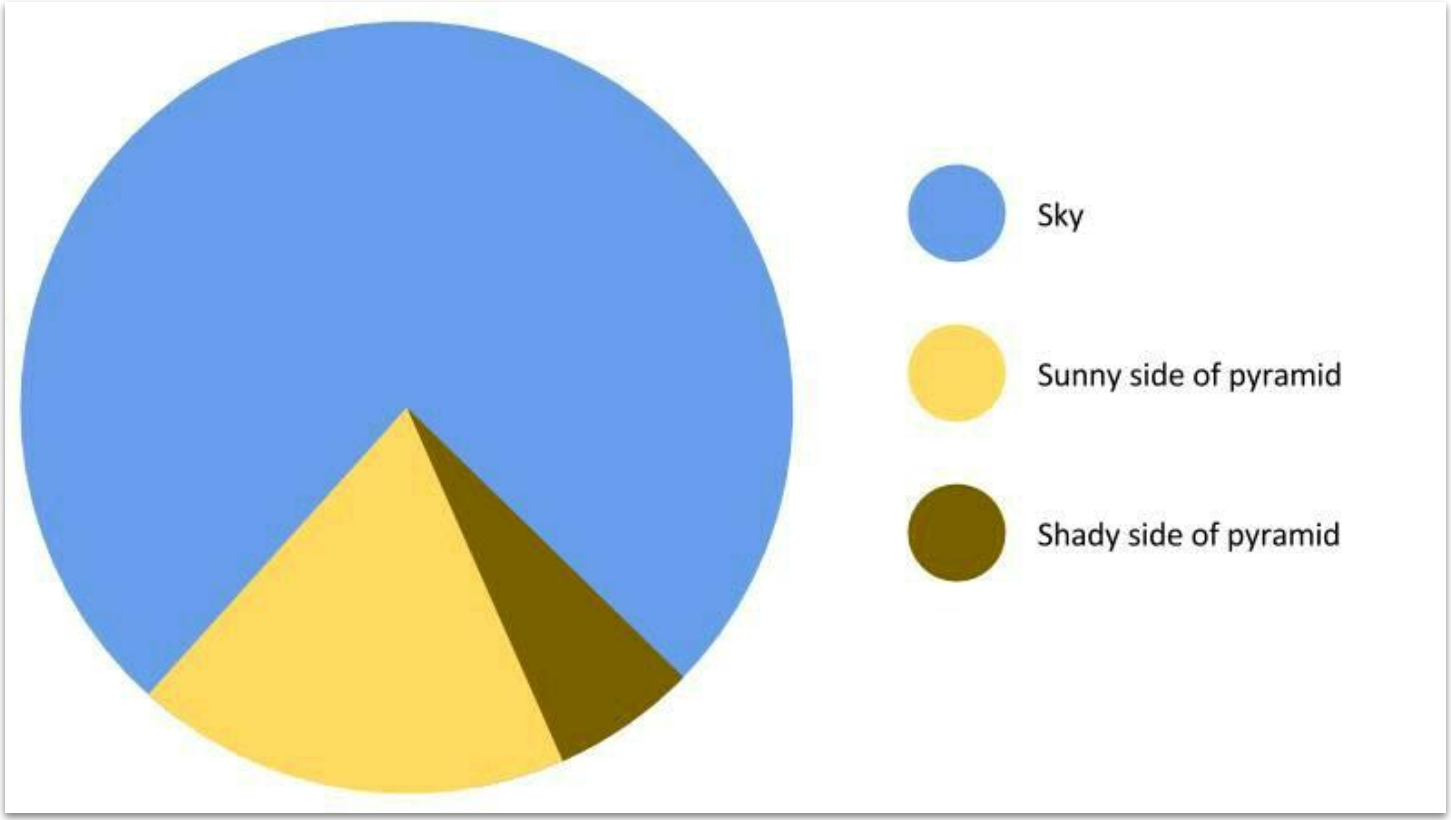






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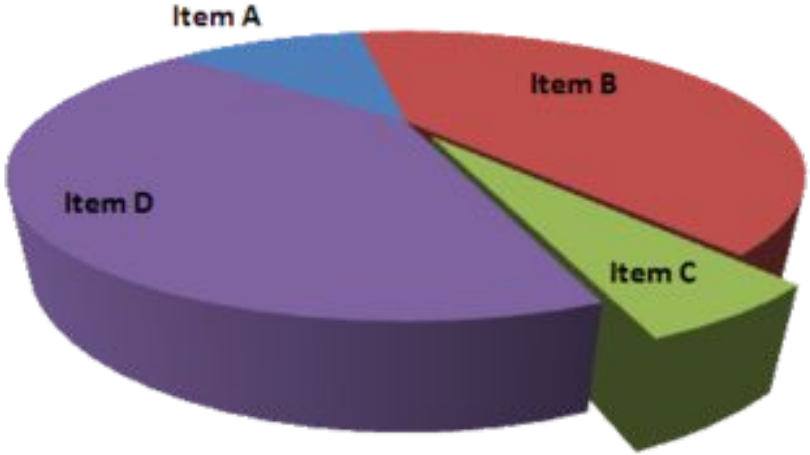
An Egyptian pie chart

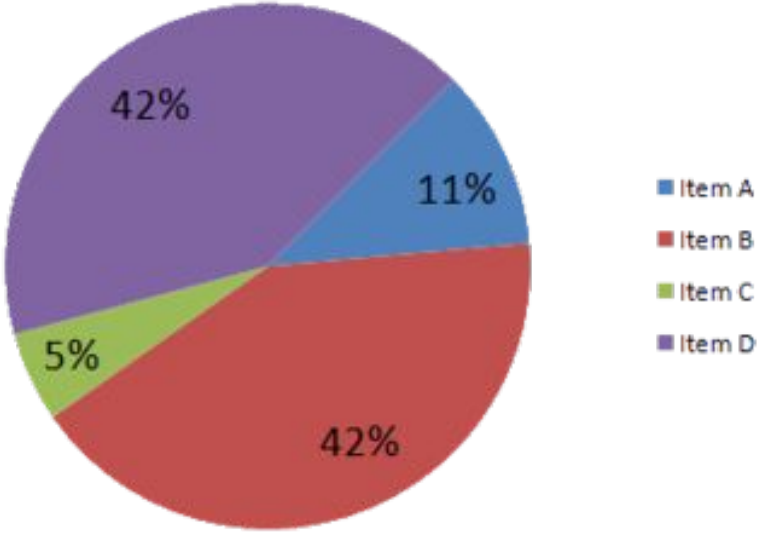


... or binoculars pointed at a pyramid?

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Beware of 3D graphics

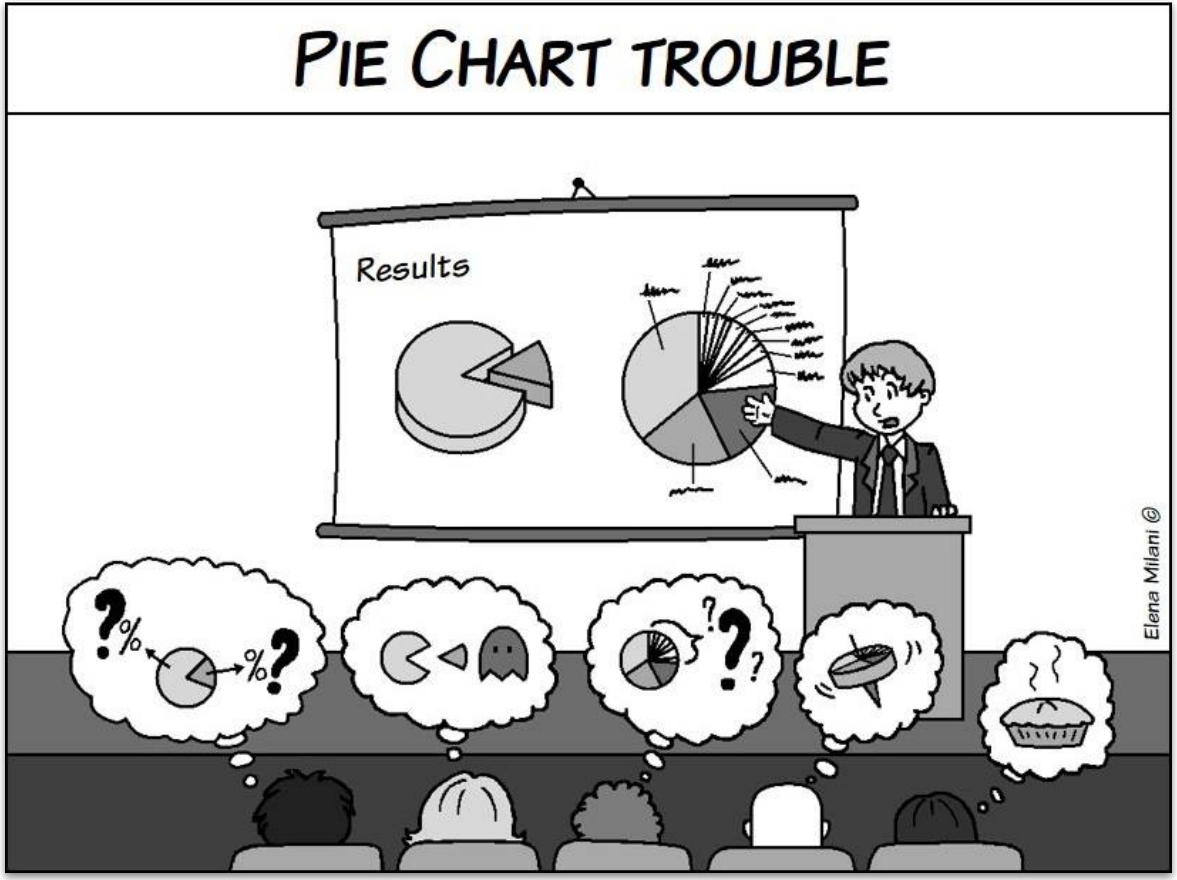




Element C appears to be at least as large as element A, when in fact, it is less than half the size

https://en.wikipedia.org/wiki/Misleading_graph

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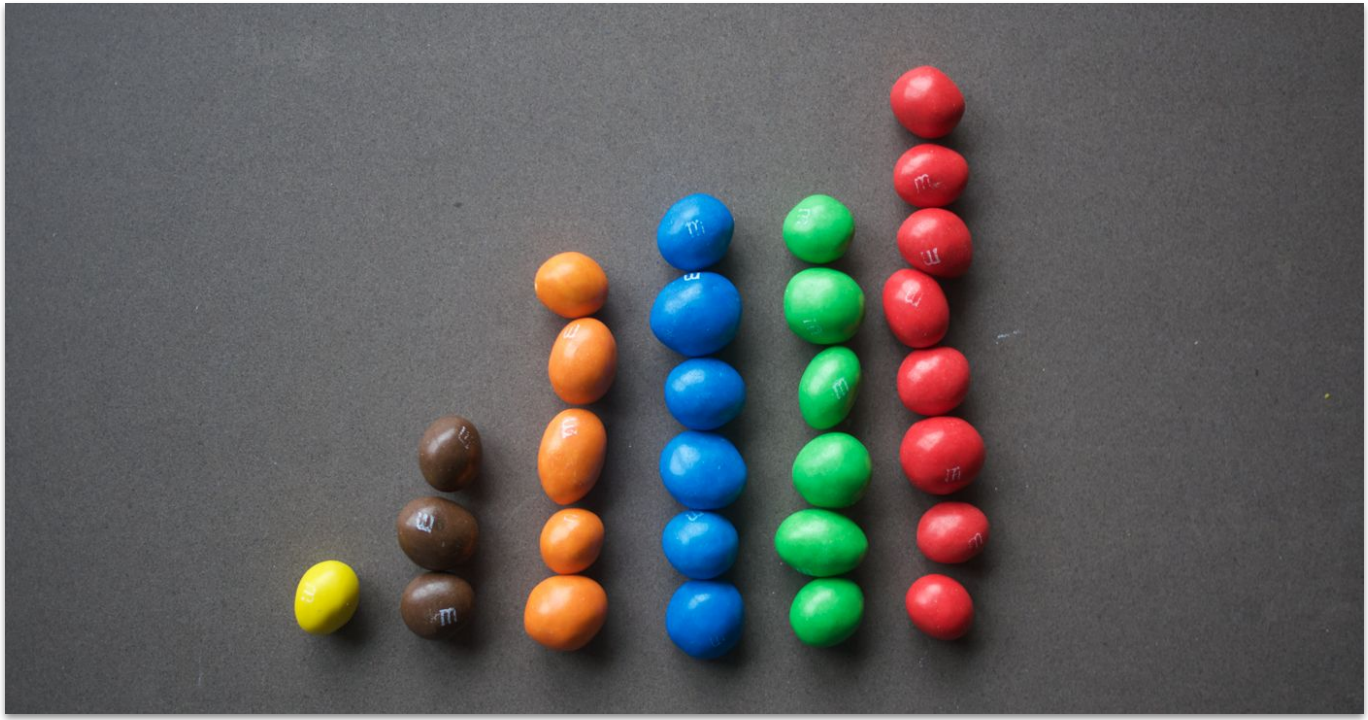
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Which colour has the most M&M's?



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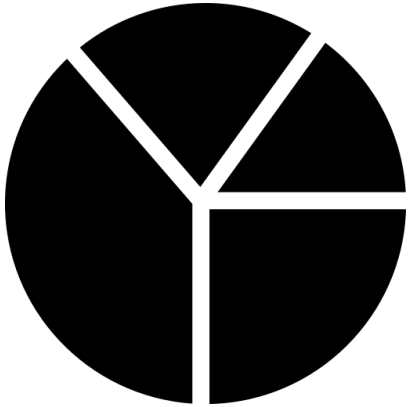
Which colour has the most M&M's?



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Pie Charts

Comparing **pie charts** of **different sizes** can often be **misleading** because people **cannot** read the **comparative** area accurately



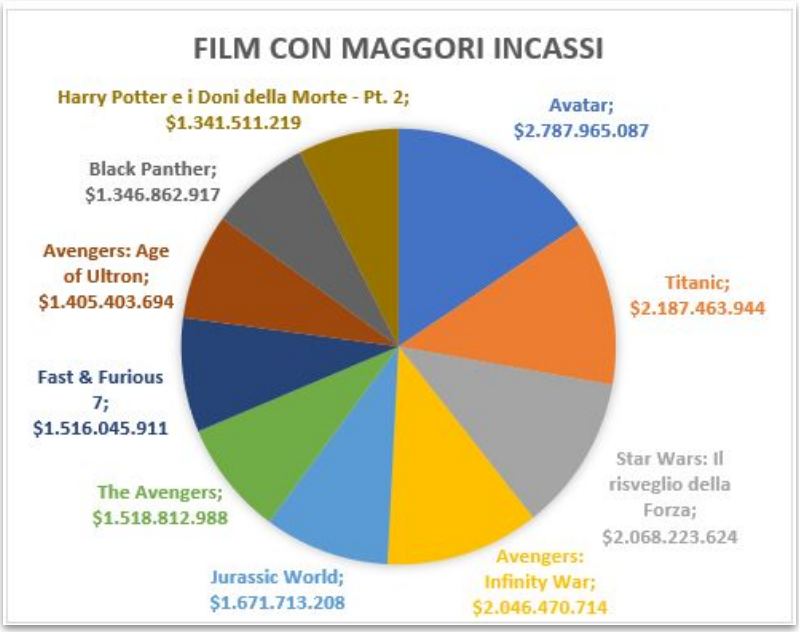
Thin **slices** are **difficult** to **distinguish** and can be **difficult** to **interpret**.

Using **percentages** as labels on a pie chart can be **misleading** when the **sample size** is **small**

The pie chart in **3D** or 2D by adding a **'slant'** makes **interpretation** even more **difficult** due to the **distorted effect** of perspective

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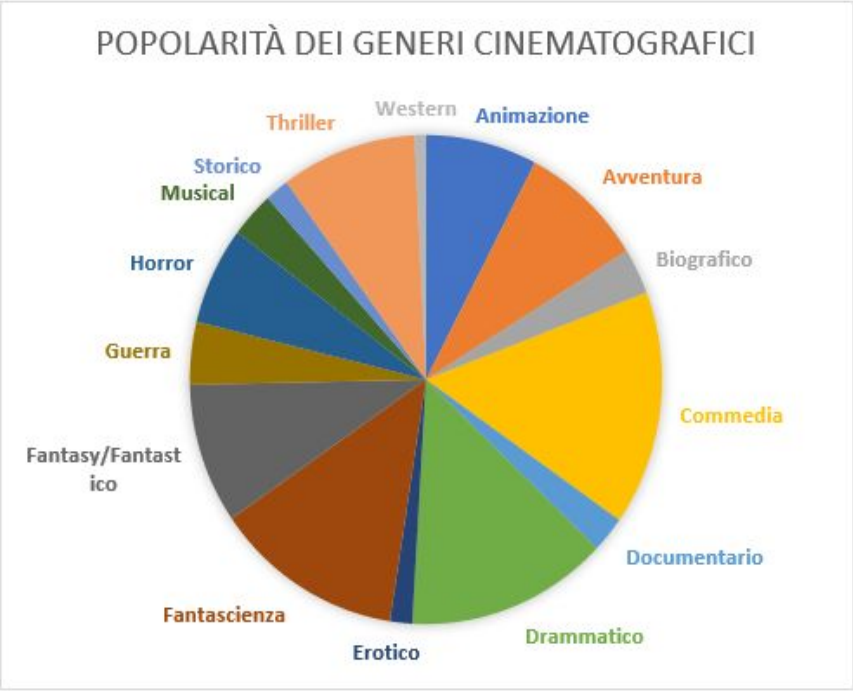
Alternatives: bar charts

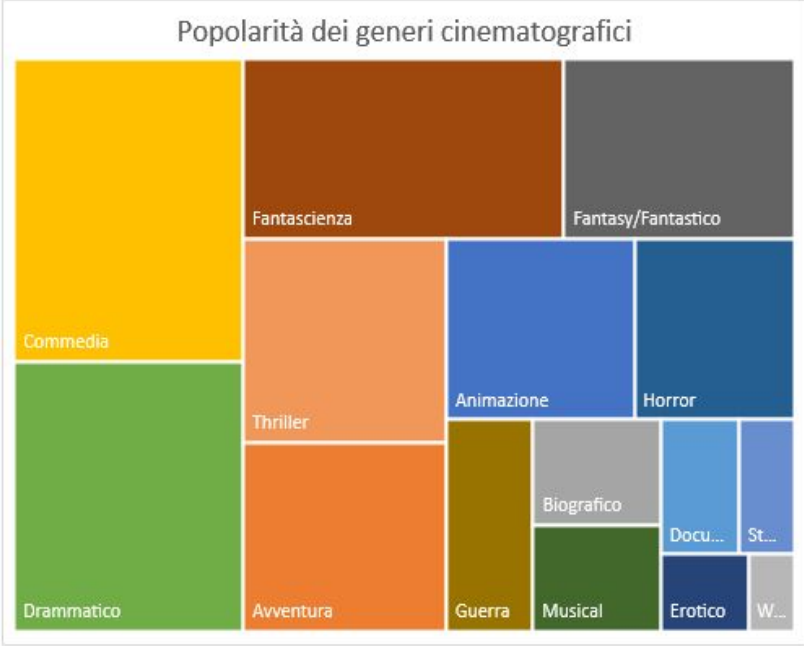




https://data-storytelling.it/visualizzazione-efficace/alternative-ai-grafici-a-torta/

Alternatives: treemap

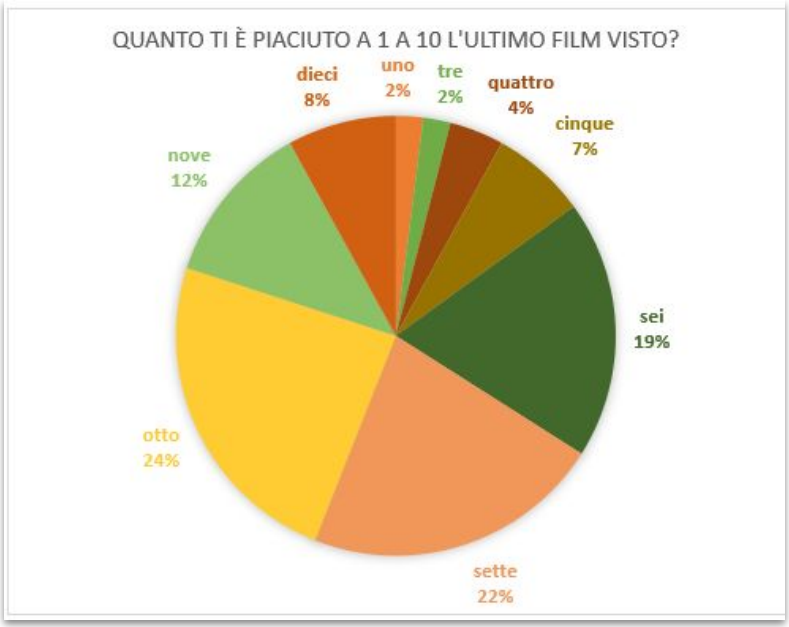


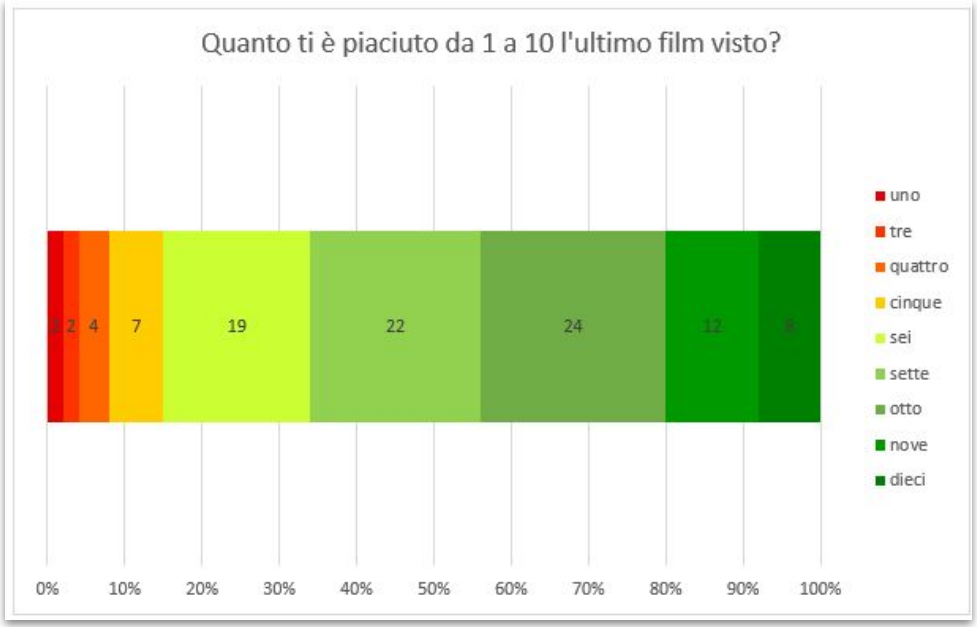


https://data-storytelling.it/visualizzazione-efficace/alternative-ai-grafici-a-torta/

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Alternative: 100% horizontal bar chart





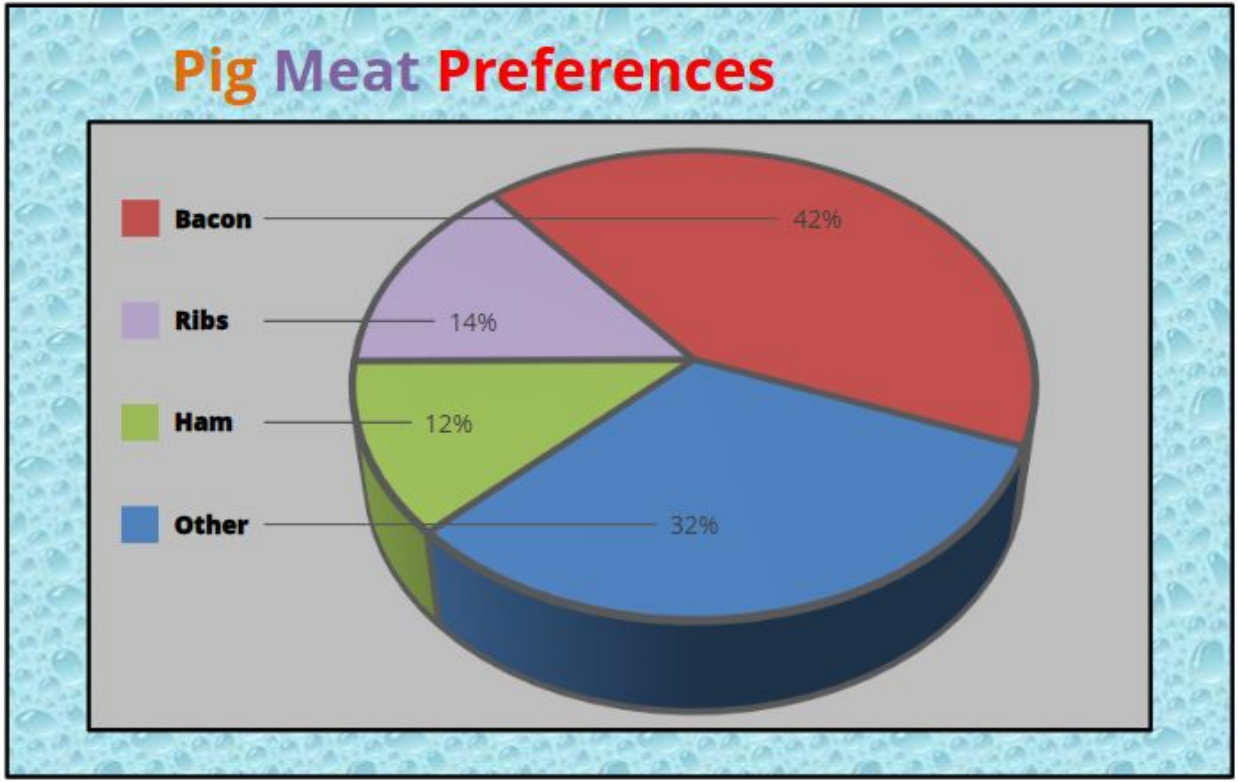
https://data-storytelling.it/visualizzazione-efficace/alternative-ai-grafici-a-torta/

Pie Charts- DarkHorse Analytics



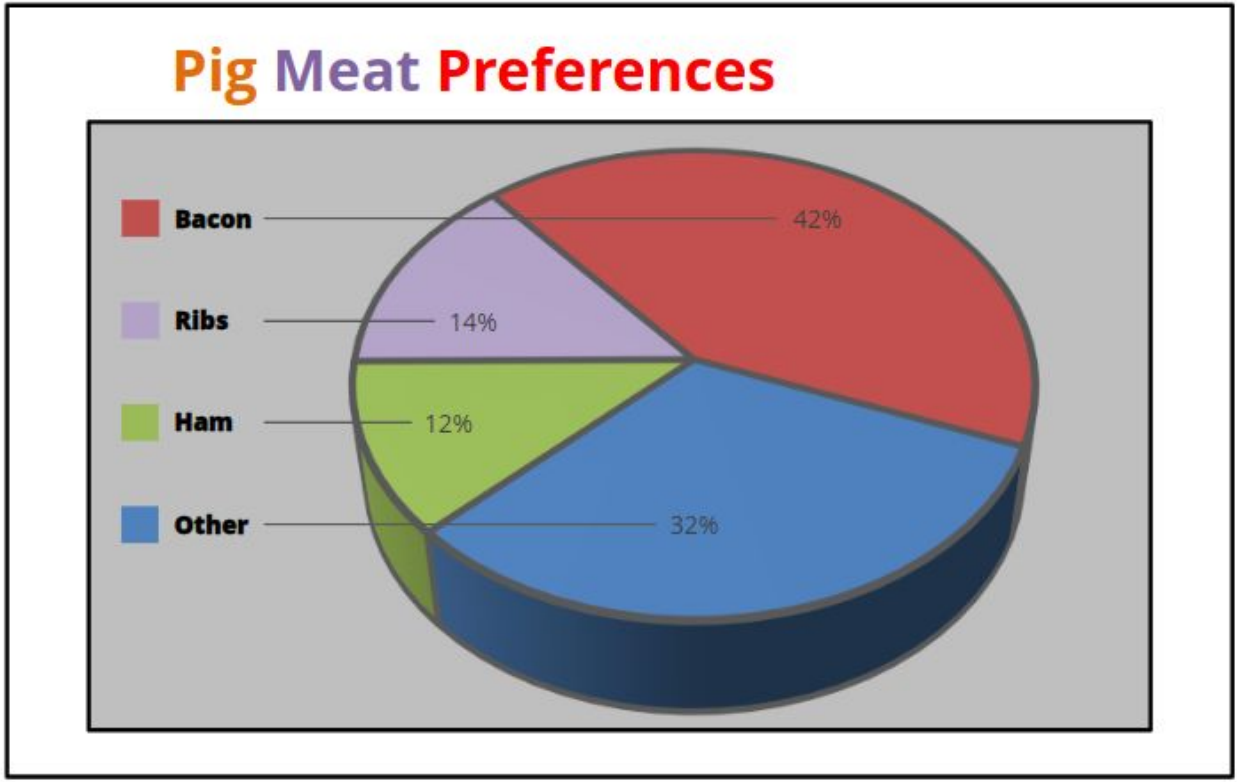
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Cake Charts - Removing Backgrounds



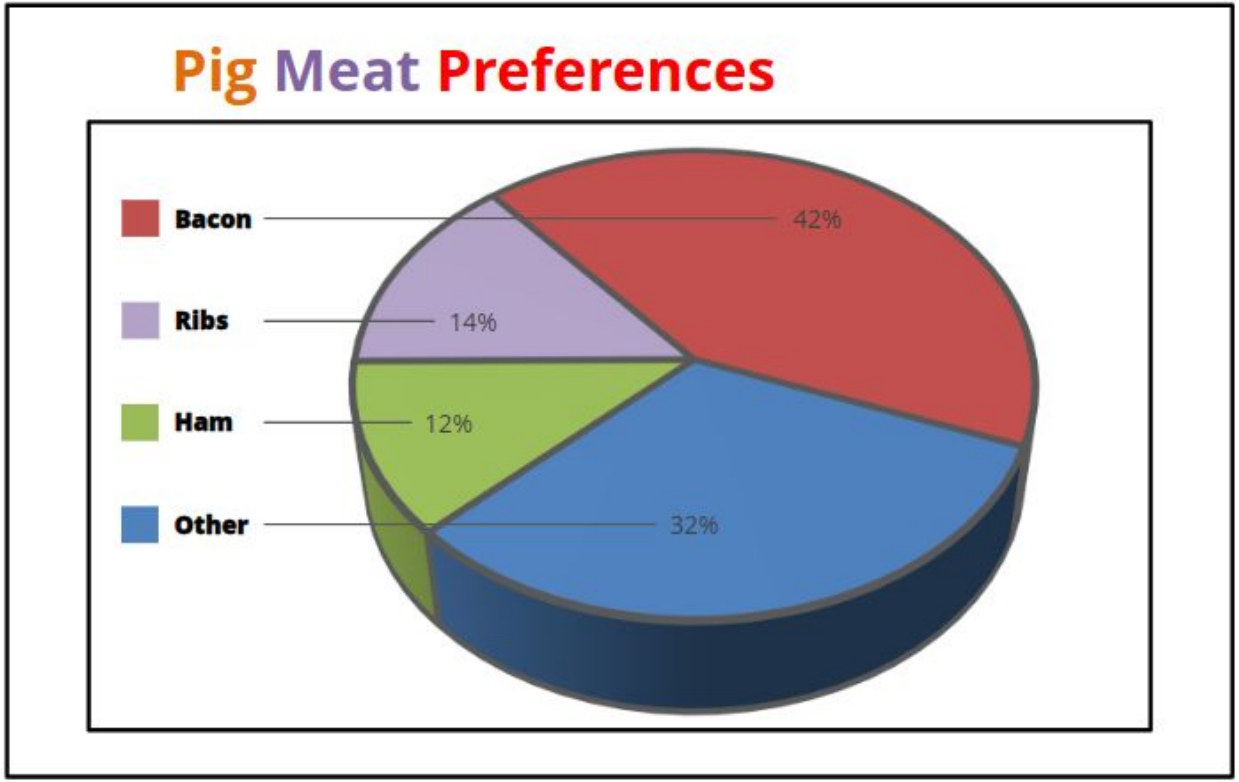
CC BY

Cake Charts - Removing Backgrounds



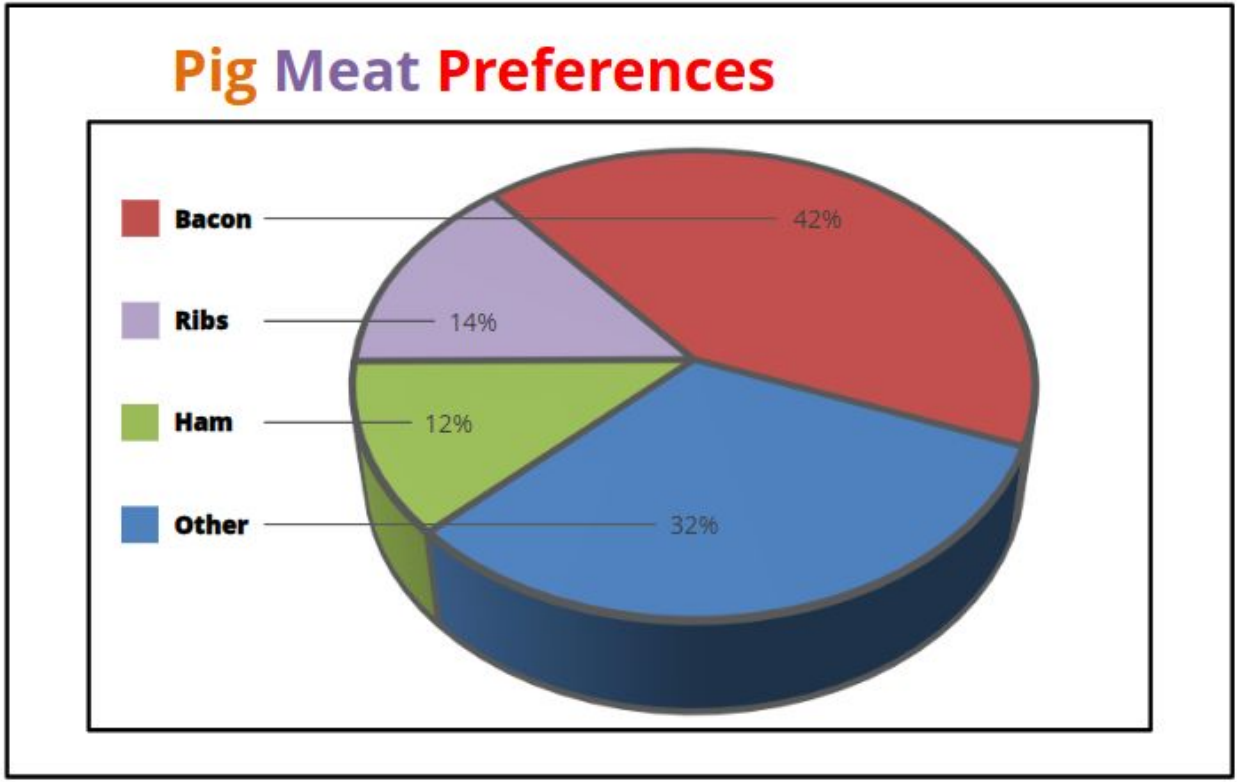
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Cake Charts - Removing Backgrounds



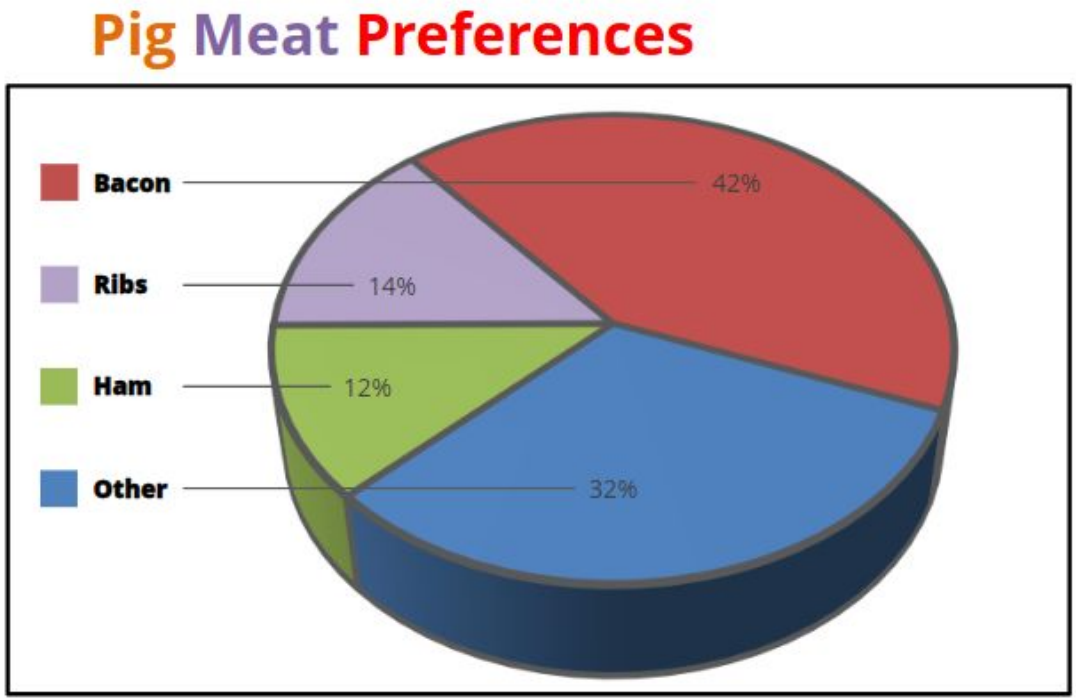
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Cake Charts - Removing Backgrounds



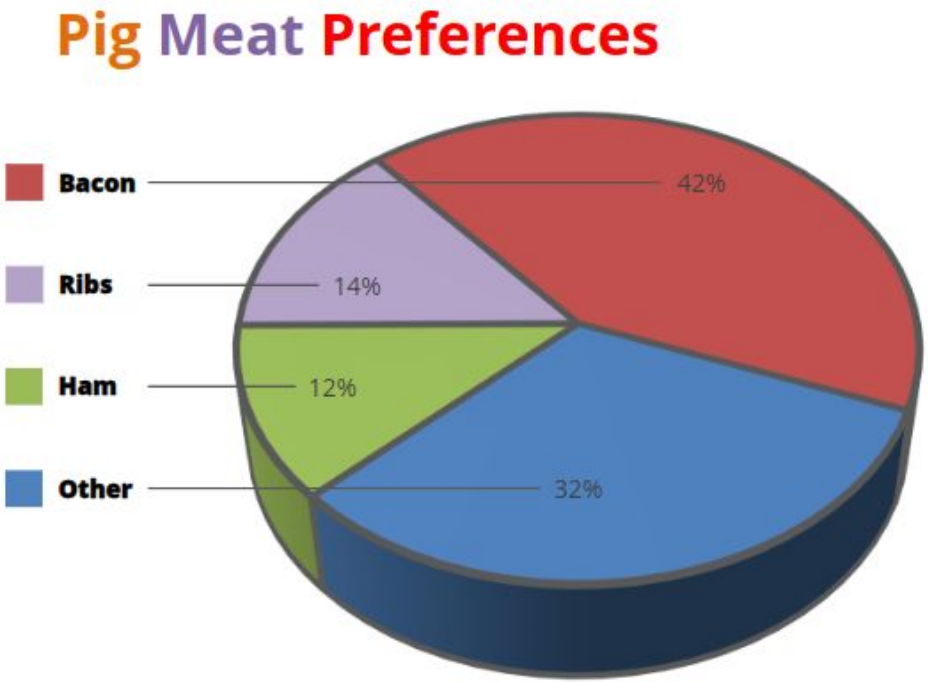
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Cake Charts - Removing Backgrounds



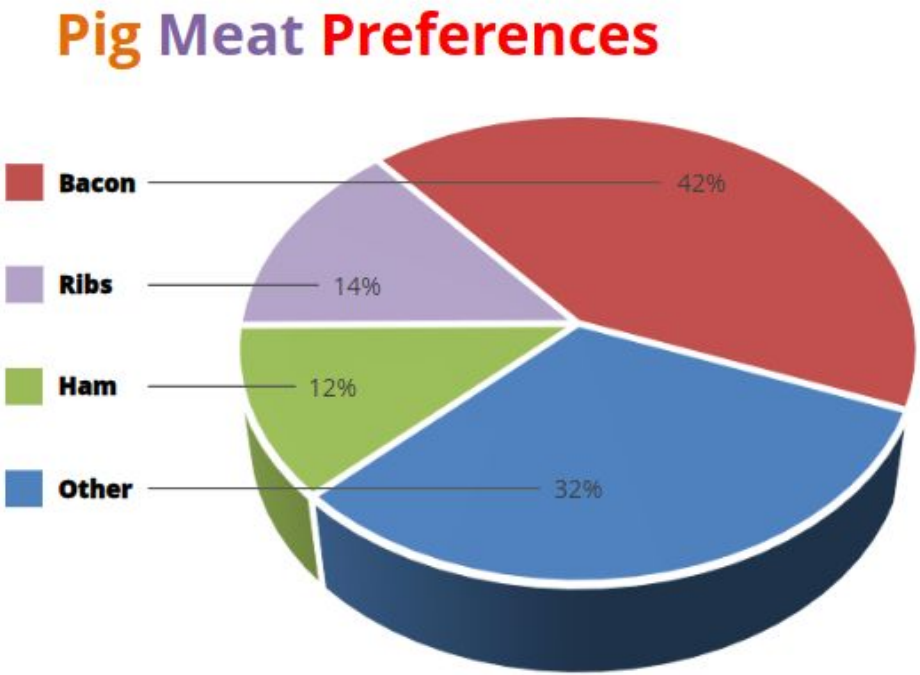
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Cake Charts - Removing Backgrounds



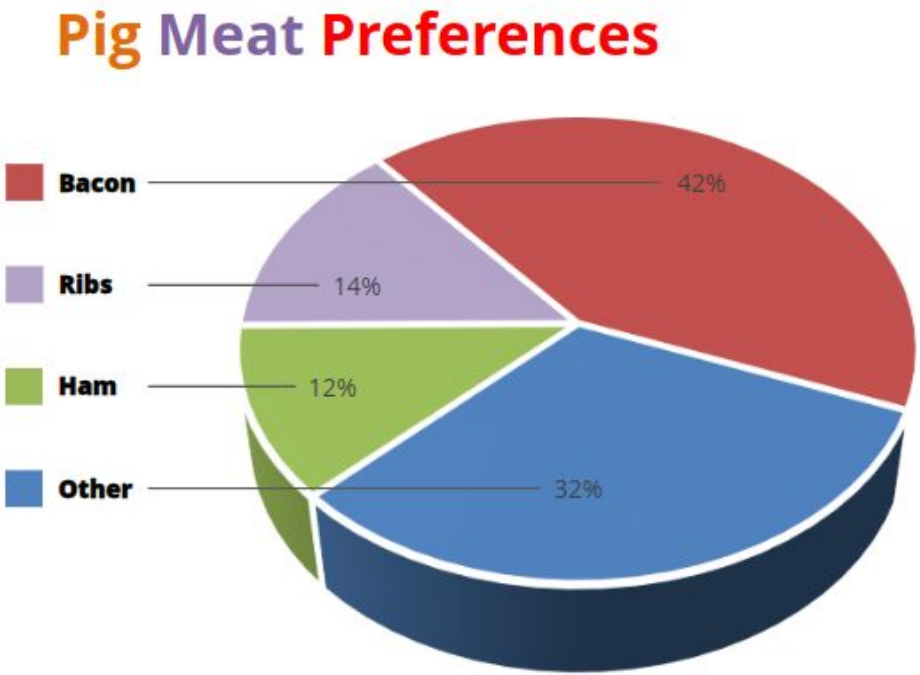
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Cake Charts - Removing Backgrounds



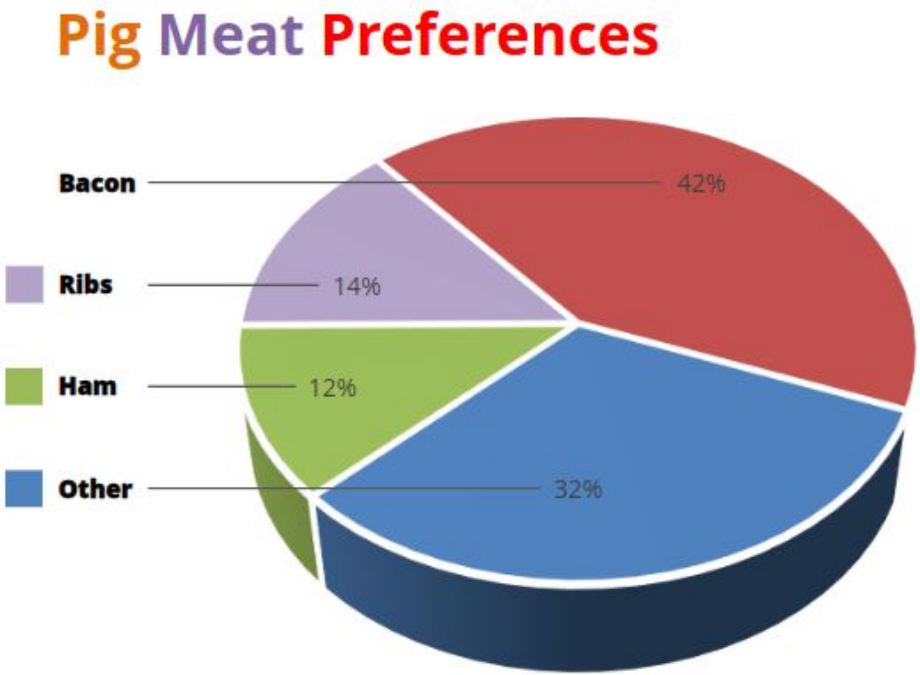
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Pie Charts - Remove Redundant Legend



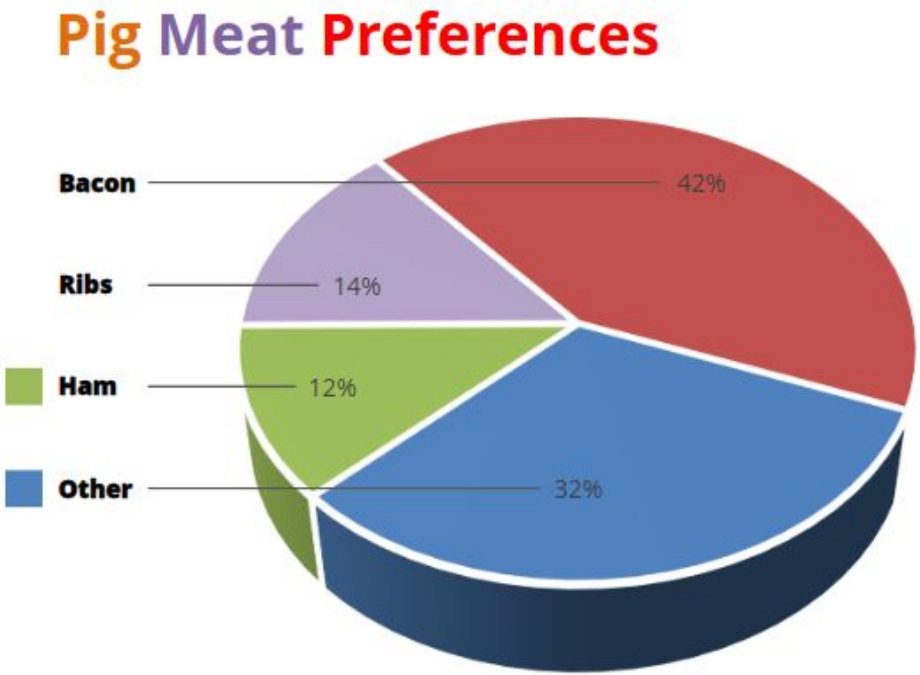
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Pie Charts - Remove Redundant Legend



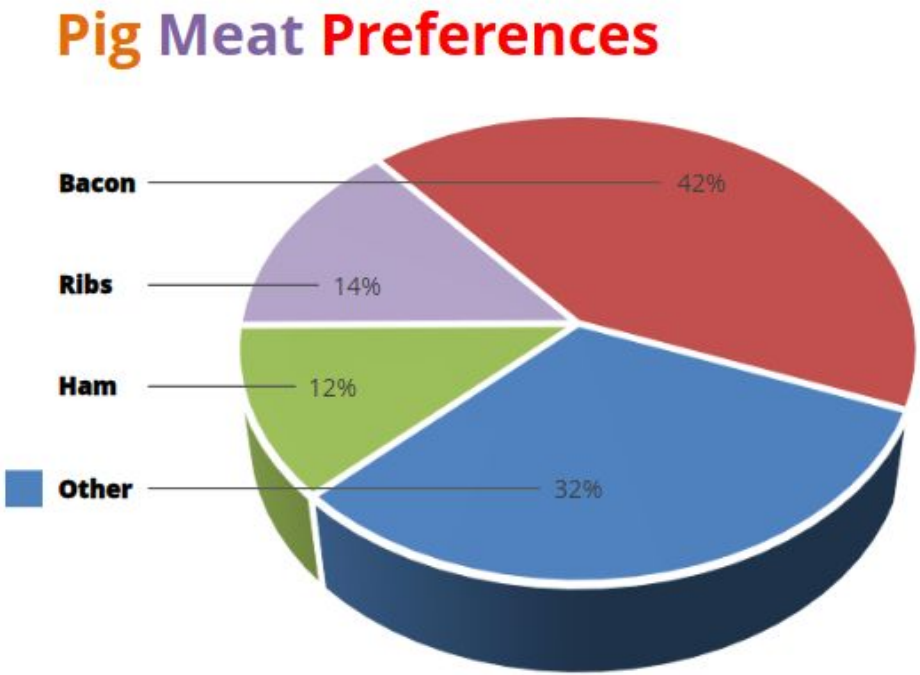
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Pie Charts - Remove Redundant Legend



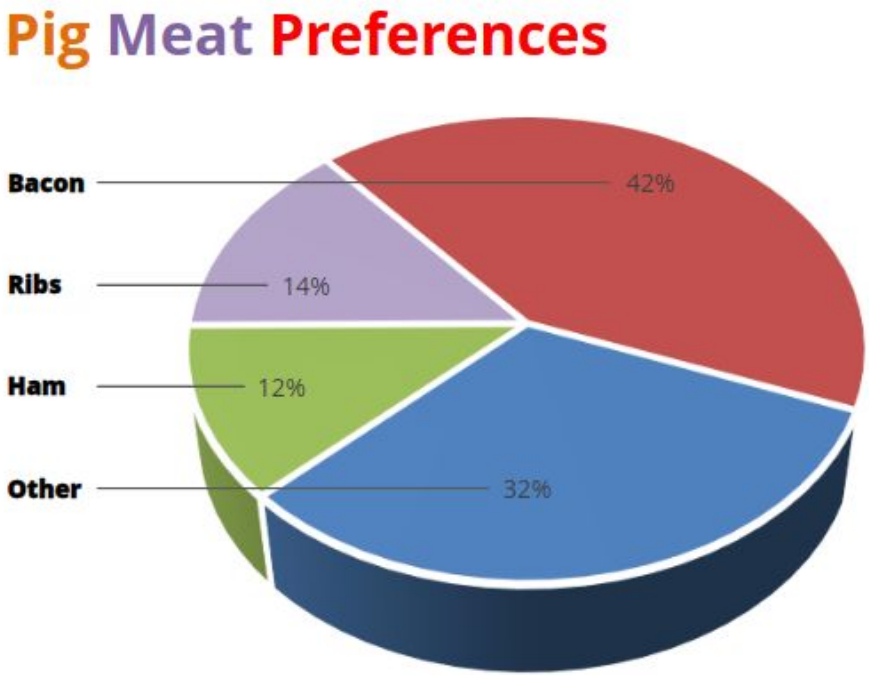
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Pie Charts - Remove Redundant Legend



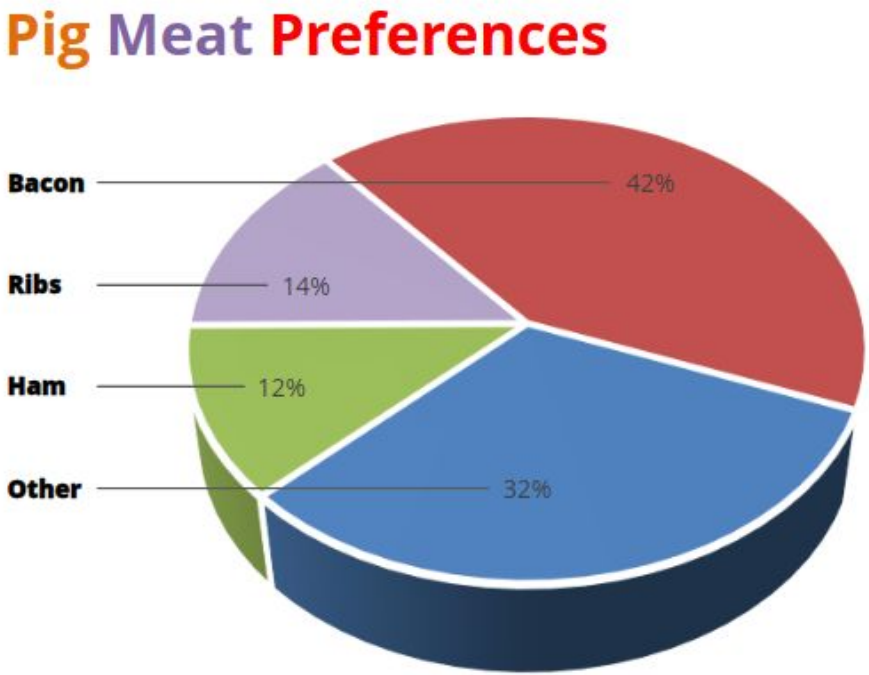
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Pie Charts - Remove Redundant Legend



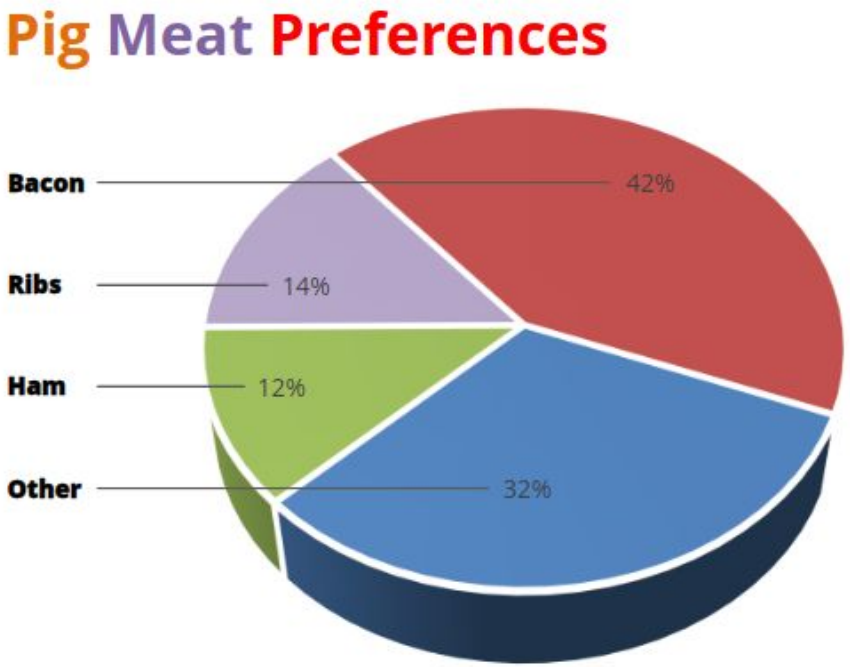
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Pie Charts - Remove 3D



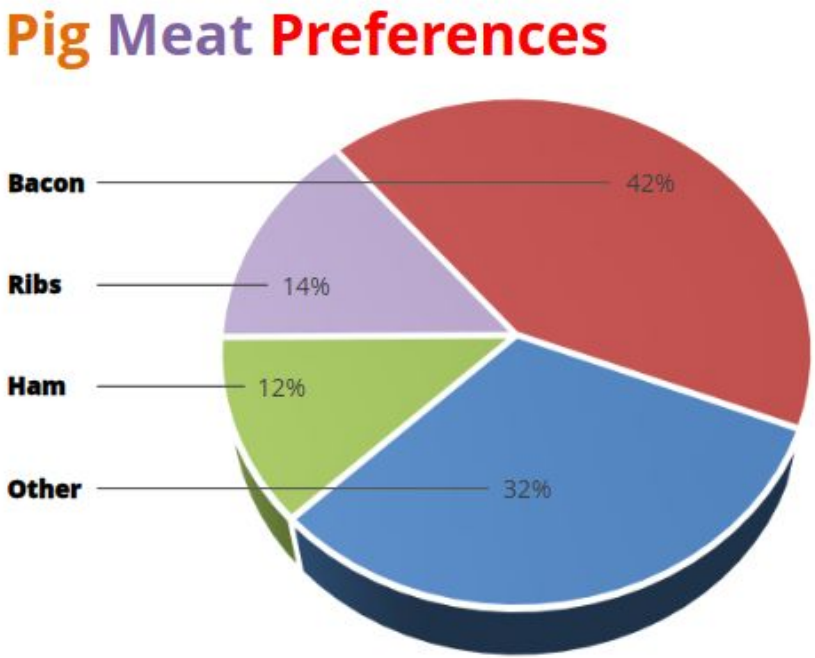
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Pie Charts - Remove 3D



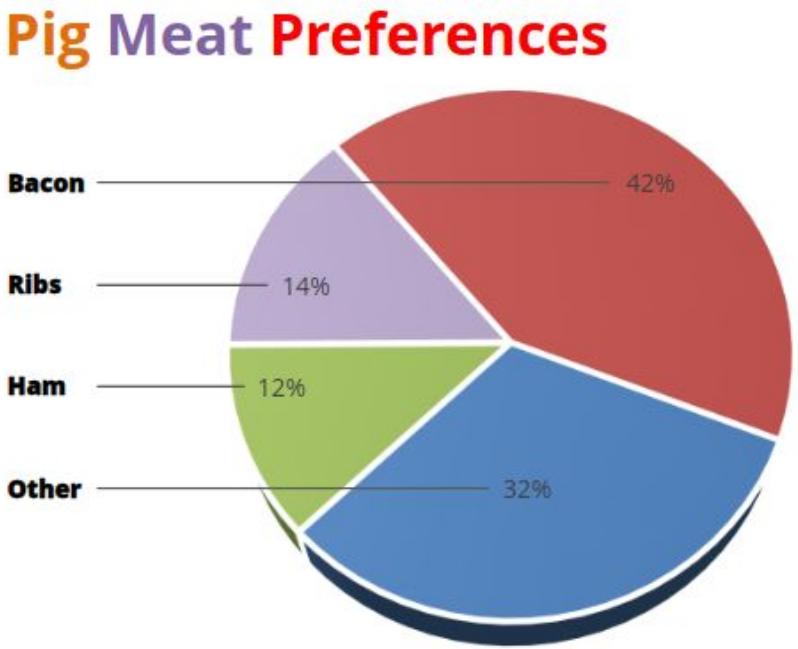
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Pie Charts - Remove 3D



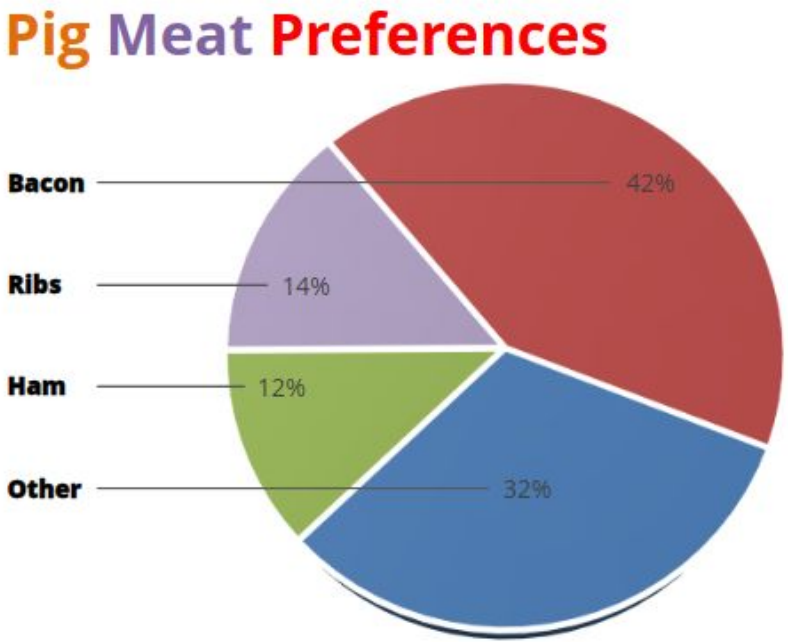
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Pie Charts - Remove 3D



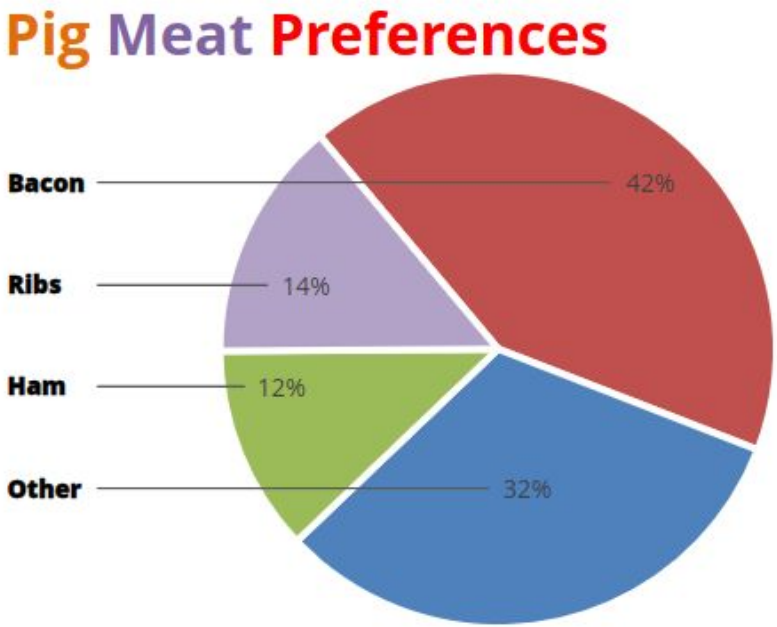
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Pie Charts - Remove 3D



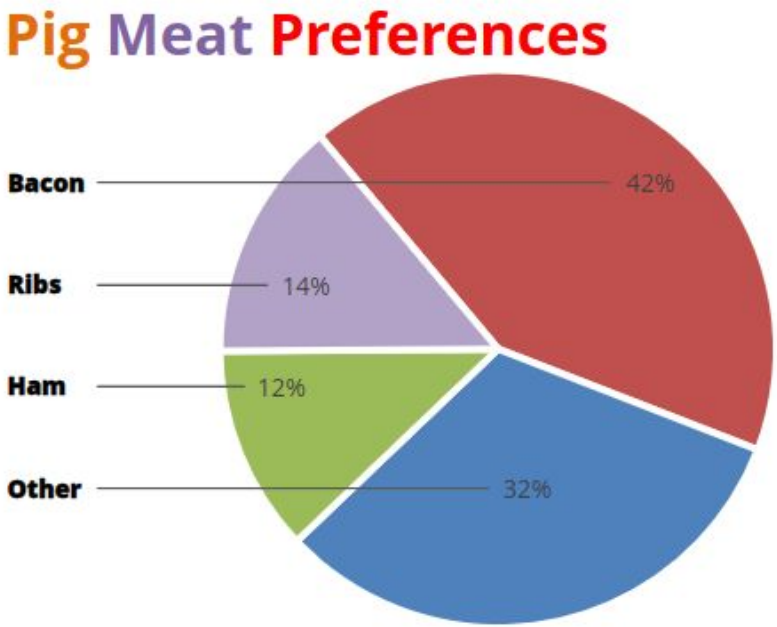
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Pie Charts - Remove 3D



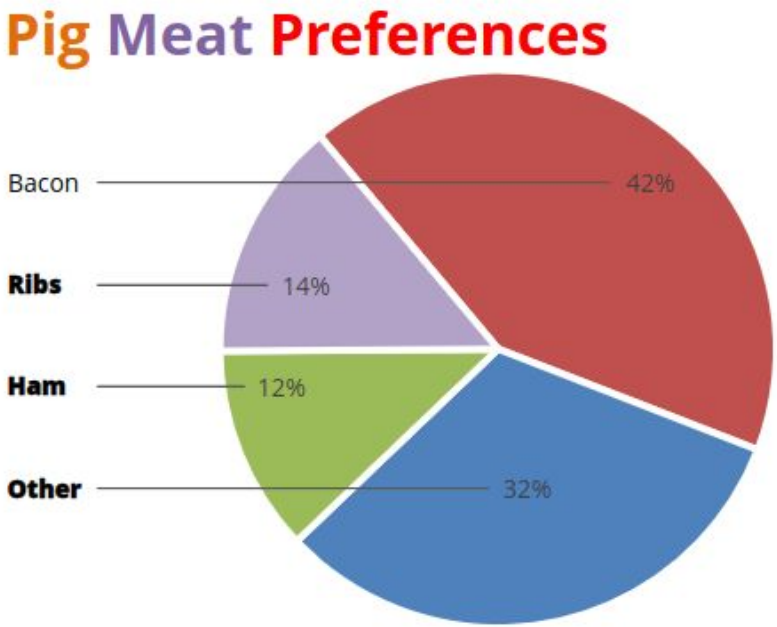
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Pie Charts - Remove Bold



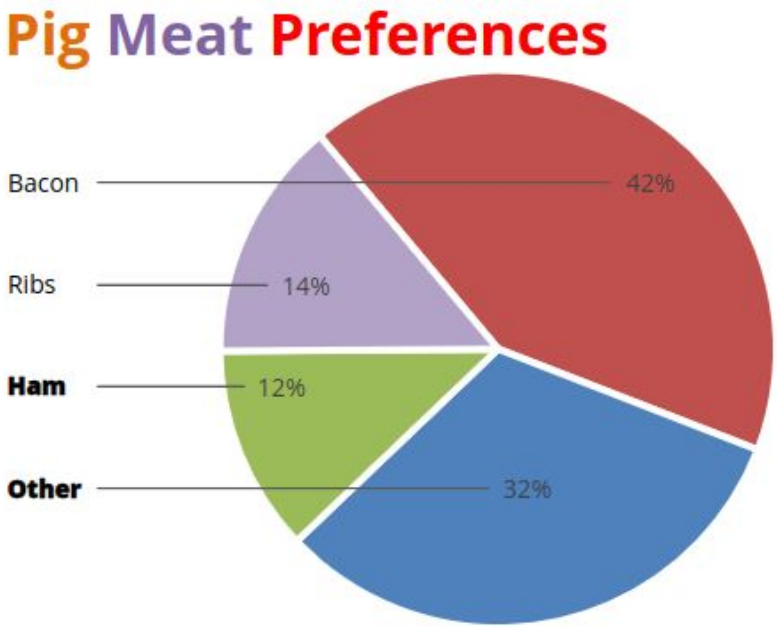
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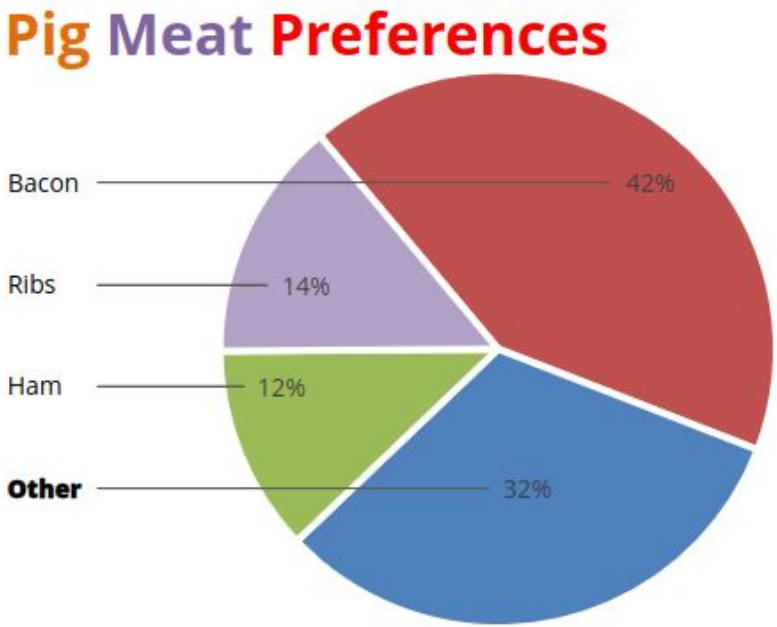
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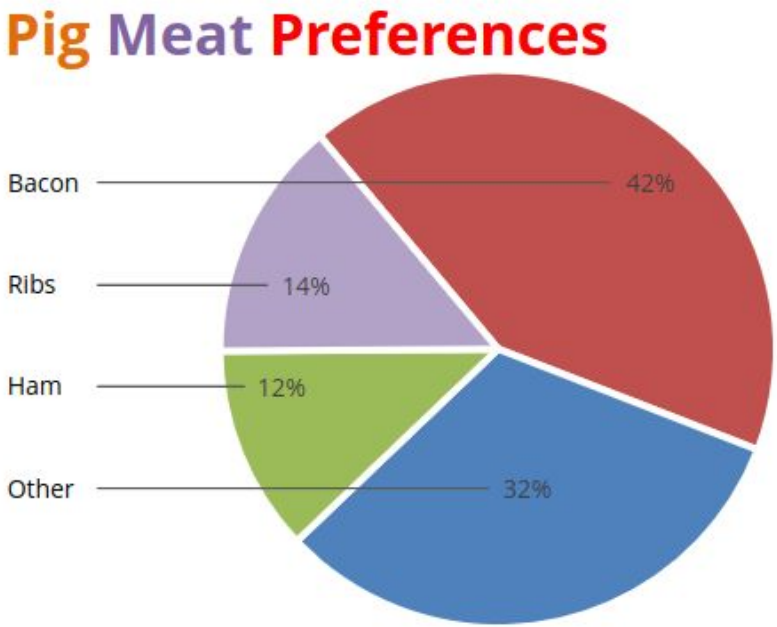
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Pie Charts - Remove Bold



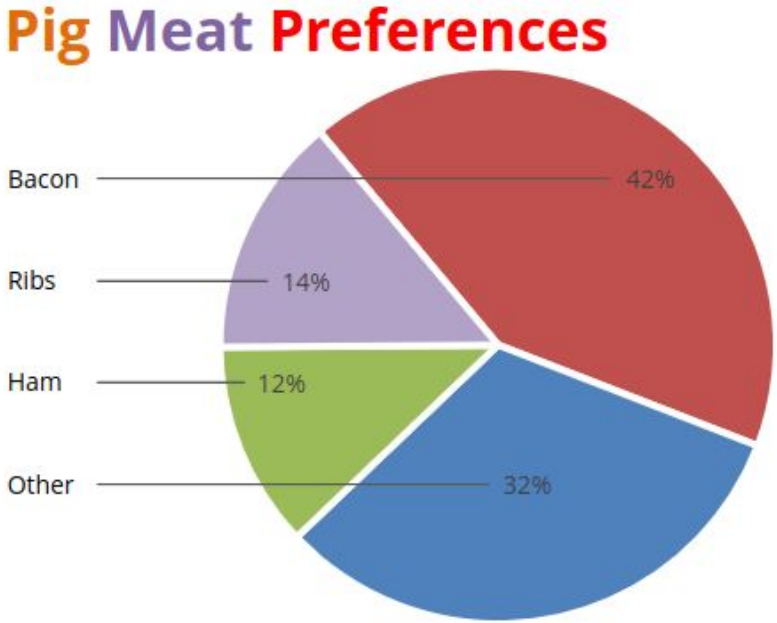
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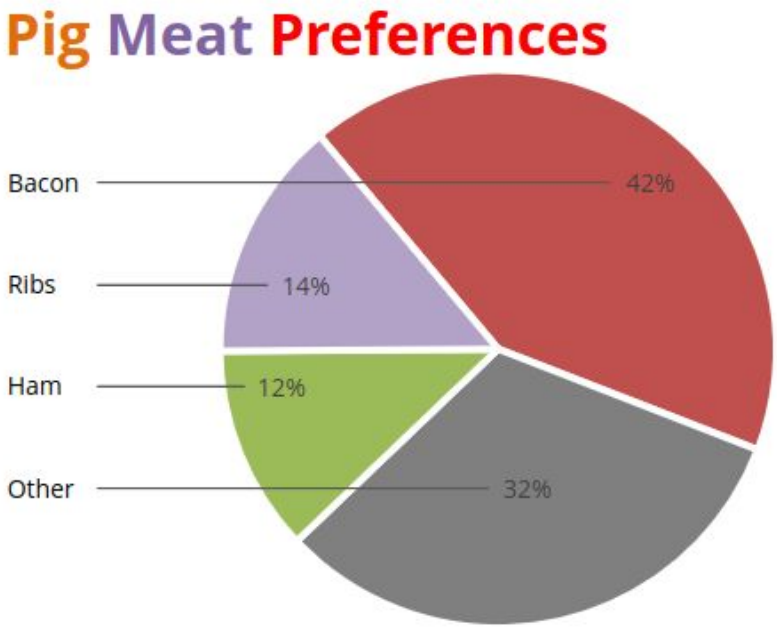
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Pie Charts - Reduce Colours



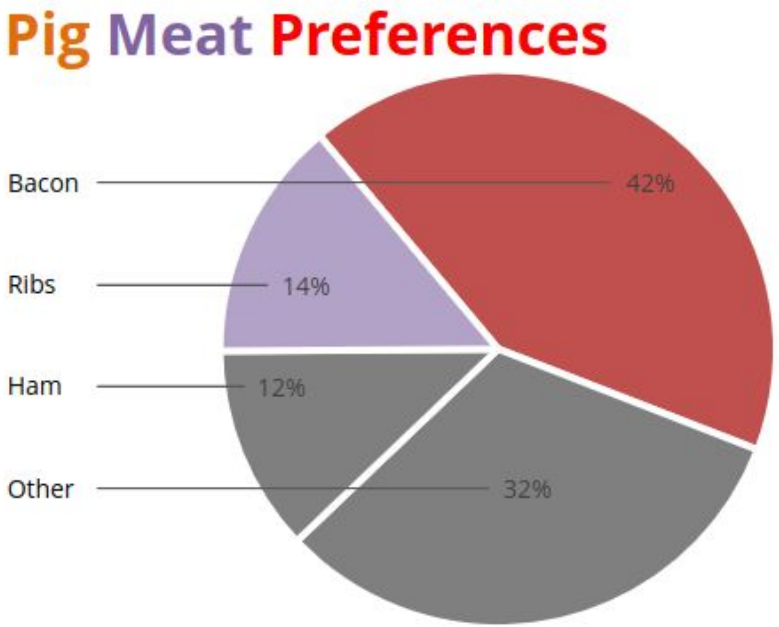
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Pie Charts - Reduce Colours



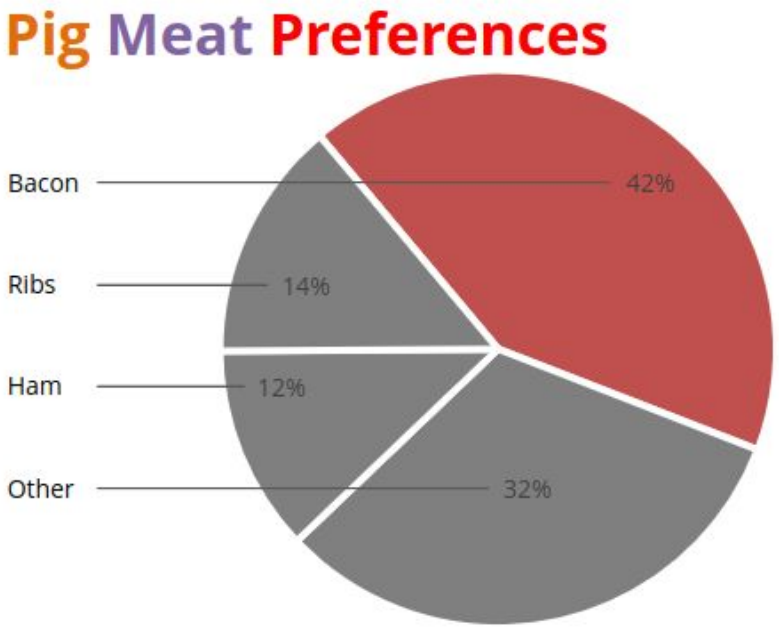
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Pie Charts - Reduce Colours



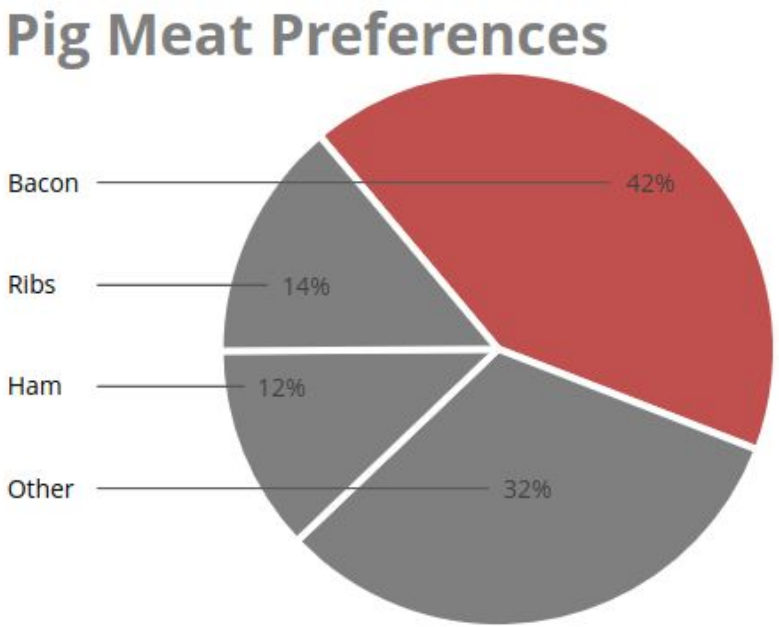
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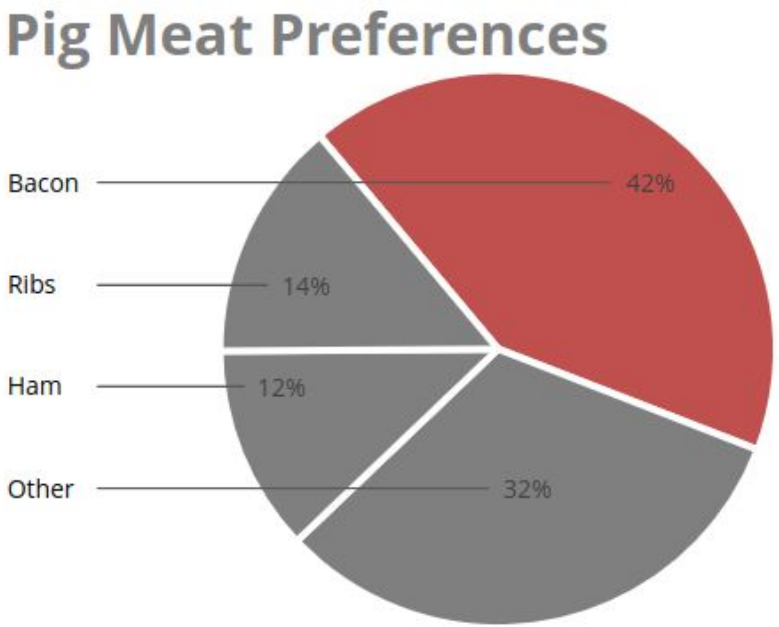
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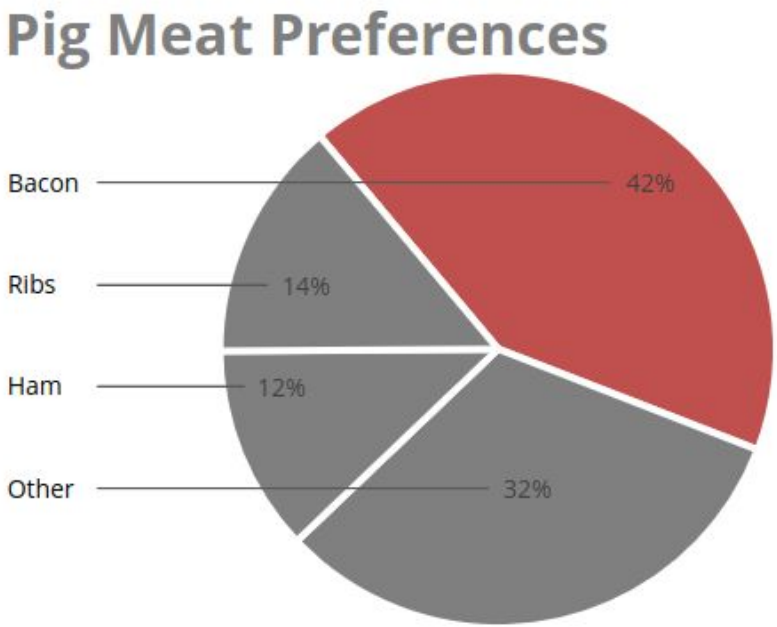
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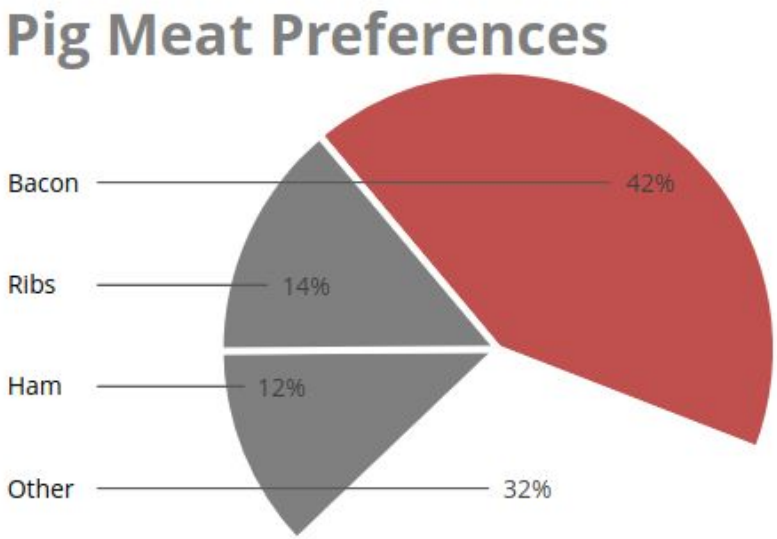
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Cake Charts - Removing Slices



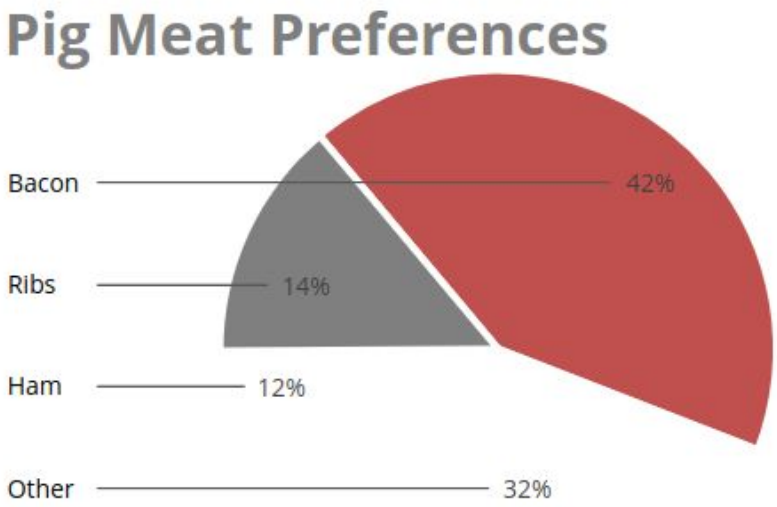
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Cake Charts - Removing Slices



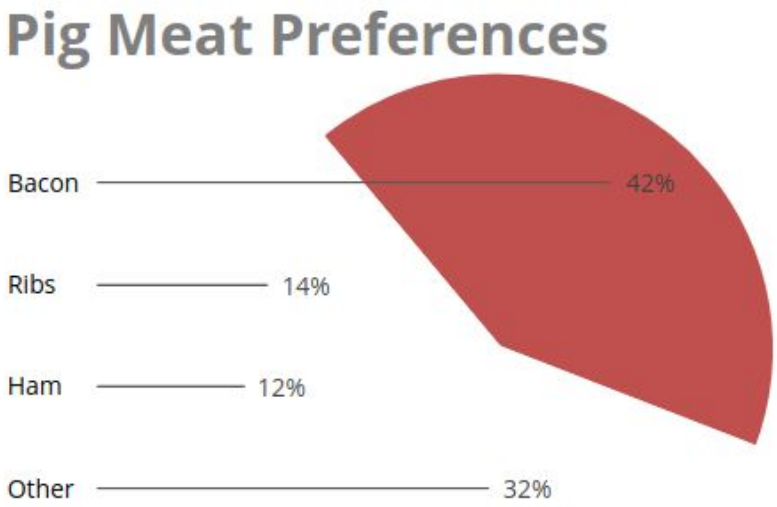
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Cake Charts - Removing Slices



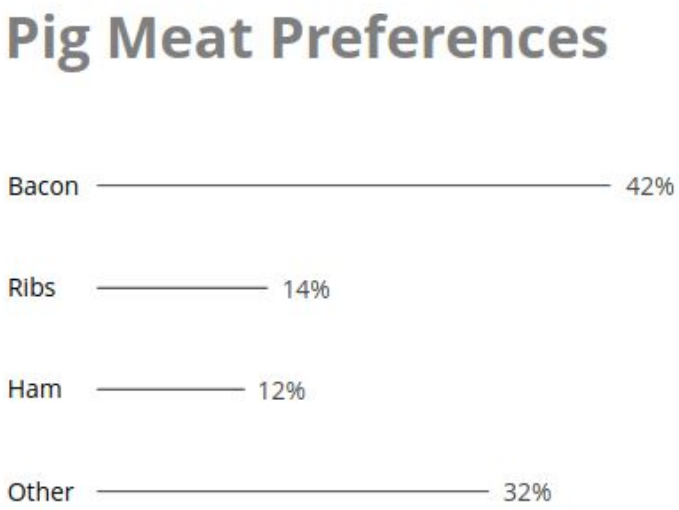
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Cake Charts - Removing Slices



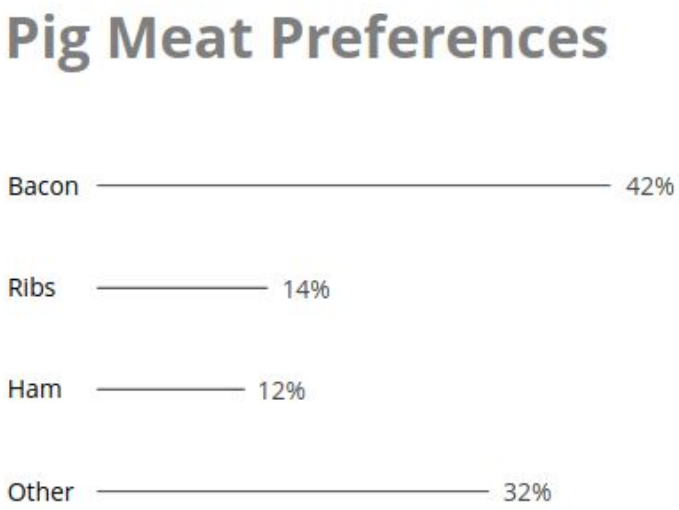
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Cake Charts - Removing Slices



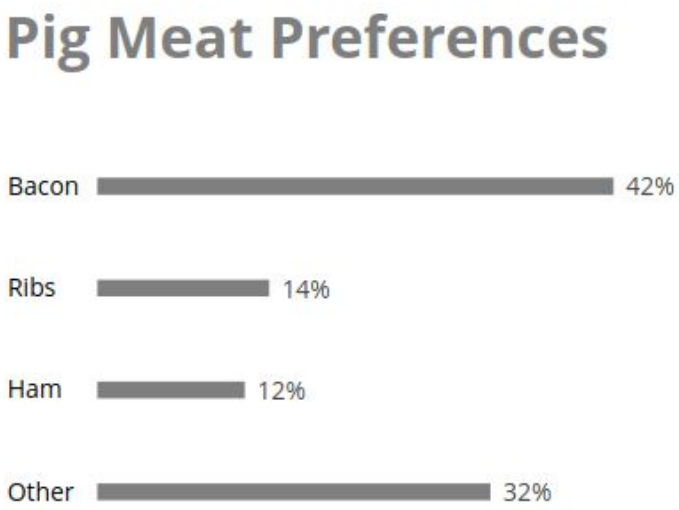
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Cake Charts - Enlarge Lines



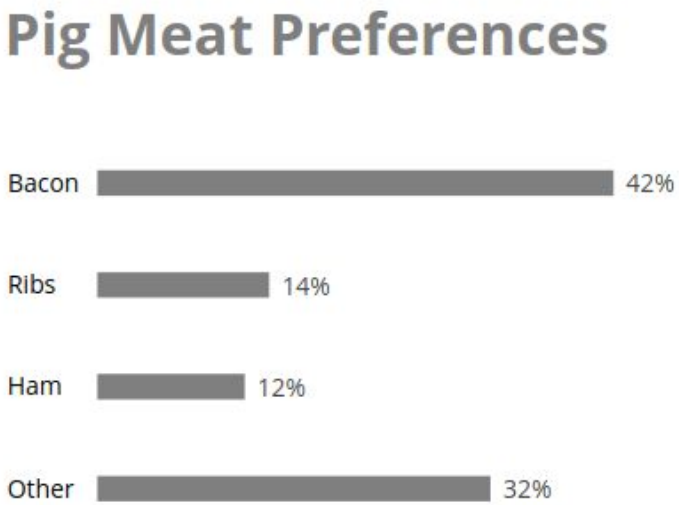
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Cake Charts - Enlarge Lines



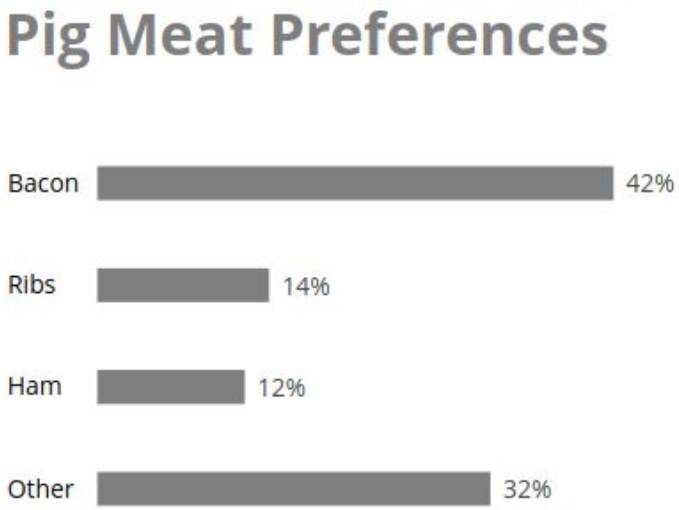
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Cake Charts - Enlarge Lines



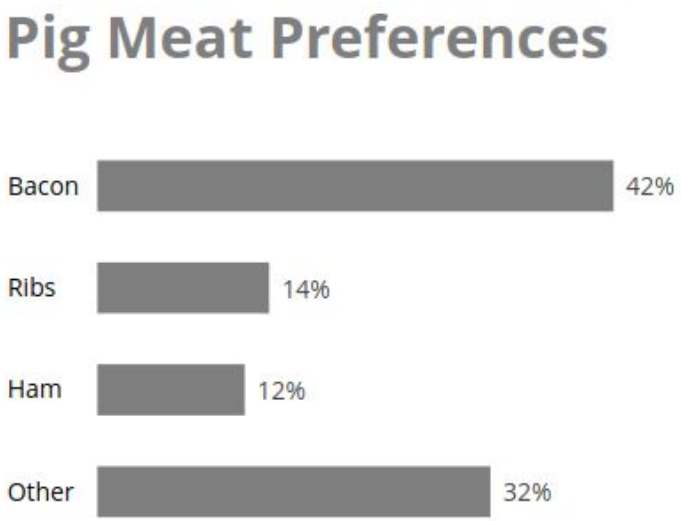
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Cake Charts - Enlarge Lines



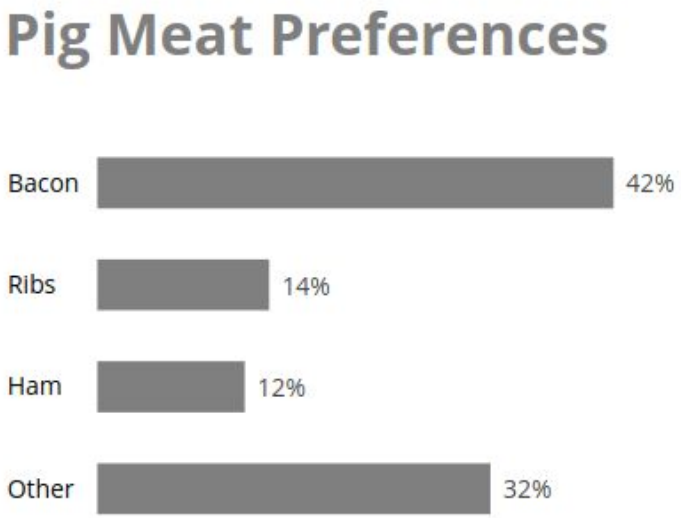
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Cake Charts - Enlarge Lines



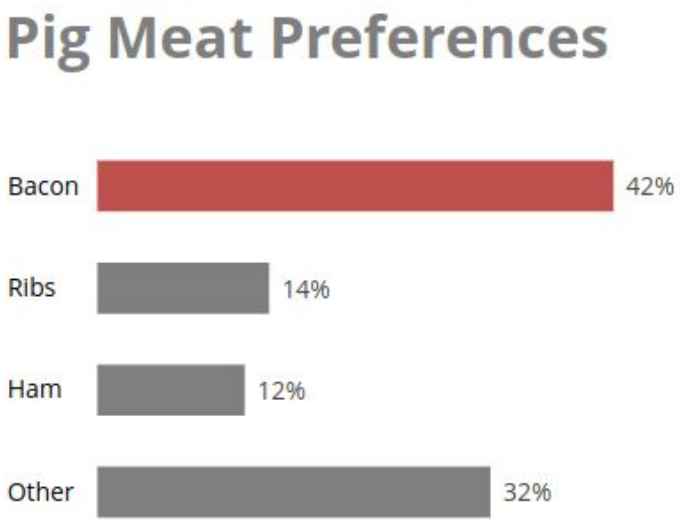
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Cake Charts - Emphasising Bacon



CC BY

Cake Charts - Emphasising Bacon



CC BY

Pie Charts



CC BY

Pie Charts

Less is **more**

(attractive)

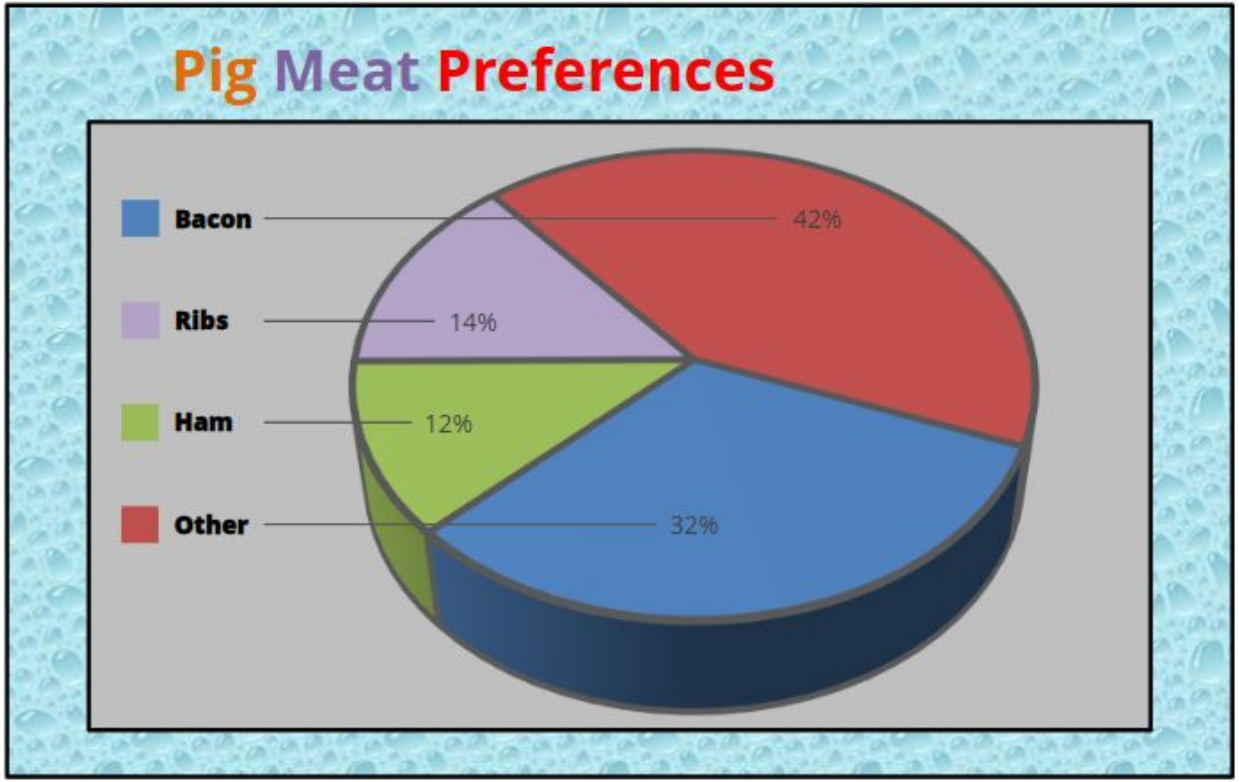
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Grafici Torta



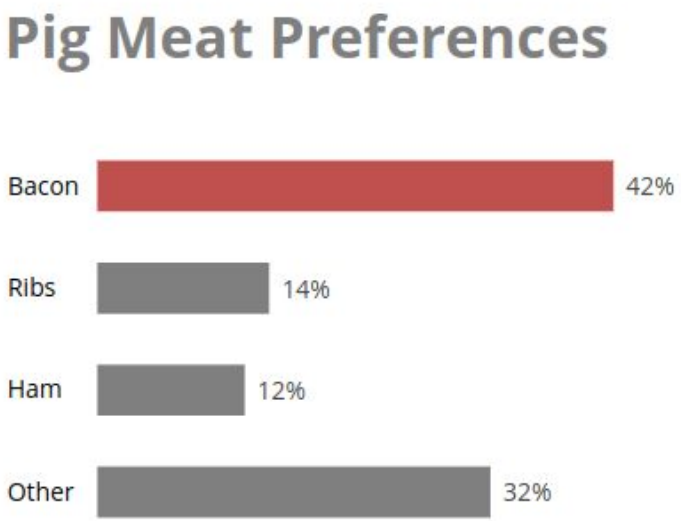
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Pie Charts - Before



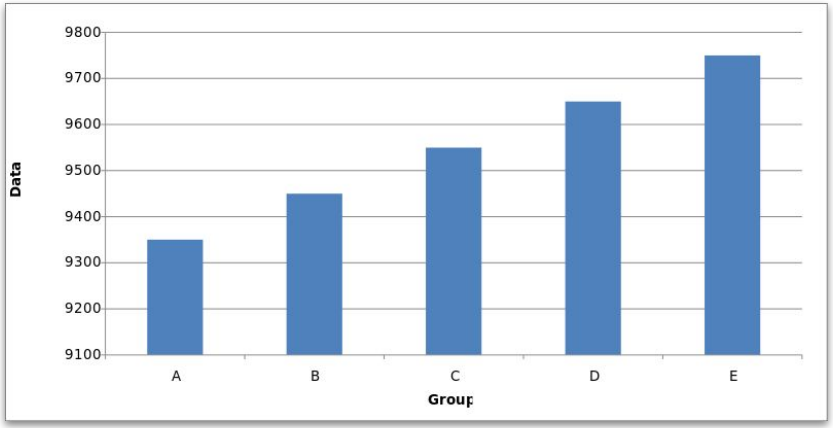
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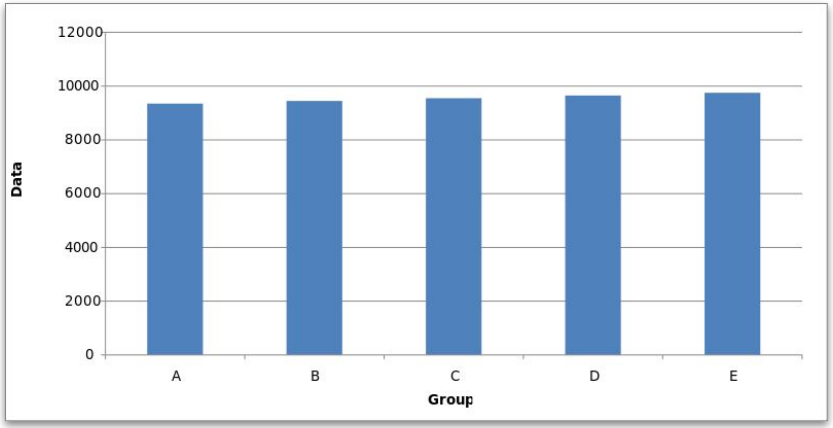
Pie Charts - After



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Truncated Graphs





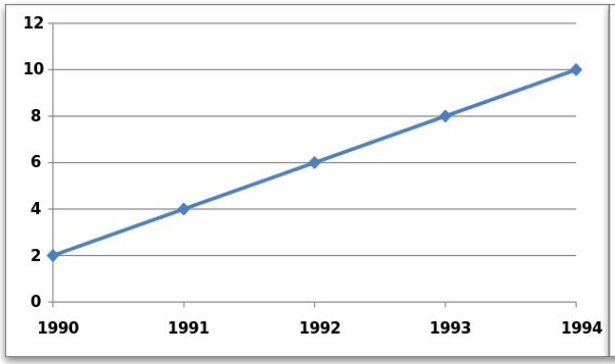
The graphs show identical data
Left: significant differences
Right: differences are not so significant

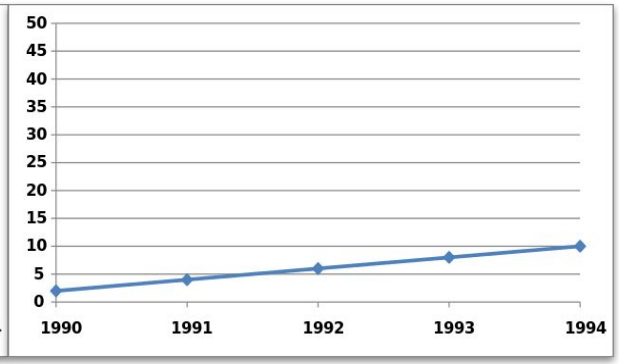
https://en.wikipedia.org/wiki/Misleading_graph

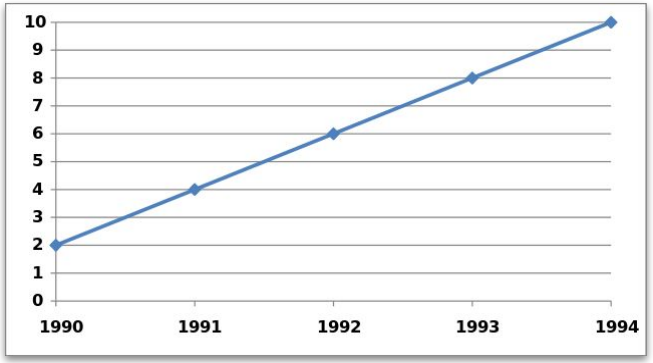
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CONFUSING BY CHANGING THE Y-AXIS





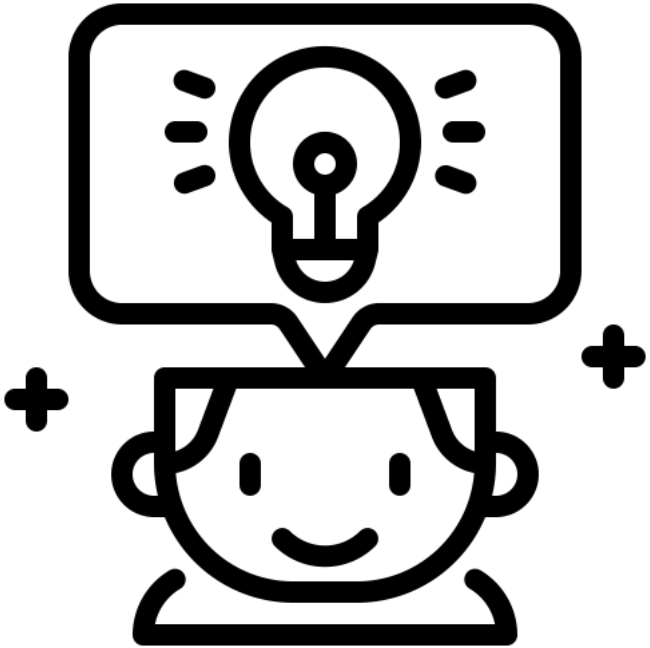


https://en.wikipedia.org/wiki/Misleading_graph

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Tasks of a good data visualisation

Answering (or discovering) questions
Making decisions
Visualising data in context
Identifying recurrences
Presenting topics
Telling a story
Inspiring

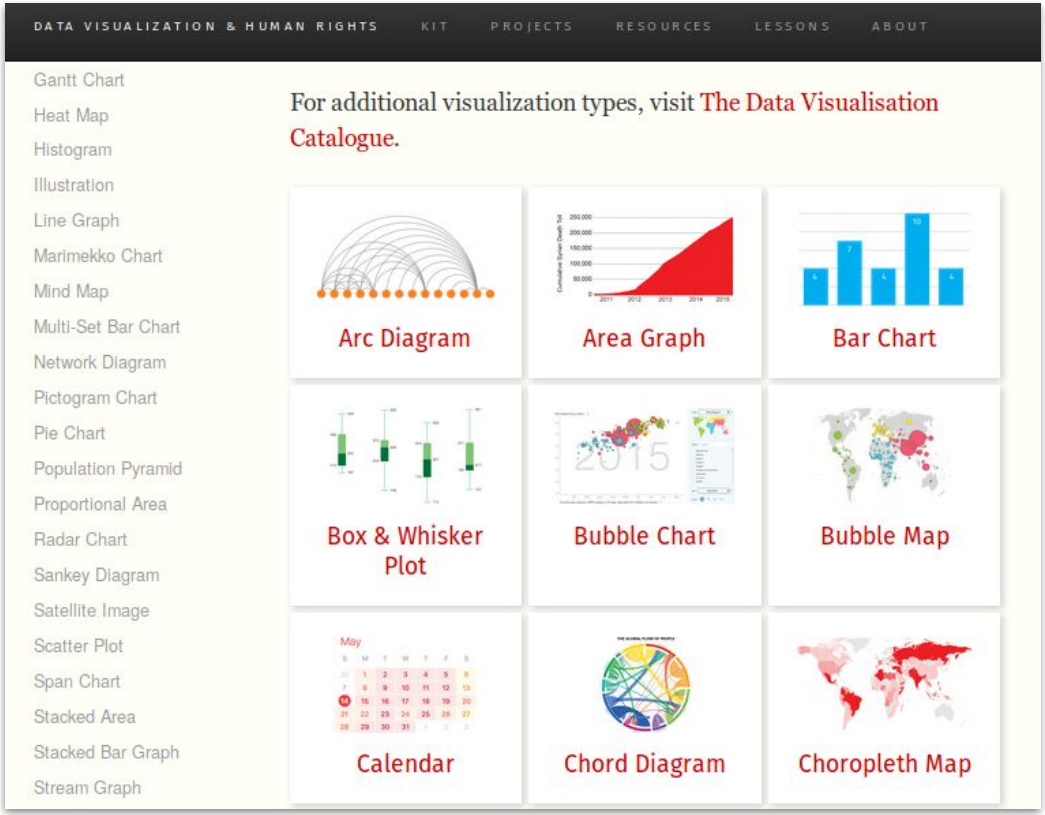




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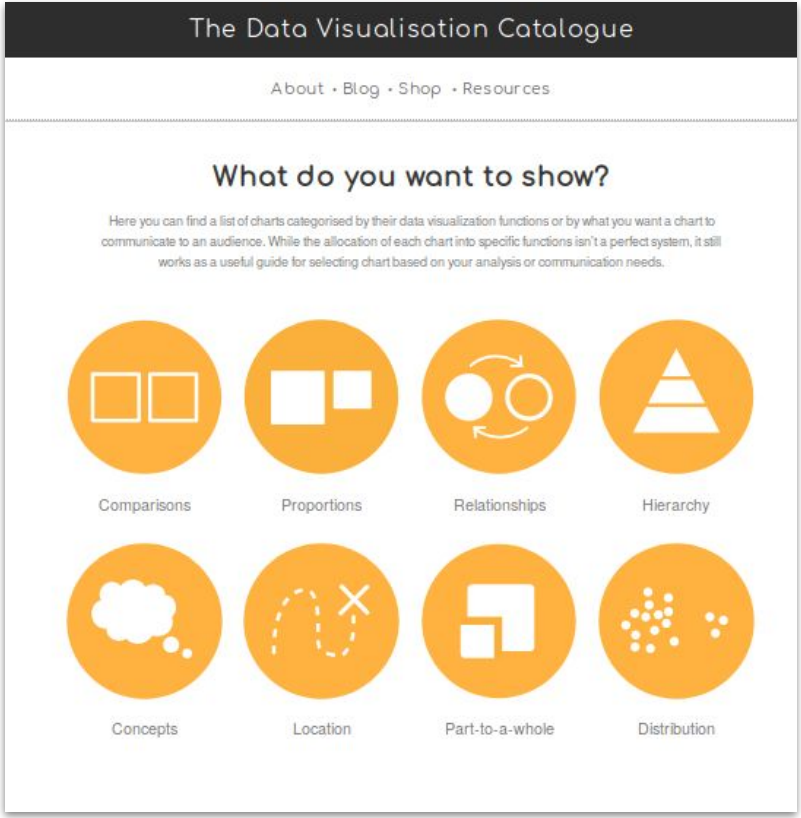
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Which charts exist?



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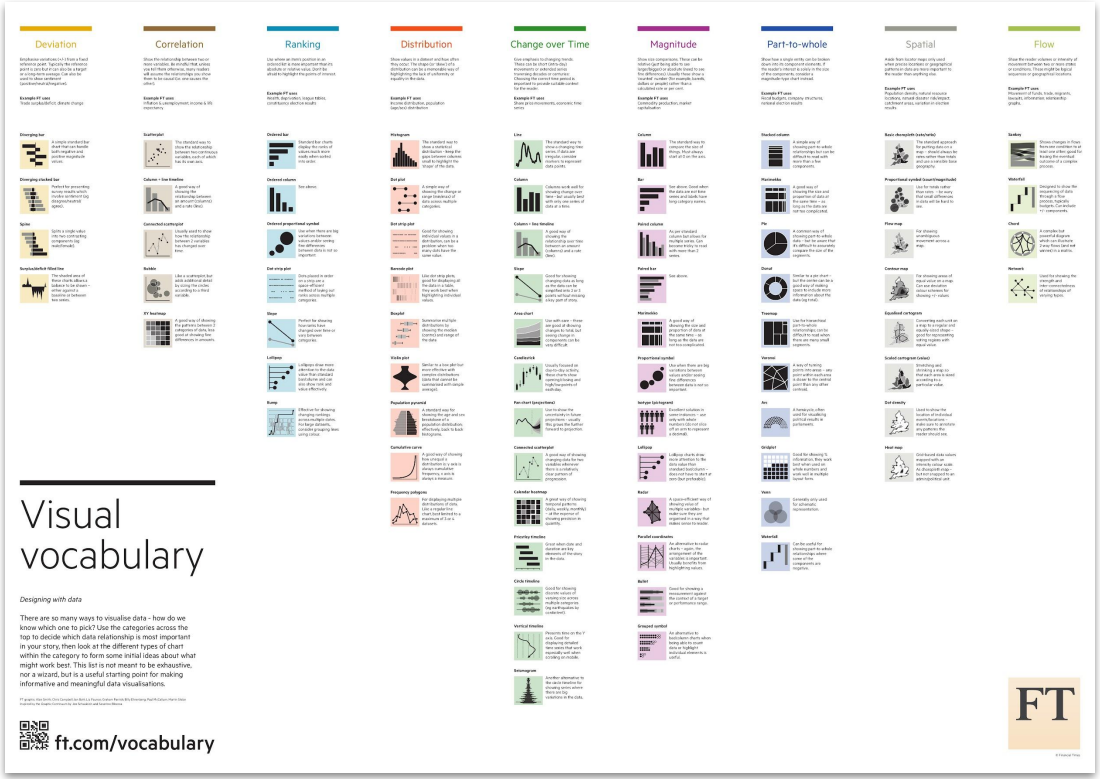
What do I have to represent?



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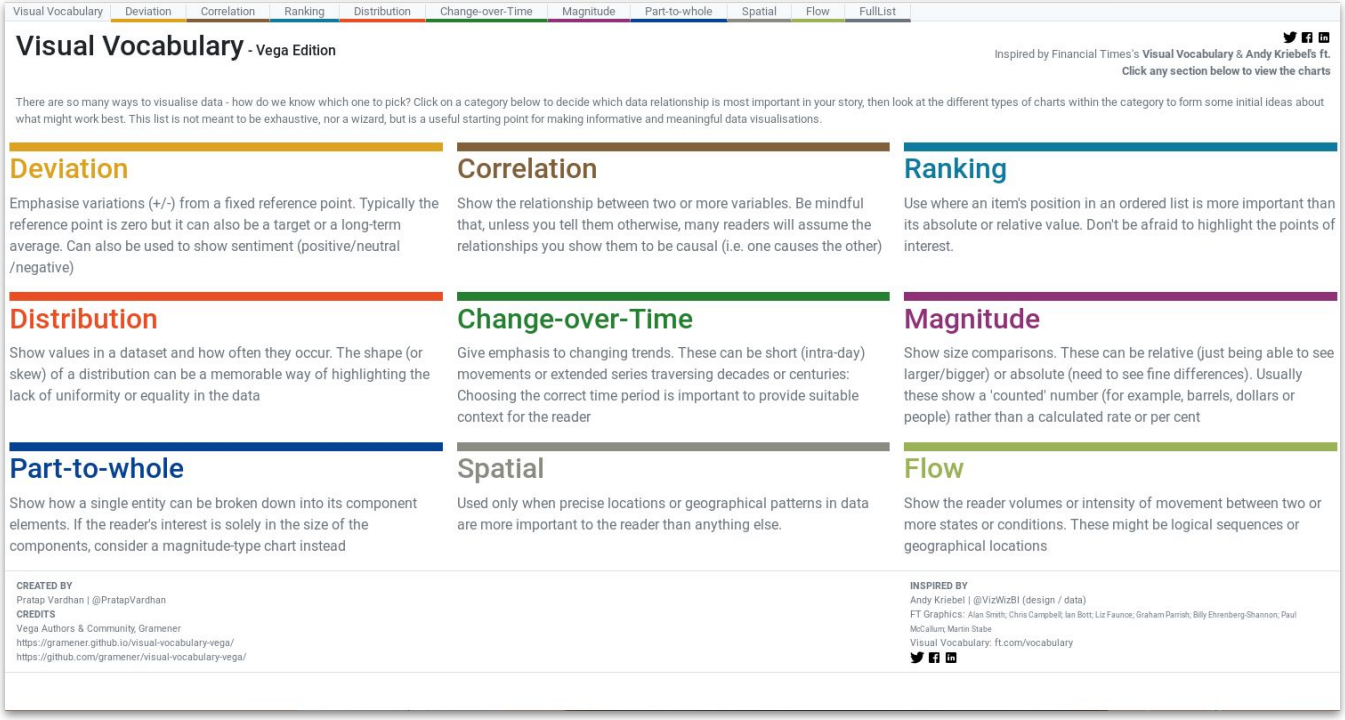
the Visual Vocabulary of Financial TImes



https://ft-interactive.github.io/visual-vocabulary/

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the Visual Vocabulary of Financial TImes - Vega Version



from Data to Viz

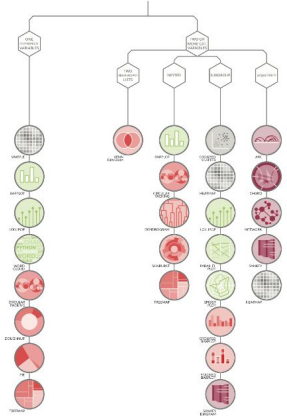
'From Data to Viz' is a classification of chart types based on input data format. It will help you find the perfect chart in three simple steps :

1. Identify what type of data you have.
2. Go to the corresponding decision tree and follow it down to a set of possible charts.
3. Choose the chart from the set that will suit your data and your needs best.

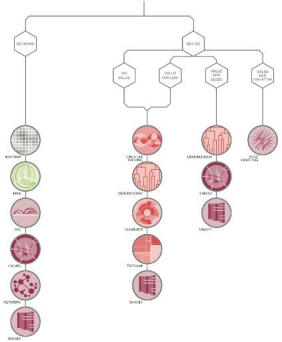
Dataviz is a world with endless possibilities and this project does not claim to be exhaustive. However it should provide you with a good starting point. For an interactive version and much more, visit:

data-to-viz.com

CATEGORIC



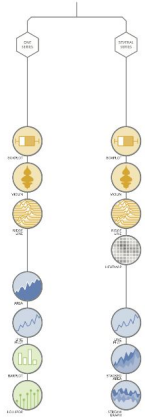
RELATIONAL



MAP



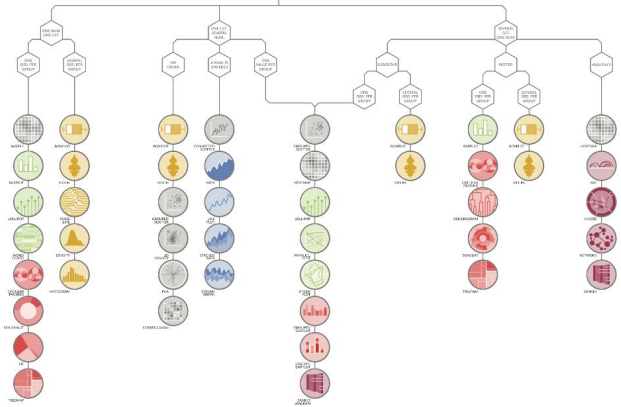
TIME SERIES



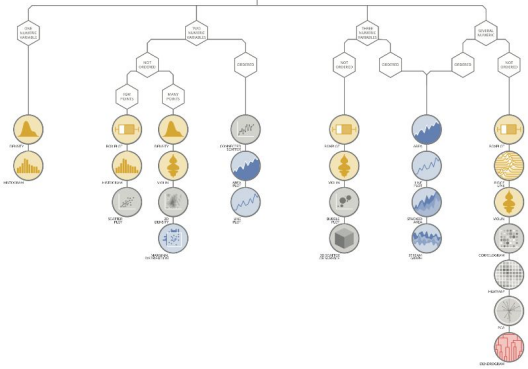
WHAT DO YOU WANT TO SHOW ?

- Distribution
- Correlation
- Ranking
- Part of a whole
- Evolution
- Maps
- Flow

CATEGORIC AND NUMERIC

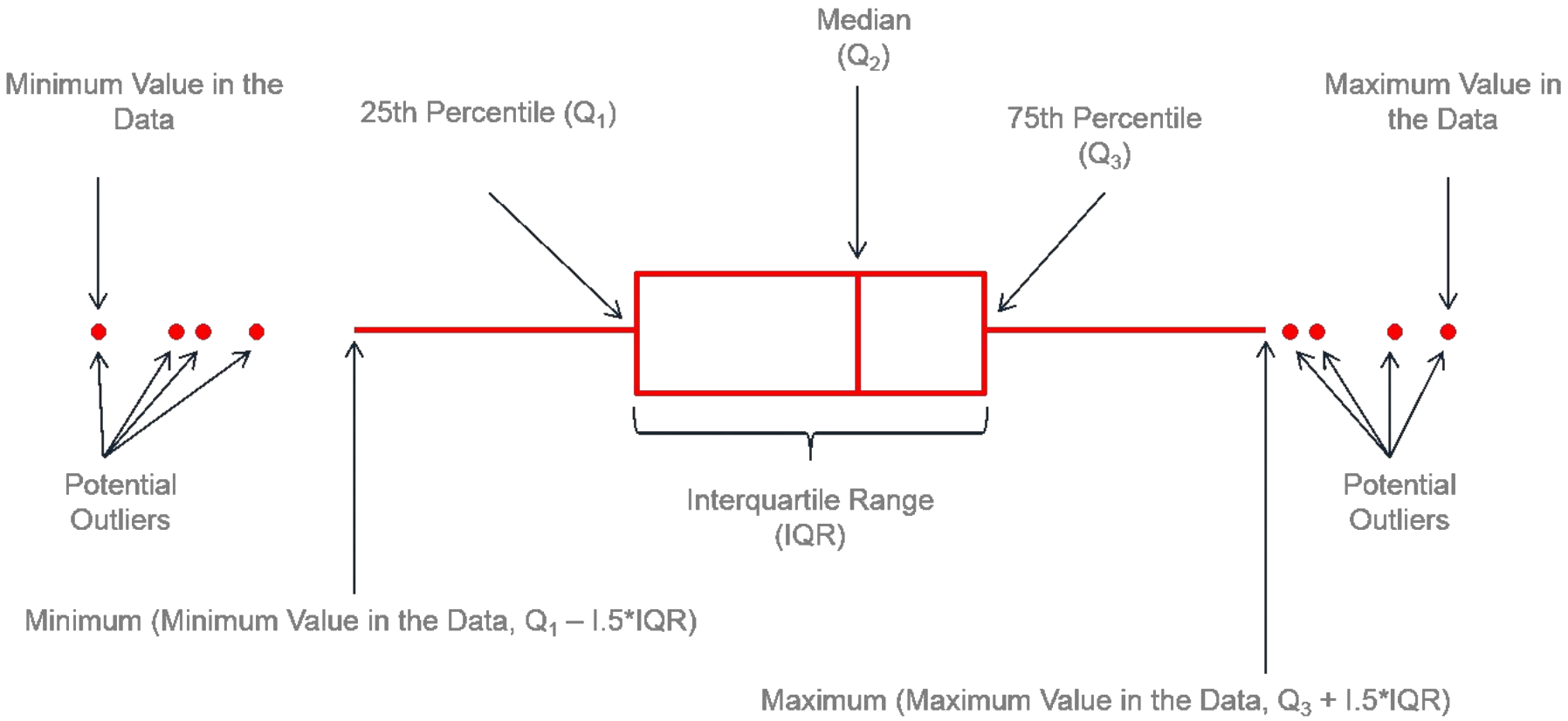


NUMERIC



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Median
(Q_2)
Minimum Value in the Data
25th Percentile (Q_1)
75th Percentile
(Q_3)
Maximum Value in the Data
Potential Outliers
Interquartile Range
(IQR)
Potential Outliers
Minimum (Minimum Value in the Data, Q_1 – 1.5*IQR)
Maximum (Maximum Value in the Data, Q_3 + 1.5*IQR)

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Minilab

Sticking our finger in the jam

- Rawgraphs
- Datawrapper
- Flourish Studio
- uMap
- KnitLab Tools
- Lapis Story



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Resources

Readings:
How Charts Lie: Getting Smarter about Visual Information - Alberto Cairo

Websites:
Colorbrewer 2 https://colorbrewer2.org
Charts Type - http://visualizingrights.org/kit/charts/index.html
DataVizCatalogue - https://datavizcatalogue.com/
Visual Vocabolary https://ft-interactive.github.io/visual-vocabulary/
Misleading graph - https://en.wikipedia.org/wiki/Misleading_graph
Datawrapper - https://datawrapper.de
Rawgraph - https://rawgraph.io
Automate Google Spreadsheets
https://napo.medium.com/automatizzare-la-raccolta-e-analisi-dei-dati-con-google-fogli-3f1eca078556

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"A chart should be like a good friend:
he doesn't need many words to make himself understood."



#30daychartchallenge
#horrorchart
#30daymapchallenge

For contacts Maurizio Napolitano - napo@fbk.eu

Special greetings
Andrea Borruso Marco Cortella Lihua Ma Raffaele Mastrolonardo Matteo Moretti Flavia Marzano Francesco Piccinelli Casagrande Giorgia Lodi Riccardo Saporiti Isaia Invernizzi The Noun Project
#horrorchart

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Data Storytelling

Maurizio Napolitano
<napolitano@fbk.eu>

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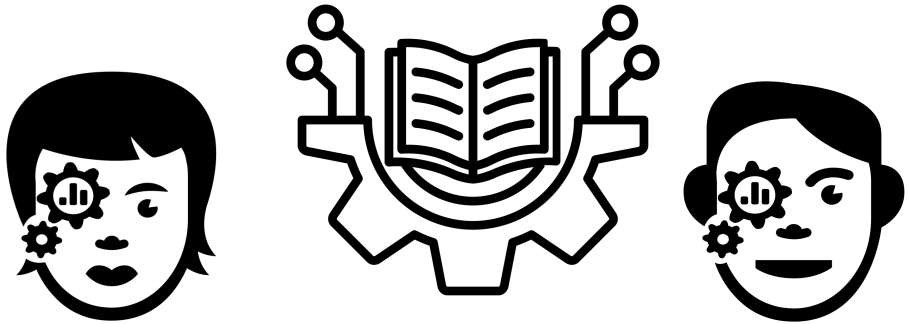
Goal of the lecture

Data storytelling uses narratives to help readers comprehend a phenomenon by representing and analyzing data

empower readers to independently identify additional narratives

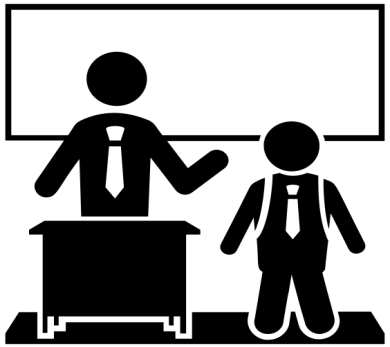
The participants will acquire these skills

data analysis and visualization skills - narrative development and web design - critical thinking and ethical reporting



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Introduction



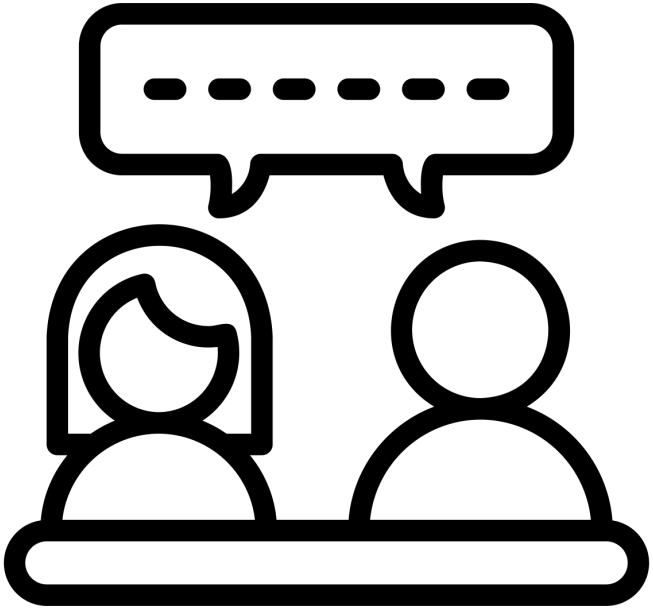
Data plays a crucial role in modern society, yet it remains exclusive as it requires specific skills for processing, understanding, and representing it effectively.

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And you?

One word to describe the data storytelling

Why are you here?

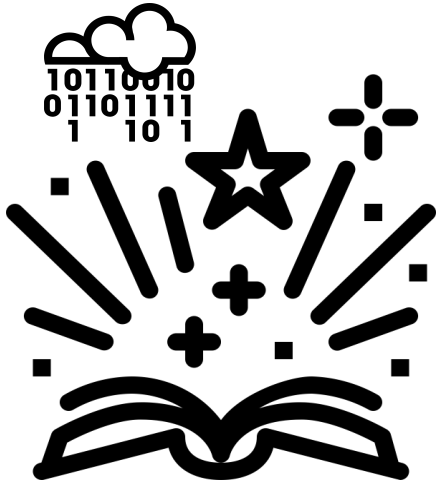


Skills to share?

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Introduction to Data Storytelling





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definitions

Narrative
a story helps the audience understand insights by transforming complex information into narrative elements.

Visual
it serves to illustrate, make the story more real and concrete. Narration and images must always be connected, that is, images support the narrative.

Data
is the basis of the narrative and supports the narrative itself.

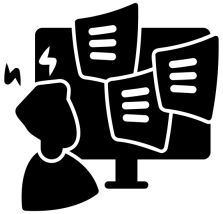
"Data Storytelling literally means **telling** a **story** through **data**. Concretely, however, it consists of reading through hundreds, thousands, millions of data, exploring the relationships between them, finding the significant patterns and, through **Data Visualization**, telling them in such a way as to provide meaningful information that can **guide decisions**, **choices**, and **actions**."

"Data storytelling aims to **democratize data**, making it **accessible** even to those **without** advanced analytical **skills**. In the past, data analysis was often considered an elitist activity, but data **storytelling** has **change**d this **perspective**."

"Human **decisions** are **influenced** by cognitive **biases**, **emotions**, and **conceptual** leaps beyond what the data can suggest. The **best** way to **educate** and **persuade** decision-makers is through stories. **Stories** built on good **data analysis**."

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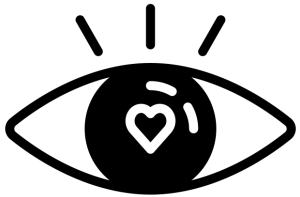
The power of data visualization



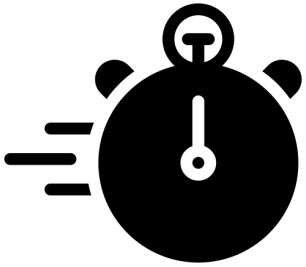
We are receiving more and more information



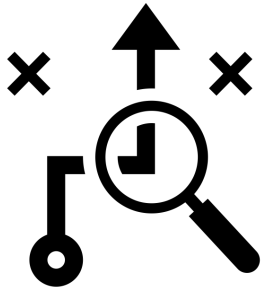
Increase the audience



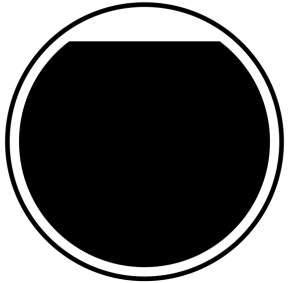
Grab attention



In a few seconds it conveys the message

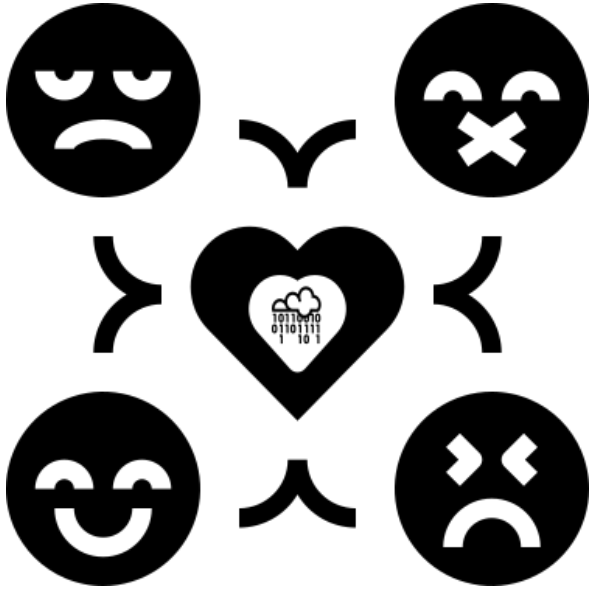


Helps locate information



We have an ever-increasing amount of data

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STORY must be at the **center**

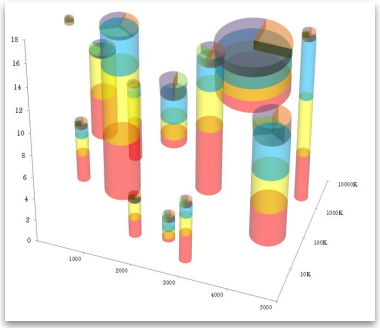
The **images**, **chart** and **data support** the narrative.



Here I'll tell you a story where 0.6%....



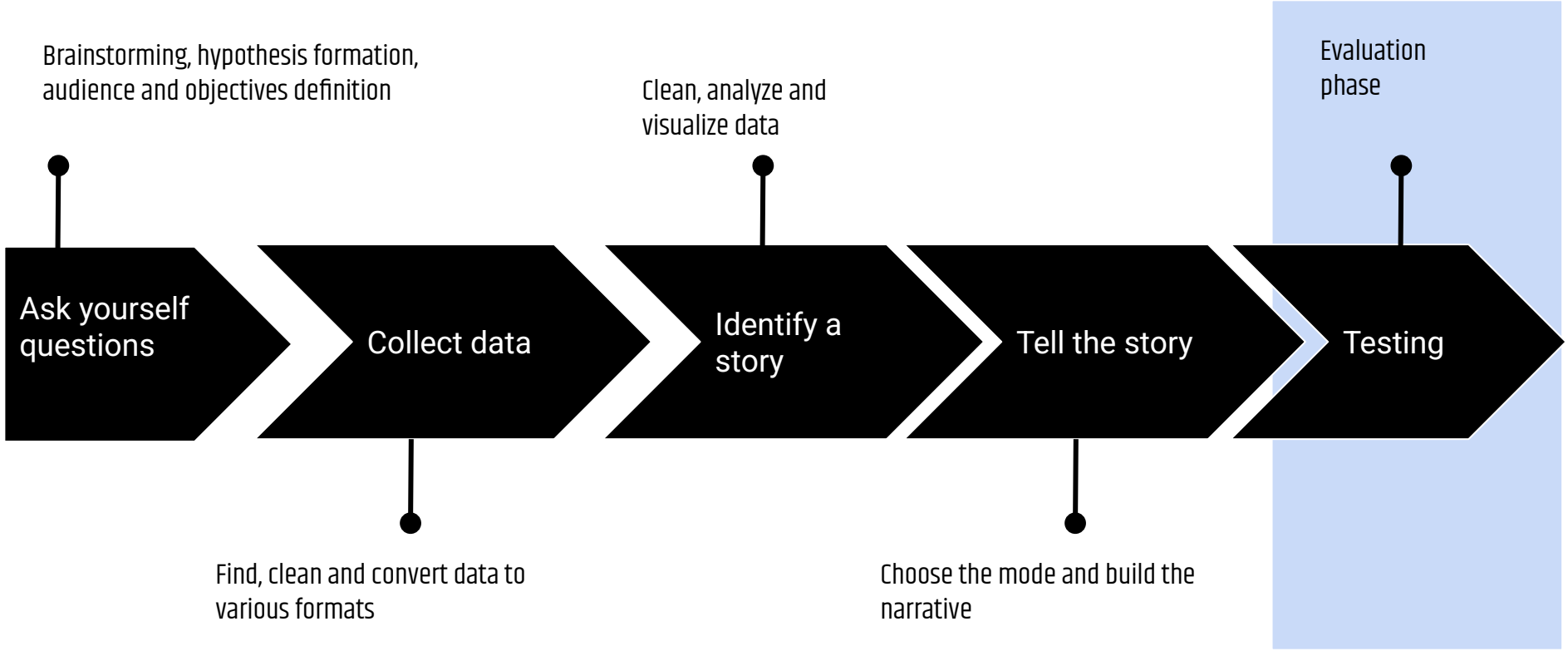
What is that graph for?



Nice! What does it mean?

Being able to convey emotions with data!!!

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a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

Ask yourself questions

Example: cycle paths in Berlin

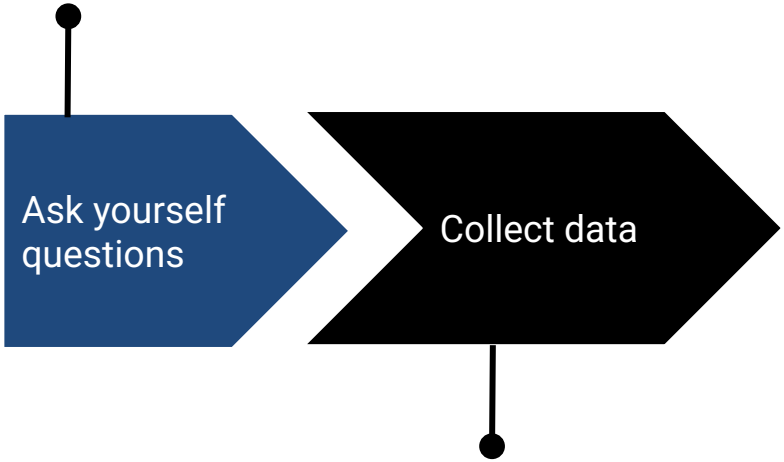
Find out how Berlin's streets and, above all, cycle paths should be designed in the future so that not only urban planners and politicians feel comfortable with them, but also the people who ride them.





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Brainstorming, hypothesis formation,
audience and objectives definition



Find, clean and convert data to various formats

You need to ask yourself some questions to identify biases in what you are looking for.
80% of this is about finding the right questions.
We must avoid data limiting our work

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Brainstorming, hypothesis formation, audience and objectives definition

Ask yourself questions

Collect data

Find, clean and convert data to various formats

Data to collect



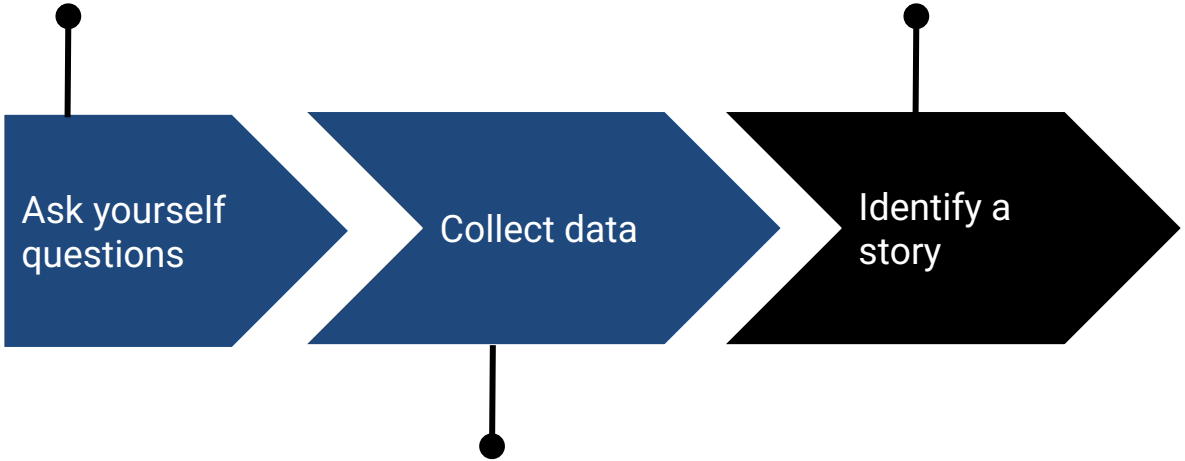
https://fixmyberlin.de/research/subjektive-sicherheit
https://interaktiv.tagesspiegel.de/lab/strassencheck-das-stoert-im-berliner-verkehr-am-meisten/

a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

Clean, analyze and visualize data



Find, clean and convert data to various formats

Statistical vocabulary

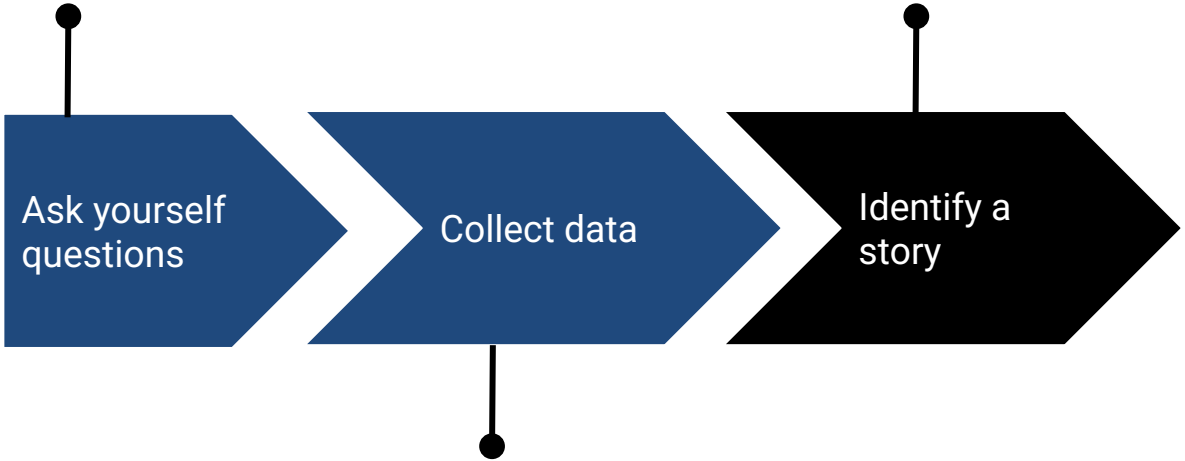
Minimum
Maximum
Average
Median
Quartile
...

a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

Clean, analyze and visualize data



Find, clean and convert data to various formats

Types of stories

Interesting facts
Surprising interactions
Changes over time
Personal experiences
Comparisons

a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

Clean, analyze and visualize data

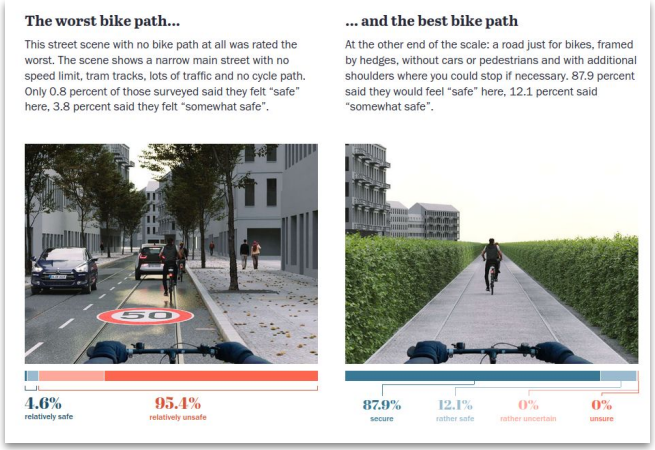
Ask yourself questions

Collect data

Identify a story

Find, clean and convert data to various formats

Surprising interactions



a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

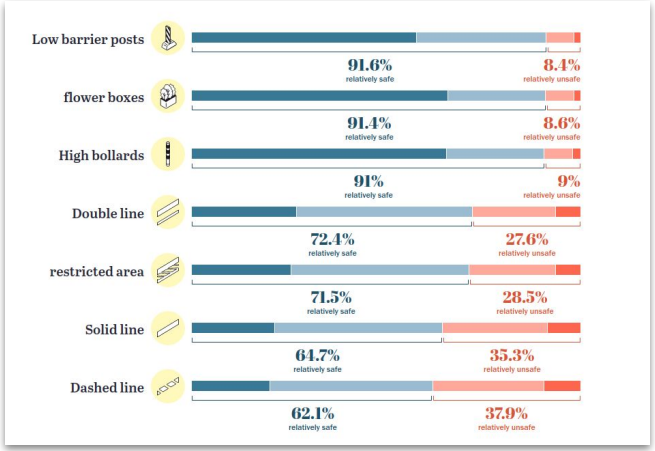
Clean, analyze and visualize data

Interesting facts

Ask yourself questions

Collect data

Identify a story



Find, clean and convert data to various formats

a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

Clean, analyze and visualize data

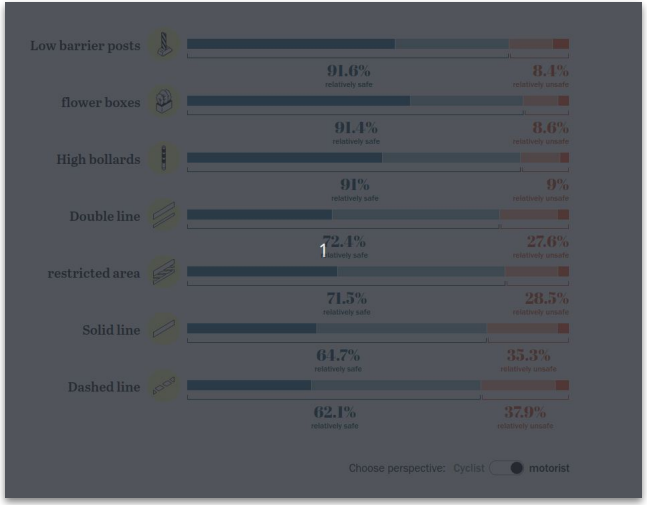
Comparisons

Ask yourself questions

Collect data

Individuare una storia

Find, clean and convert data to various formats



a process to follow

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Brainstorming, hypothesis formation, audience and objectives definition

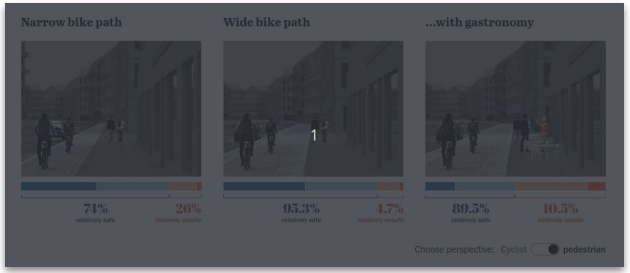
Clean, analyze and visualize data

Comparisons

Ask yourself questions

Collect data

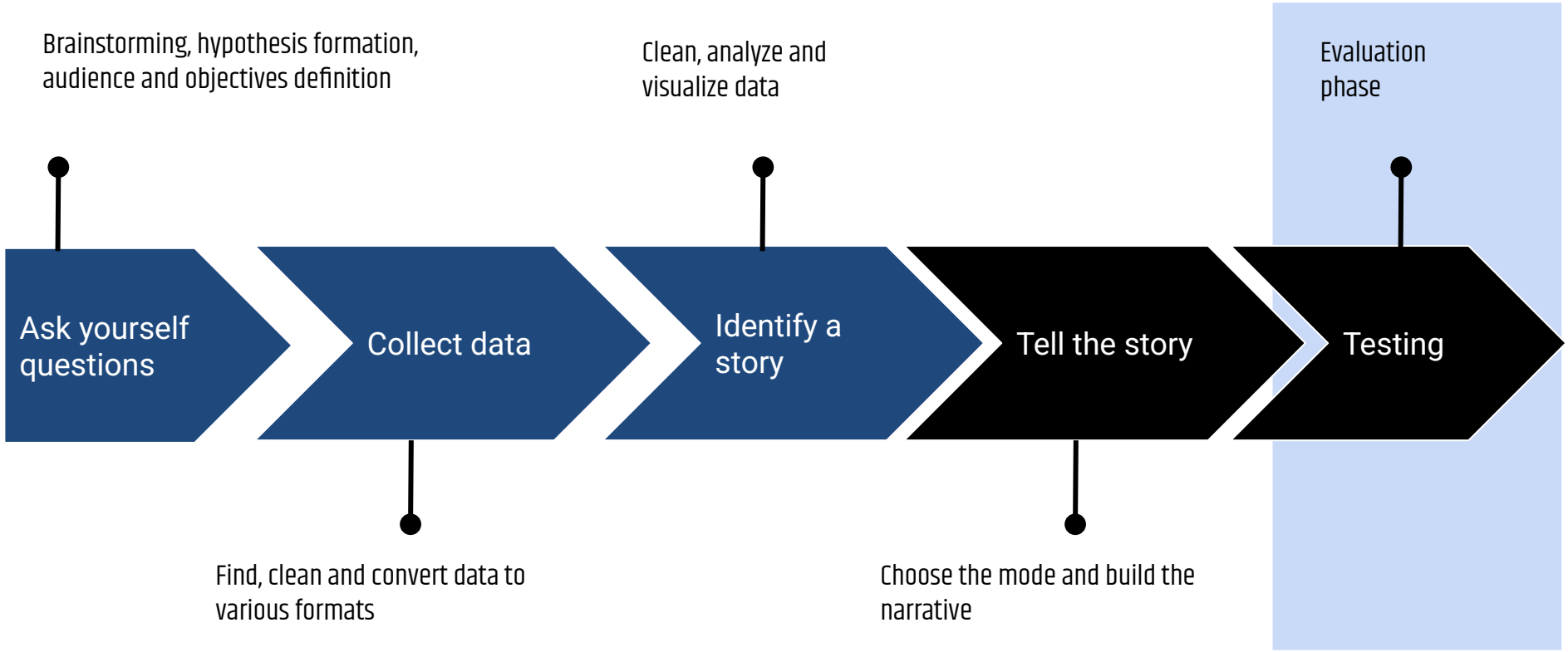
Identify a story



Find, clean and convert data to various formats

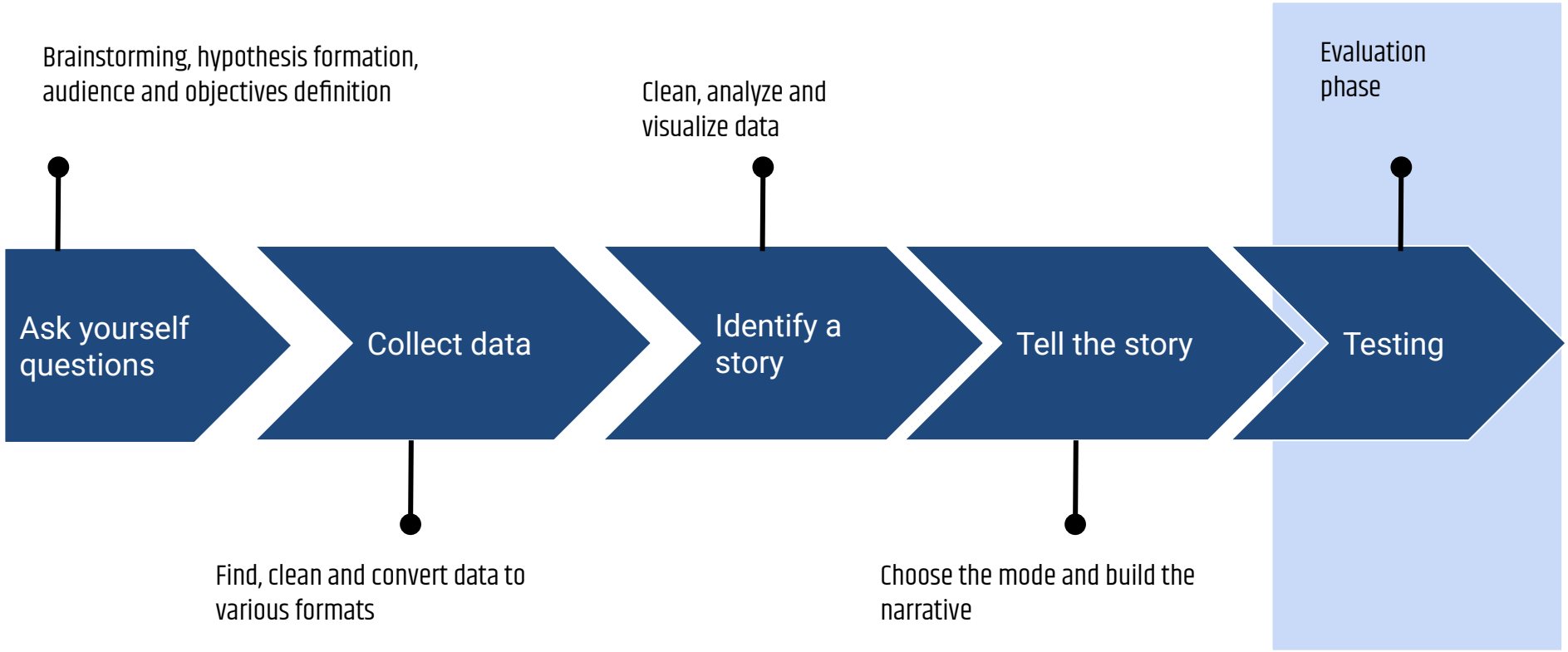
a process to follow

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un processo da seguire

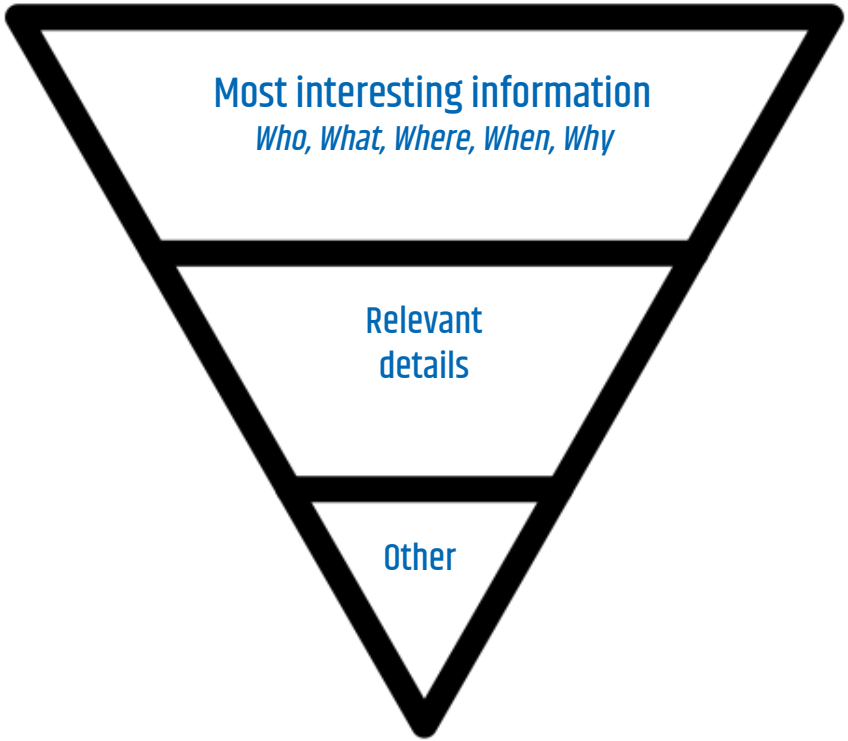
CC BY



a process to follow

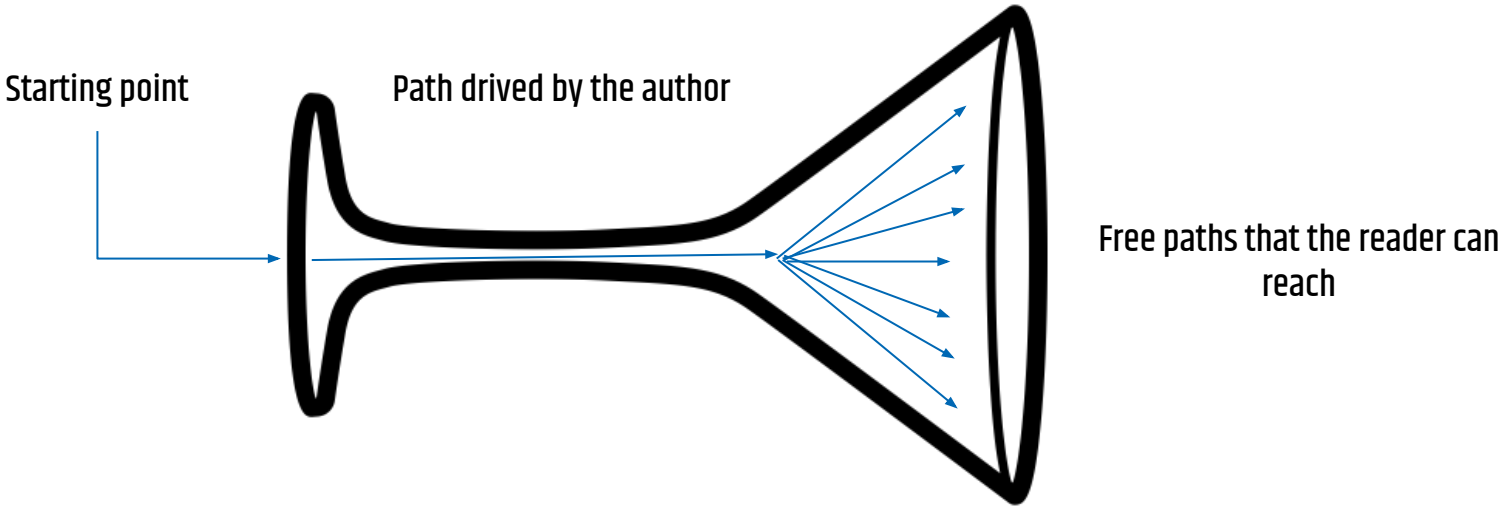
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The inverse pyramid



CC BY

The overturned martini glass



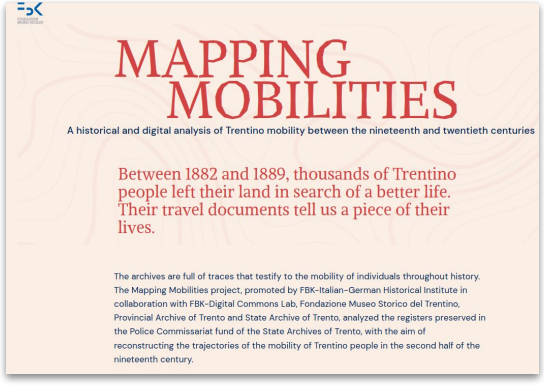
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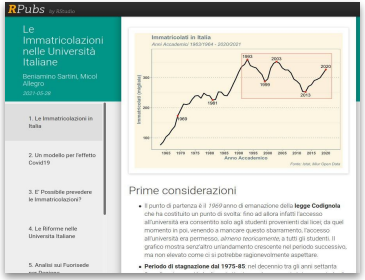


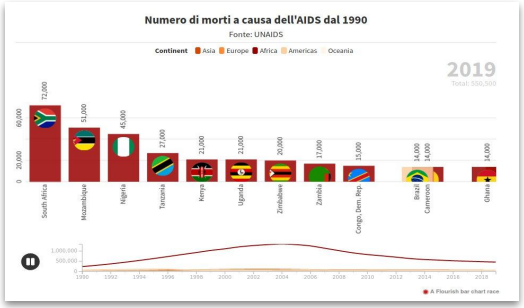




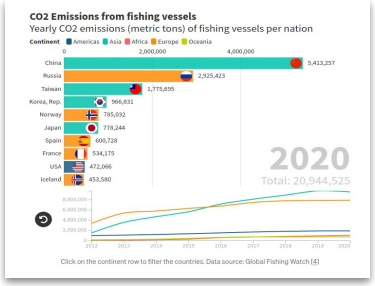
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Some students project works

knight lab

Storytelling

We build easy-to-use tools that can help you tell better stories.



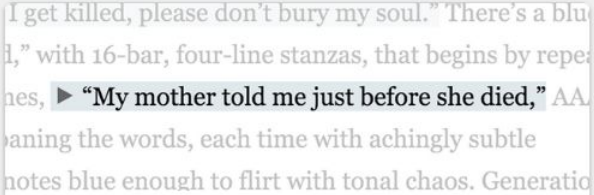
Juxtapose JS

Easy-to-make frame comparisons.



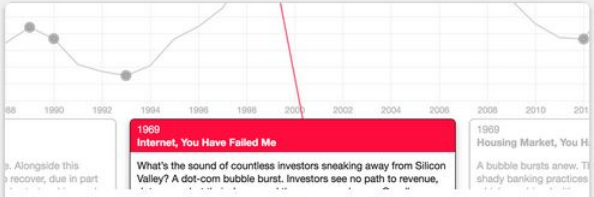
Scene VR

Easy-to-make VR stories.



Soundcite JS

Seamless inline audio.



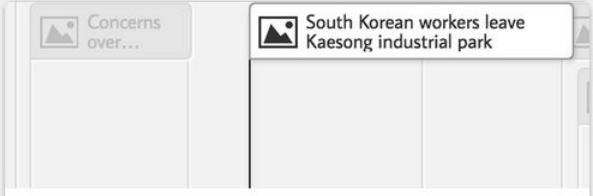
Storyline JS

Tell the story behind the numbers.



StoryMap JS

Maps that tell stories.



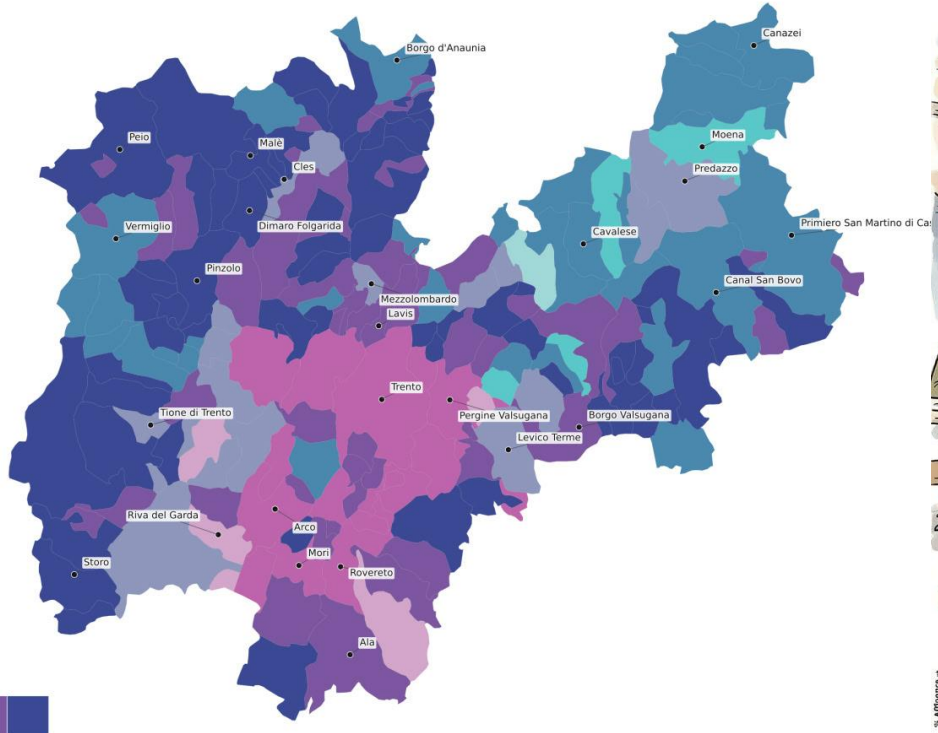
Timeline JS

Easy-to-make, beautiful timelines.

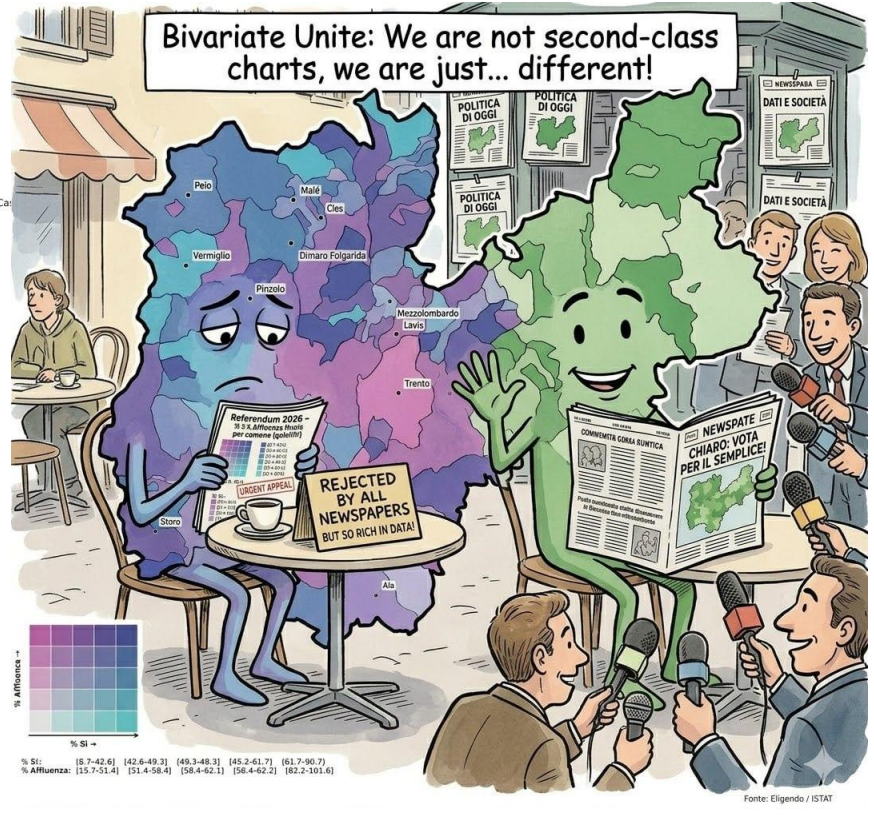
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UNIVERSITY OF TRENTO

Referendum 2026 – Bivariata: % Sì × % Affluenza finale per comune (terzili)



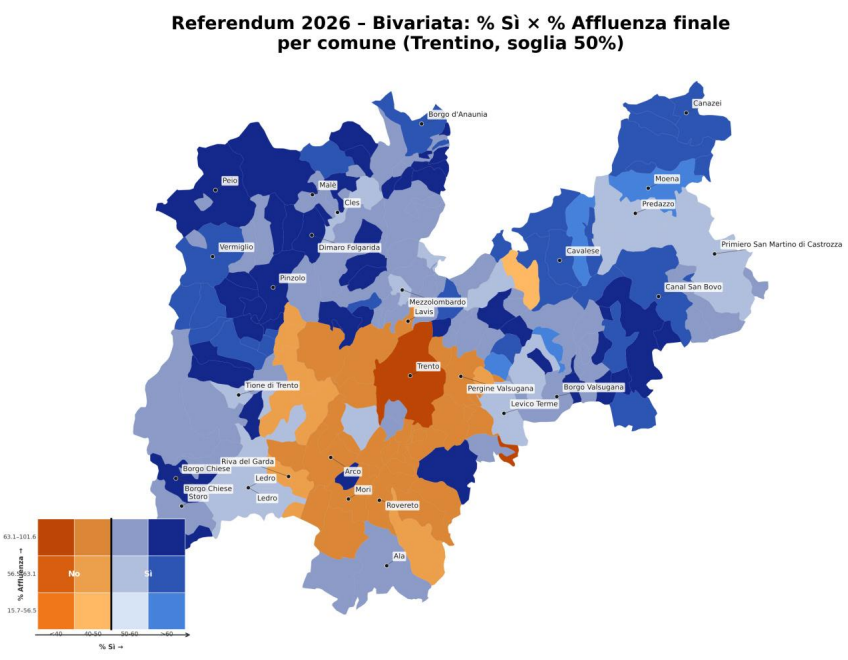
% Sì: [8.7–47.1] [47.1–57.3] [57.3–90.7]
% Affluenza: [15.7–56.5] [56.5–63.1] [63.1–101.6]



Fonte: Eligendo / ISTAT

Fonte: Eligendo / ISTAT

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16

Primo piano

La mappa affluenza esito del referendum

Referendum 2026 - Bivariata: % Sì × % Affluenza finale per comune (Trentino, soglia 50%)

Il «No» ha mobilitato più del «Sì»

La mappa che incrocia affluenza e voto rompe la dicotomia città-valli

La ricerca

L'analisi frutto del lavoro di tre «hacker civici» Napolitano: «Migliori visualizzazioni dei dati ci fanno capire meglio»

di *Simone Casciano*

La fotografia di un Trentino «spaccato in due» frutto di una prima analisi del voto referendario non è del tutto sbagliata, ma imprecisa. Se si prendono due colori e si colorano indistintamente con l'uno o l'altro a seconda del voto vincente, si fa un lavoro corretto, ma che lascia fuori tanto. Sfumature cromatiche che sono però anche dati importanti che raccontano la realtà ben più complessa del territorio. Una realtà che prova a ritrarre invece la mappa bivariata rappresentata in questo articolo che incrocia la quota di voto e l'affluenza, con una distinzione cromatica efficace tra aree a prevalenza «Sì» (blu) e «No» (arancione), graduata per partecipazione. Il prezioso lavoro è frutto della collaborazione tra tre ricercatori e «hacker civici». Raffaele Mastrolonardo ha recuperato i dati del voto di Eligendo e i confini dei Comuni dall'Istat, Andrea Bonasso ha avuto l'idea di costruire la mappa bivariata e Maurizio Napolitano ha realizzato l'adattamento del Trentino.

Un mosaico complesso

La geografia del voto referendario in Trentino restituisce un'immagine tutt'altro che uniforme e, soprattutto, smonta alcune narrazioni semplificate. L'incrocio tra affluenza e orientamento di voto evidenzia una frattura territoriale netta, ma anche più complessa di quanto suggerisca la tradizionale contrapposizione tra città e valli. Il dato più evidente è la concentrazione del «No» nell'area centrale del Trentino, con un nucleo compatto che comprende Trento e si estende lungo l'asse dell'Adige fino a Rovereto e all'Alto Garda. Qui prevalgono tonalità arancioni anche intense, segno non solo di una maggioranza contraria al quesito, ma spesso accompagnata da livelli medio-alti di partecipazione. È un elemento politicamente rilevante: nelle aree più urbanizzate e densamente popolate, il voto contrario non è stato passivo o marginale, ma espresso con una mobilitazione significativa. All'opposto, gran parte delle aree periferiche e montane si colloca nel campo del «Sì», con una diffusione ampia di blu soprattutto nelle valli del nord e dell'est. Tuttavia, questo dato va letto insieme all'affluenza: molte di queste zone mostrano livelli di partecipazione più bassi o intermedi. Ne emerge un quadro in cui il consenso al «Sì» è spesso meno mobilitato, più diffuso territorialmente ma meno intenso sul piano della partecipazione. Questa combinazione suggerisce una prima chiave di lettura: il «No» si concentra dove la partecipazione è più alta, mentre il «Sì» è più presente dove il coinvolgimento elettorale è mediamente più debole. Non si tratta di una regola assoluta, ma di una tendenza chiaramente visibile nella mappa. C'è poi un secondo elemento che merita attenzione: la non totale sovrapposizione tra città e «No». Se è vero che Trento rappresenta il baricentro del voto contrario, anche alcune aree vallive mostrano dinamiche meno lineari, con risultati più equilibrati o addirittura con una presenza significativa del «No». Questo rafforza una lettura già emersa in altri contesti: l'affluenza non è un semplice dato quantitativo, ma un indicatore qualitativo della capacità della politica di attivare consenso. Le aree in cui si registra maggiore partecipazione sono anche quelle in cui il messaggio politico – in questo caso il rifiuto del quesito – è passato con più forza. In sintesi, la mappa racconta un dato politico chiaro: il terreno decisivo non è solo quello del consenso, ma anche quello della partecipazione.

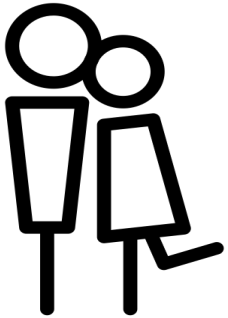
Hacker civico Maurizio Napolitano

«Mappe per capire meglio»

«L'idea del lavoro è quella di visualizzare bene i dati e poi lasciare che sia il lettore a interpretarli – spiega Maurizio Napolitano – Questo doppio indicatore permette valutazioni più informate». Per un appassionato di dati come Napolitano le mappe sono una passione, ma anche una «croce». «Ogni mappa ha i suoi "pregiudizi" a cui dobbiamo stare attenti. Per esempio, usando i confini dei Comuni per colorare il voto non possiamo rappresentare Moldonia di Campiglio, il cui territorio si trova per metà nel comune di Tre Ville e per metà in quello di Pinzolo.

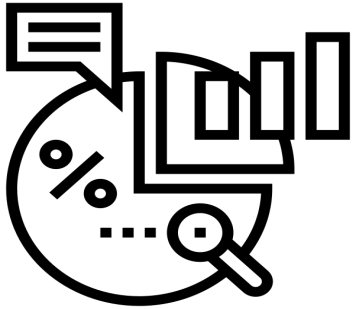
C'è poi il tema territorio-popolazione. Il comune di Daone è grande quanto la città di Palermo, ma con una popolazione estremamente più bassa, però su queste mappe appare enorme. Questa mappa aiuta ad avere una visualizzazione migliore, ma poggia anche sulle conoscenze del territorio da parte del lettore». Secondo Napolitano «servono più mappe, che mostrino dati diversi oppure, meglio ancora, mappe interattive che permettano di visualizzare informazioni differenti».

L'analisi | La coordinatrice di Unitin Pinelli: «Noi giovani abbiamo trainato il referendum»



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Examples

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